

Sri Lanka Institute of Information Technology



Assignment 1

Year 2 Semester 1

Done by

IT number :-IT23363120

Name :- A.M.K.S Attanayake

IE2012 - Systems and Network Programming

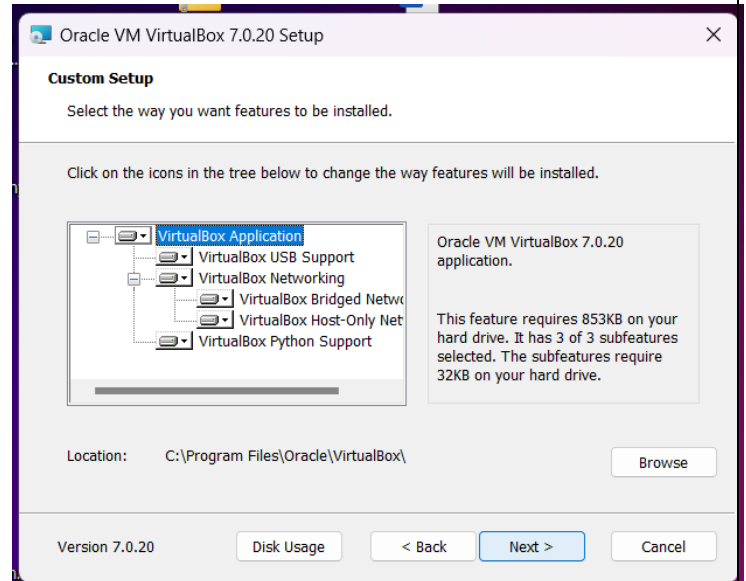
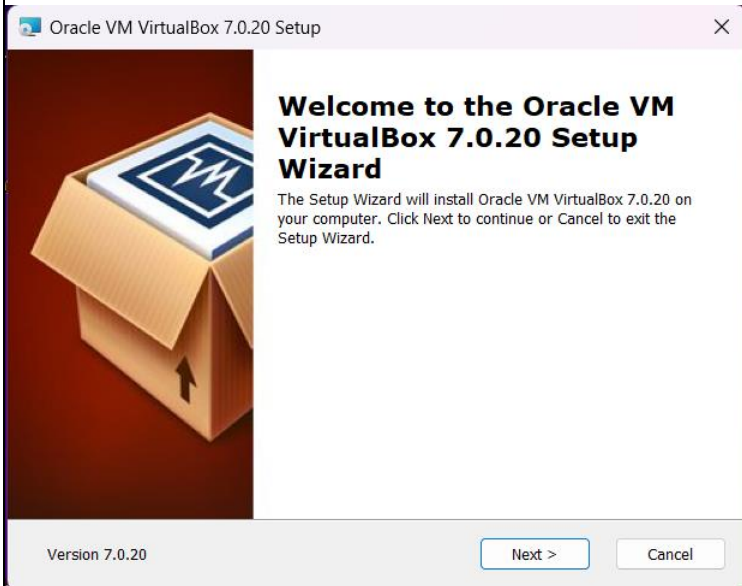
B.Sc. (Hons) in Information Technology Specializing in Cyber Security

Table of Contents

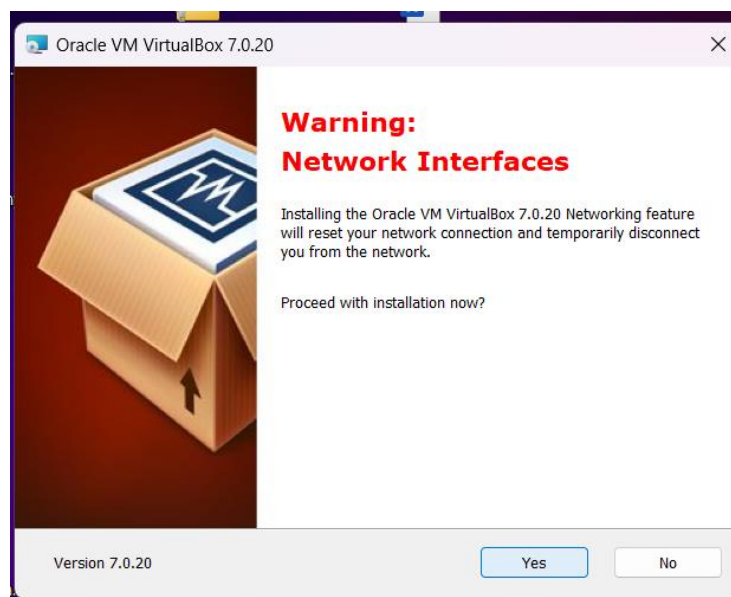
1.Virtual Machine Setup:	3
2.Command Line Introduction	9
3.System Information and User Management...	13
4.Configuring DHCP server	18
5.DNS (Domain Name System)	32
6.NTP(Network Transfer Protocol)	41
7.Shell scripting and security	44
8.SSH(Secure Shell)	54
9.Iptables and ACLs.....	60
10.Best Practices.....	65

1.Virtual Machine Setup:

Installing the Oracle virtual box machine.

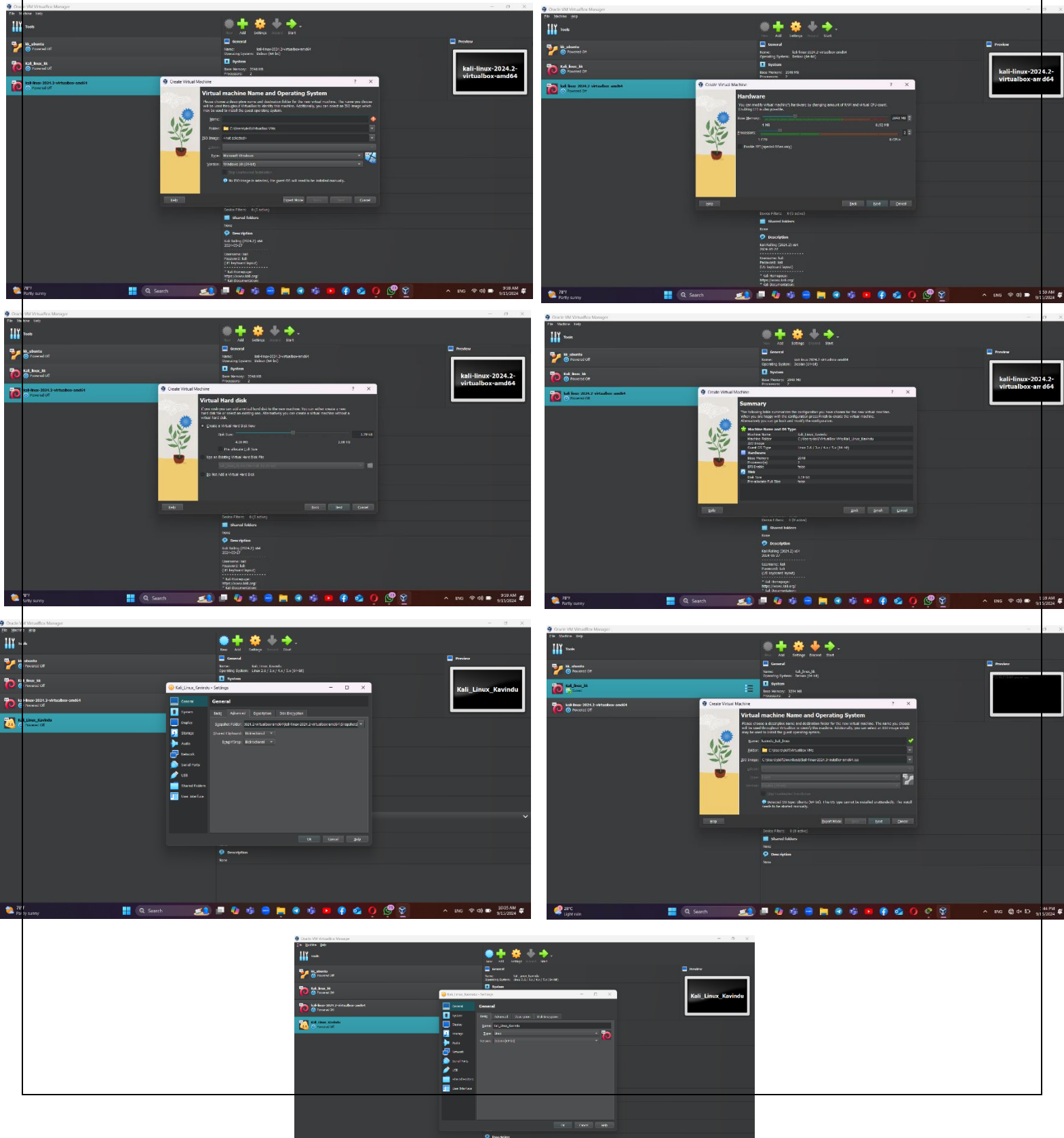


Finishing the installation.

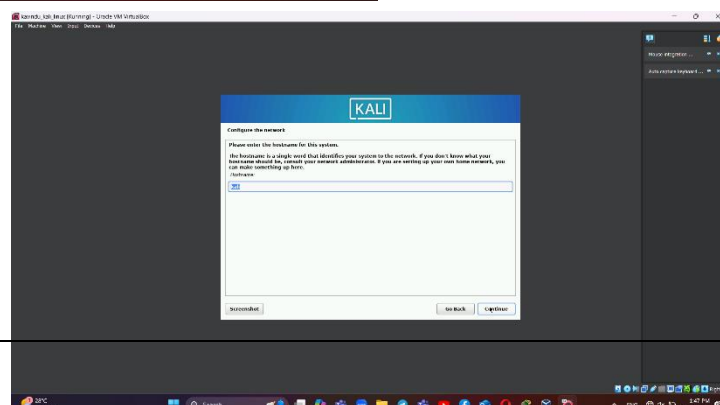
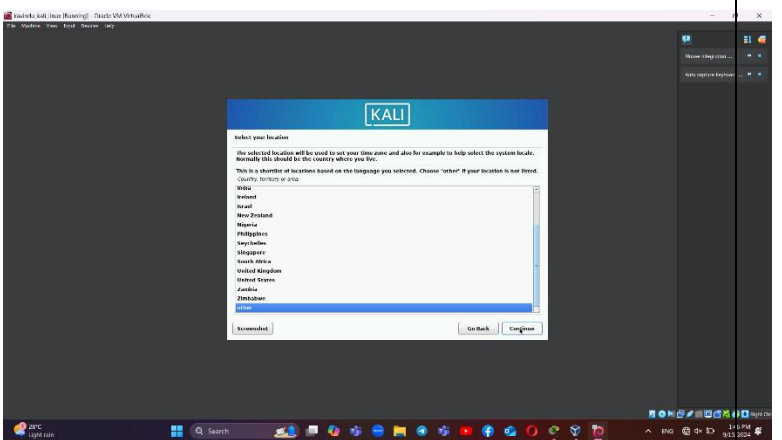
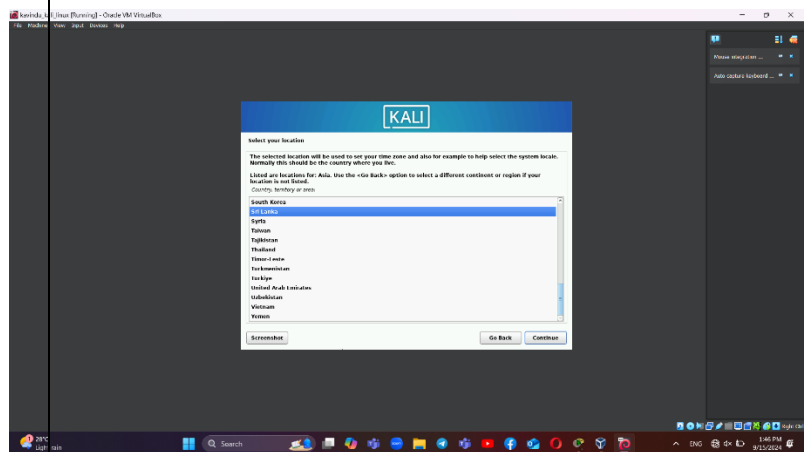
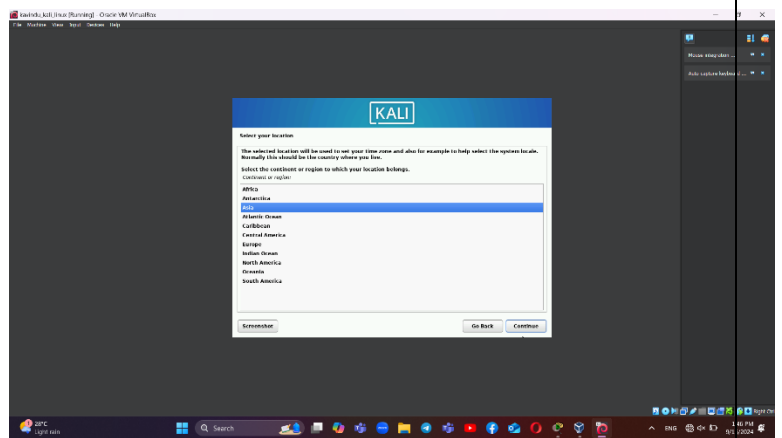
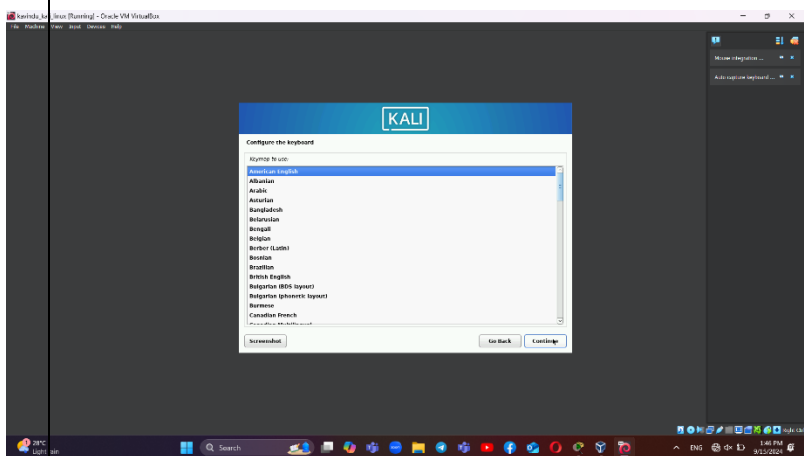
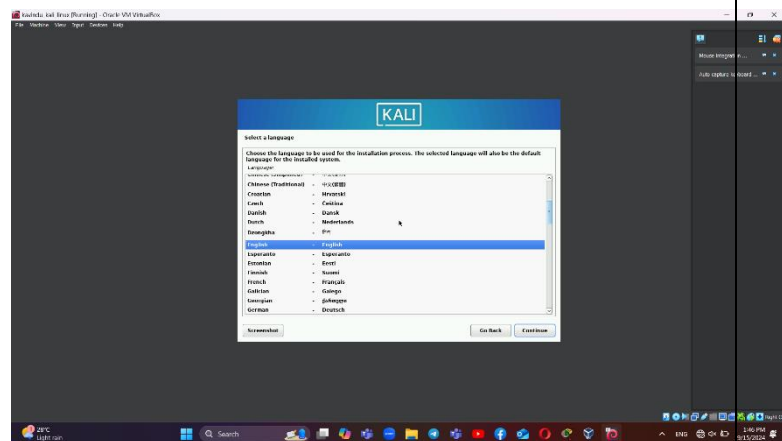
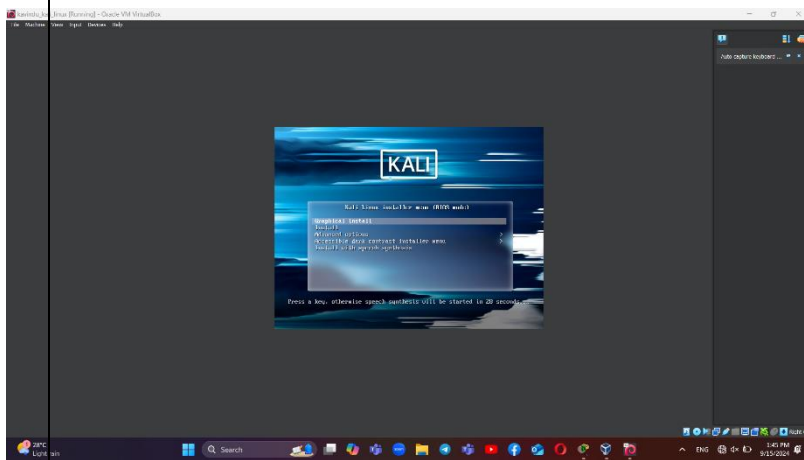


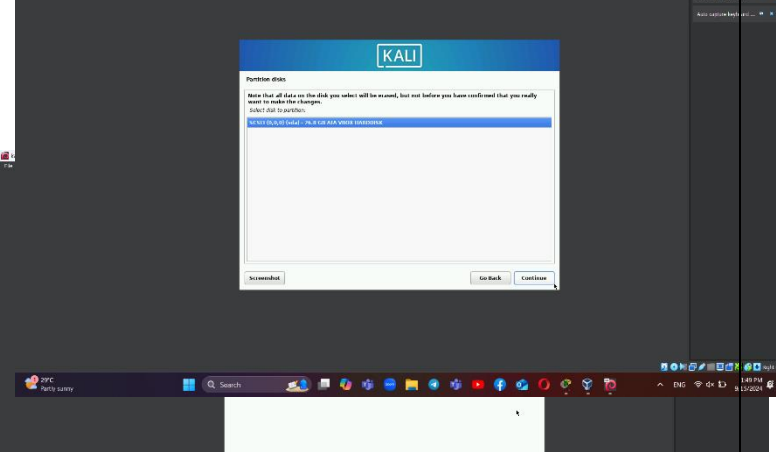
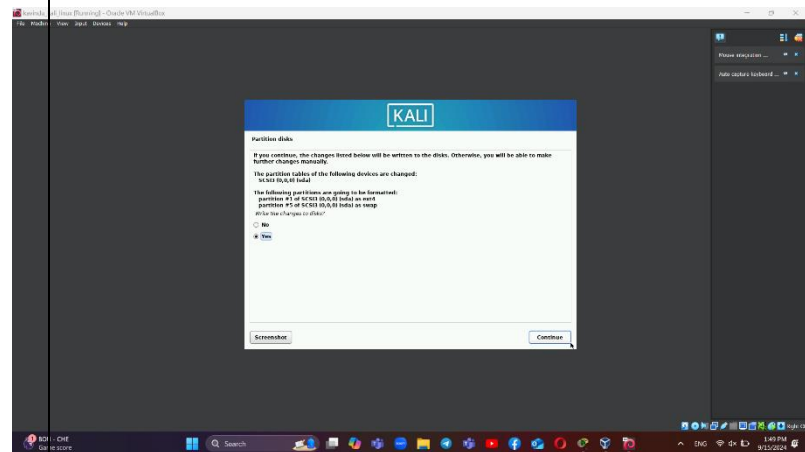
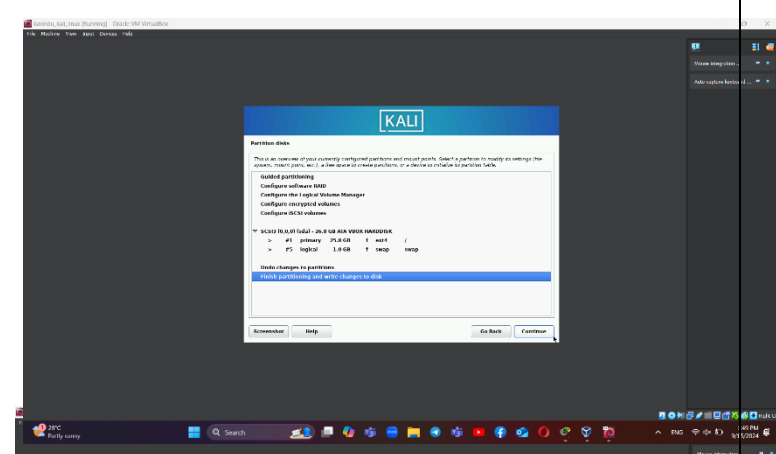
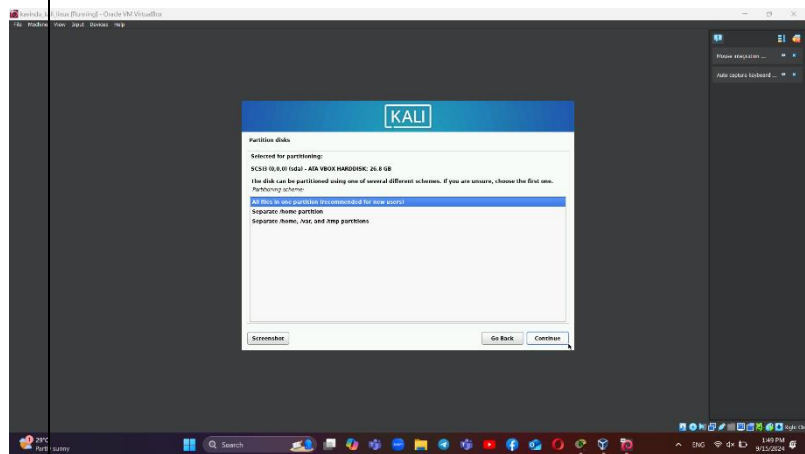
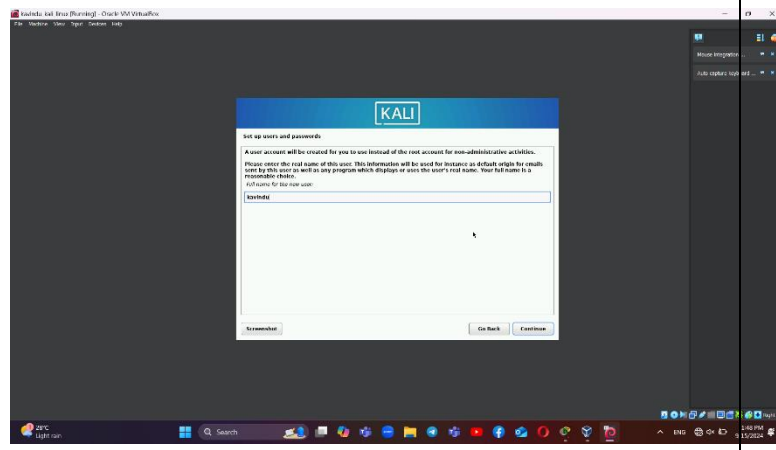
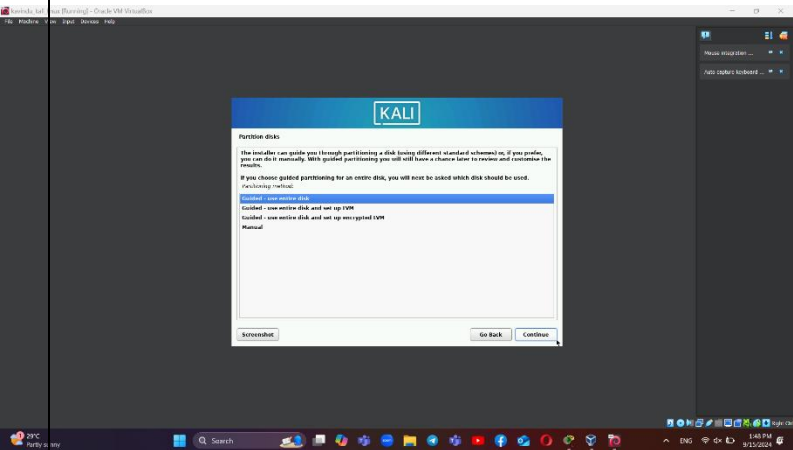
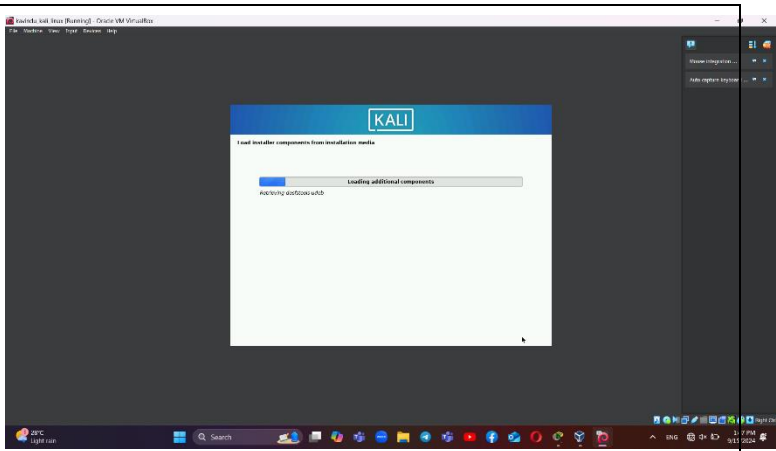
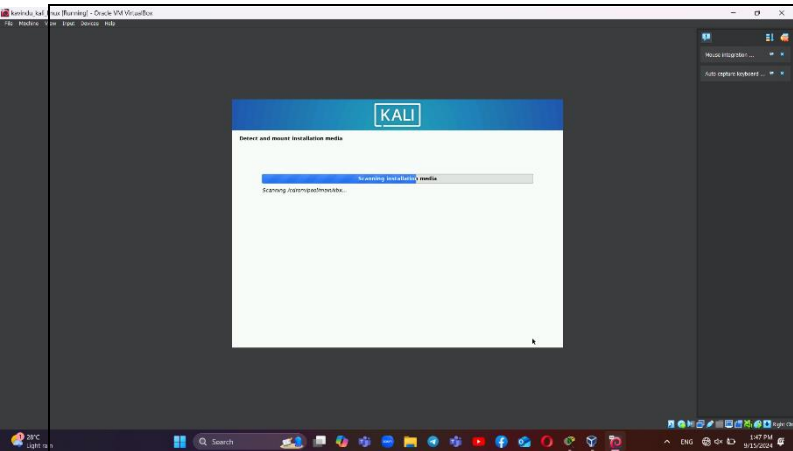
Installing Linux distribution to the VM . Here I choose to install kali Linux.

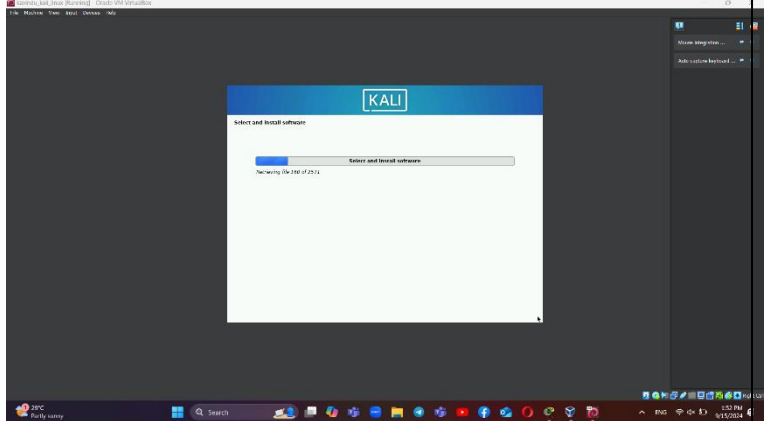
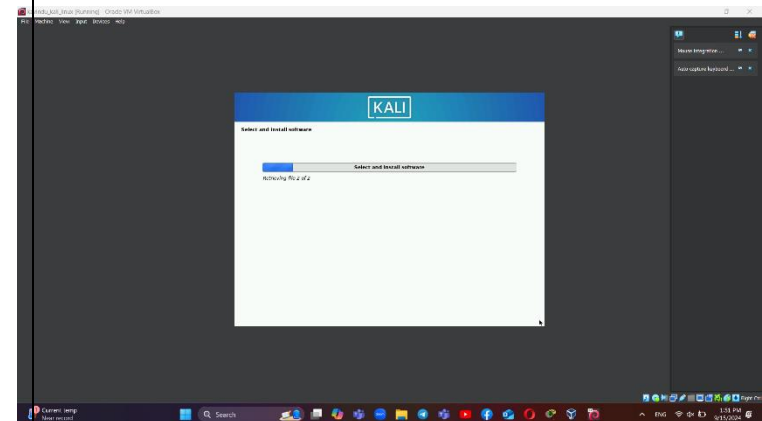
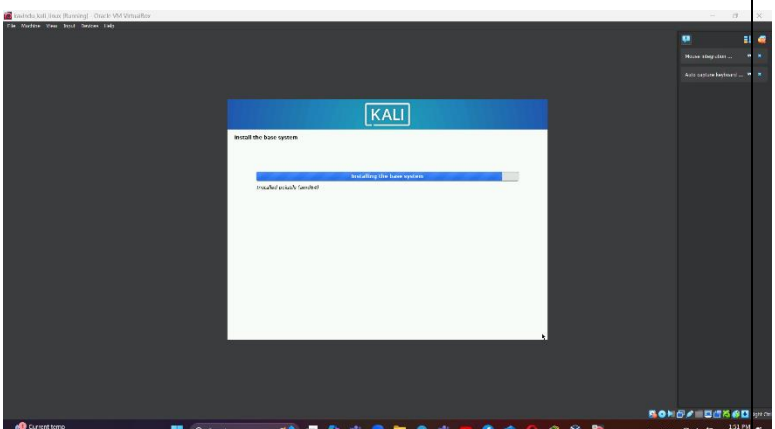
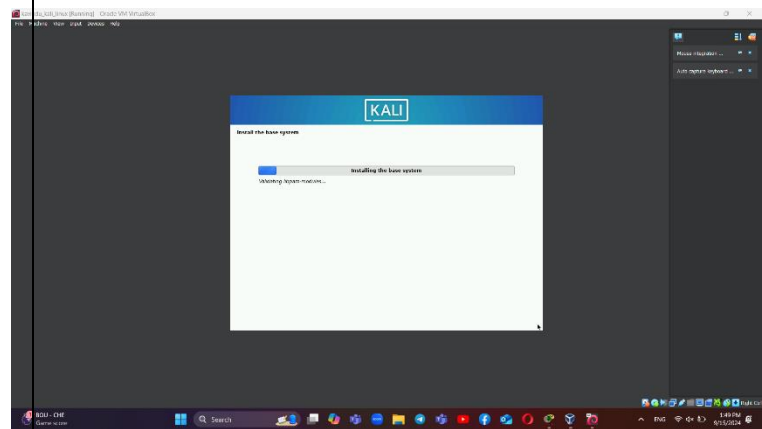
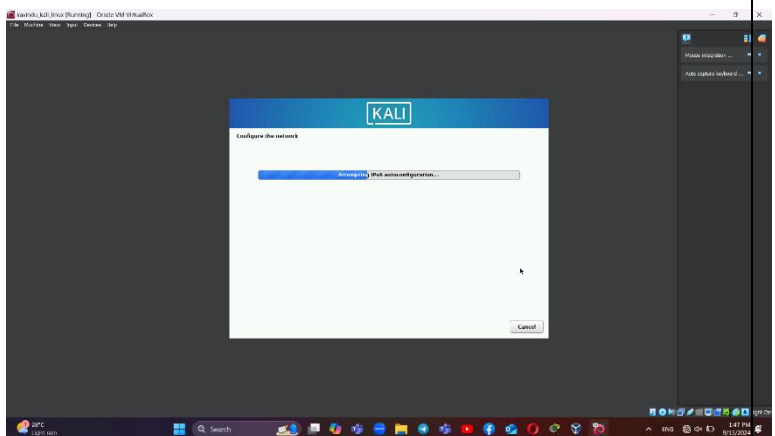
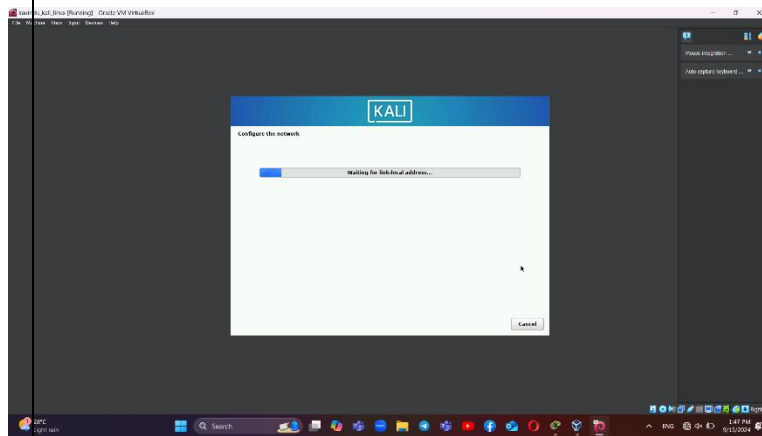
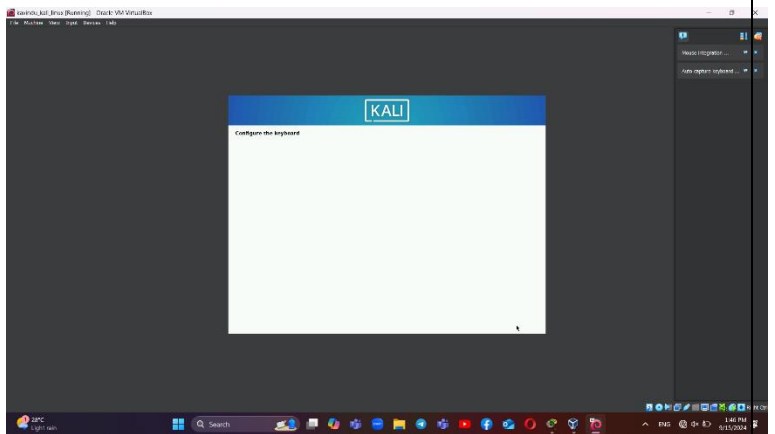
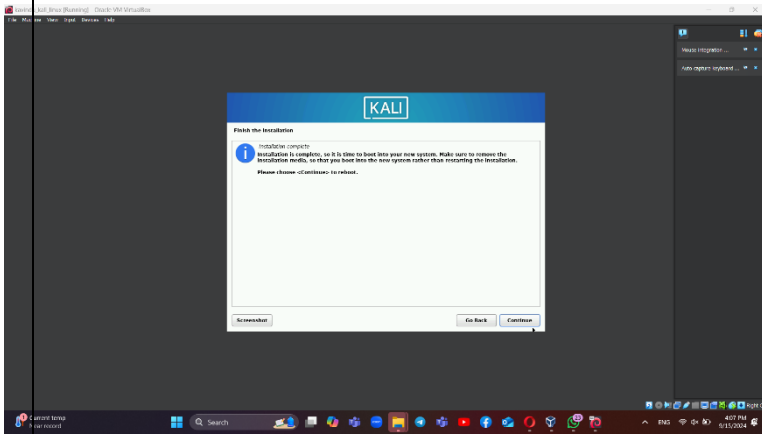
Here I allocated necessary resources.

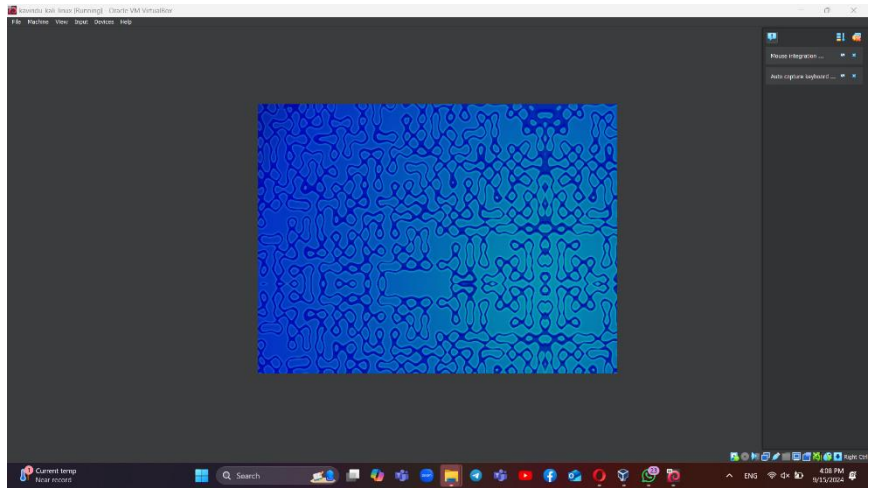
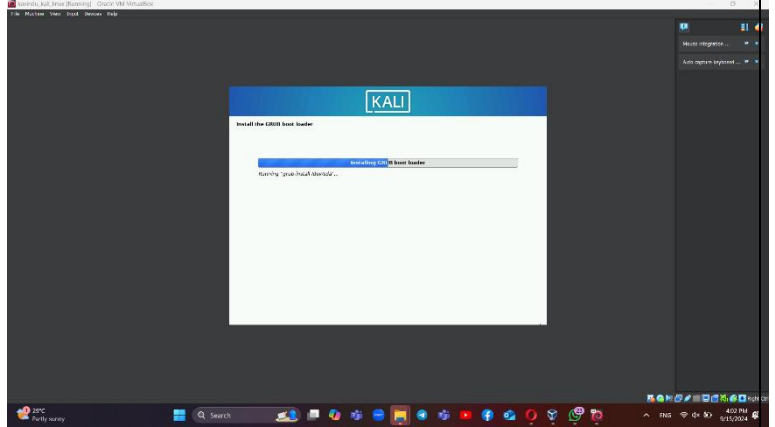
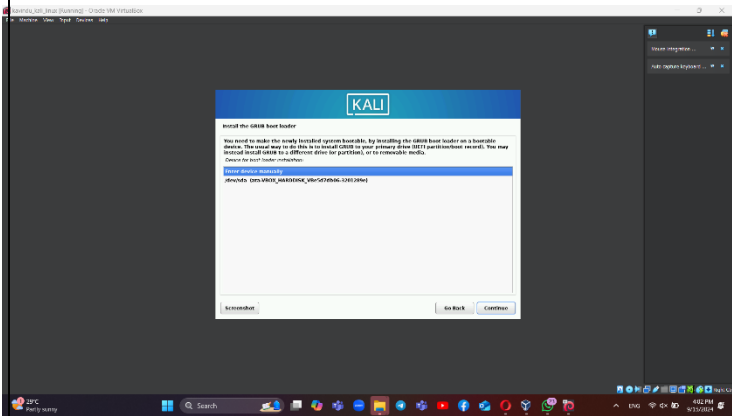


Running the pre installed kali and setting up.







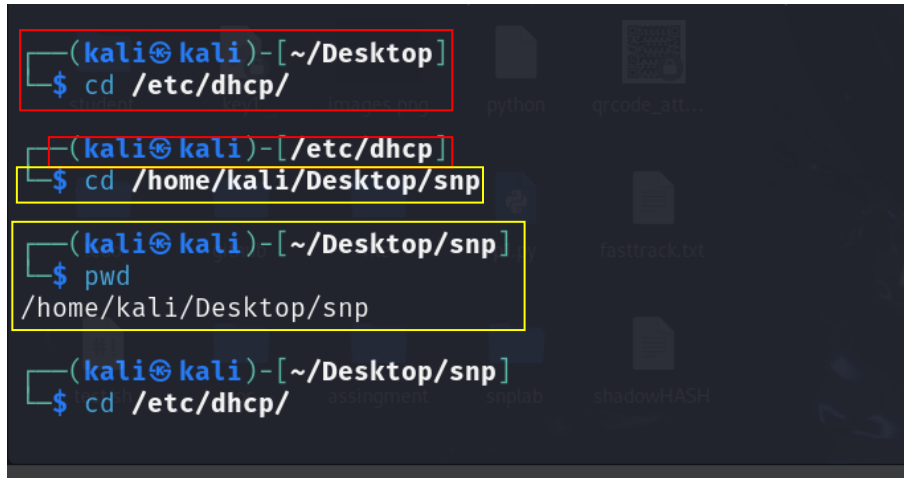


2.Command Line Introduction

- **cd** (Change Directory): Move to a different directory.

cd /path/to/directory

cd /home/user/Documents



```
(kali㉿kali)-[~/Desktop]
$ cd /etc/dhcp/

(kali㉿kali)-[/etc/dhcp]
$ cd /home/kali/Desktop/snp

(kali㉿kali)-[~/Desktop/snp]
$ pwd
/home/kali/Desktop/snp

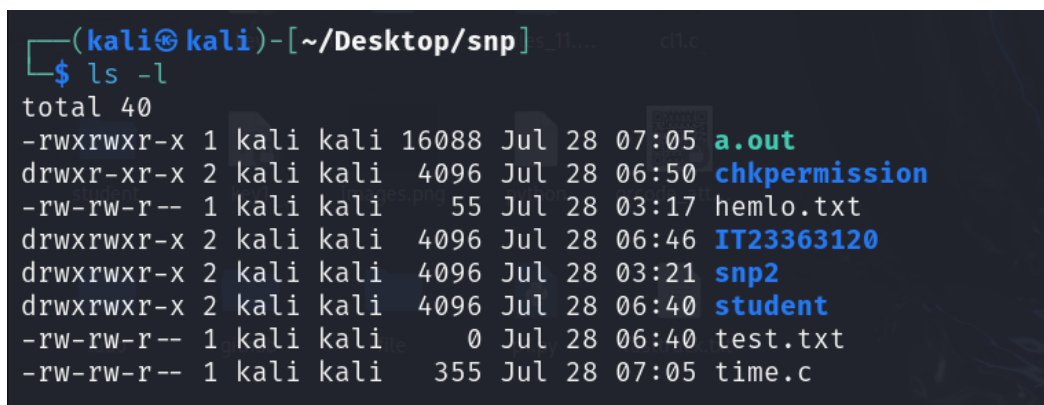
(kali㉿kali)-[~/Desktop/snp]
$ cd /etc/dhcp/
```

- **ls** (List): Display the contents of a directory.

ls *# List files and directories in the current directory*

ls -l *# List with detailed information (permissions, size, date, etc.)*

ls -a *# List all, including hidden files (those starting with a dot)*



```
(kali㉿kali)-[~/Desktop/snp]
$ ls -l
total 40
-rwxrwxr-x 1 kali kali 16088 Jul 28 07:05 a.out
drwxr-xr-x 2 kali kali 4096 Jul 28 06:50 chkpermission
-rw-rw-r-- 1 kali kali 55 Jul 28 03:17 hemlo.txt
drwxrwxr-x 2 kali kali 4096 Jul 28 06:46 IT23363120
drwxrwxr-x 2 kali kali 4096 Jul 28 03:21 snp2
drwxrwxr-x 2 kali kali 4096 Jul 28 06:40 student
-rw-rw-r-- 1 kali kali 0 Jul 28 06:40 test.txt
-rw-rw-r-- 1 kali kali 355 Jul 28 07:05 time.c
```

- **pwd** (Print Working Directory): Show the current directory you are in.

pwd

```
(kali㉿kali)-[~/Desktop/snp]
$ pwd
/home/kali/Desktop/snp
```

- **mkdir** (Make Directory): Create a new directory.

mkdir new_directory

```
(kali㉿kali)-[~/Desktop/snp]
$ mkdir snp_assingment

(kali㉿kali)-[~/Desktop/snp]
$ mkdir dir1

(kali㉿kali)-[~/Desktop/snp]
$ ls
a.out  chkpermission  dir1  hemlo.txt  IT23363120  snp2  snp_assingment  student  test.txt  time.c
```

- **rmdir** (Remove Directory): Remove an empty directory.

rmdir empty_directory

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4

kali@kali: ~/Desktop
File Actions Edit View Help
$ rmdir dir1

(kali㉿kali)-[~/Desktop/snp]
$ ls
a.out  chkpermission  hemlo.txt  IT23363120  snp2  snp_assingment  student  test.txt  time.c
```

- **rm -r** (Remove recursively): Remove a directory and all its contents.

rm -r folder_name

```
(kali㉿kali)-[~/Desktop/snp]
$ touch kk.text

(kali㉿kali)-[~/Desktop/snp]
$ ls
a.out  chkpermission  hemlo.txt  IT23363120  kk.text  snp2  snp_assingment  student  test.txt  time.c

(kali㉿kali)-[~/Desktop/snp]
$ rm -r kk.text

(kali㉿kali)-[~/Desktop/snp]
$ ls
a.out  chkpermission  hemlo.txt  IT23363120  snp2  snp_assingment  student  test.txt  time.c
```

- **cp** (Copy): Copy files or directories.

cp file1.txt /path/to/destination

cp -r directory_name /path/to/destination # Copy a directory recursively

```
(kali@kali)-[~/Desktop/snp]
$ touch snp_aasingment.txt

(kali@kali)-[~/Desktop/snp]
$ cp snp_aasingment.txt new.txt

(kali@kali)-[~/Desktop/snp]
$ ls
a.out  chkpermission  hemlo.txt  IT23363120  new.txt  snp2  snp_aasingment.txt  snp_assingment  student  test.txt  time.c

(kali@kali)-[~/Desktop/snp]
$ mv new.txt /home/kali/Desktop/new2.txt
```

- **mv** (Move): Move or rename files.

mv file.txt /new/location/

mv oldname.txt newname.txt

```
(kali@kali)-[~/Desktop/snp]
$ mv new.txt /home/kali/Desktop/new2.txt

(kali@kali)-[~/Desktop/snp]
$ cd ../

kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali: ~/Desktop
File Actions Edit View Help
(kali@kali)-[~/Desktop]
$ cd ../

(kali@kali)-[~/Desktop]
$ ls
app  application_lab  assignment2  assignment2.tar.gz  assingment  bandit  banditpassword.txt  cli.c  cli1  cli.c  fasttrack.txt  file  github  images.png  key  key1  ludo  ludo_final  natas12_form.html  natas12_getpass.php  new2.txt  pi.py  python  qrcode_attack.png  ser1.c  shadowHASH  sliit  snp  snplab  sos  student  test  test.sh  textfiles  textfiles_11.tar.gz  UCSC  week1
```

- **cat** (Concatenate): Display the contents of a file.

cat filename.txt

```
(kali@kali)-[~/Desktop]
$ cat banditpassword.txt
-level 0→ 1 : ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If
level 1→ 2 : 263JGJPfgU6LtdEvfgWU1XP5yac29mFx
level 2→ 3 : MNk8KNH3Usiio41PRUEoDFPqfxLPLSmx
level 3→ 4 : 2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ
level 4→ 5 : 4oQYVPkxZ00E005pTW81FB8j8lxXGUQw
level 5→ 6 : HWasnPhtq9AVKe0dmk45nxy20cvUa6EG
level 6→ 7 : morbNTDkSW6jIlUc0ymOdMaLn0lFVAaj
level 7→ 8 : dfwvzFQi4mU0wfNbFoE9RoWskMLg7eEc
level 8→ 9 : 4CKMh1JI91bUIZZPXQdGanal4xvAg0JM
level 9→ 10 : FGUW5iLLVJrxX9kMYMmlN4MgbpfMiqey
level 10→ 11 : dtR173fZKb0RRsDFSGsg2RwnpNVj3qRr
level 11→ 12 : 7x16WNeHIi5YkIhWsfFIqoognUTyj9Q4
level 12→ 13 : F05dwFsc0cbaIiH0h8J2eUks2vdTDwAn
level 13→ 14 : MU4VWeTyJk8R0of1qqmcBPALh7lDCPvS
level 14→ 15 : 8xCjnmgoKbGLhHFAZLGE5Tmu4M2tKJQo
level 15→ 16 : kSkvUpMQ7lBYyCM4GBPvCvT1BfWRY0Dx
level 16→ 17 : file in desktop
level 17→ 18 : cGwPmaKXVwDUNgPAVJbWYUgHVn9z13j8
level 18→ 19 :
level 19→ 20 : 0qXahG8ZjOVmn9Ghs7iOWsCfZyXOUbYO
level 20→ 21 : EeoULMCra2q0dSkYj561DX7s1CpBuOBt
level 21→ 22 : tRae0UfB9v0UzbCdn9cY0gQnds9GF58Q
```

- **touch:** Create an empty file.

touch newfile.txt

```
(kali@kali)-[~/Desktop/snp]
$ touch snp_aasingment.txt
```

```
(kali@kali)-[~/Desktop/snp]
$ ls
a.out  chkpermission  hemlo.txt  IT23363120  new.txt  snp2  snp_aasingment.txt  snp_aasingment  student  test.txt  time.c
```

- **rm** (Remove): Delete files.

rm filename.txt

rm -f filename.txt # Force delete (no confirmation)

- **nano or vim:** Edit files using a terminal-based text editor.

nano file.txt # Opens the file in the Nano editor for editing

vim file.txt # Opens the file in the Vim editor

```
(kali@kali)-[~/Desktop]
$ sudo vim /home/user/backup_documents.sh

(kali@kali)-[~/Desktop]
$ sudo vim /home/user/backup_documents.sh
```

3. System Information and User Management

1. `uname -a`

Displays detailed information about the kernel and operating system.

`uname -a`

```
(kali㉿kali)-[~/Desktop]
$ uname -a
Linux kali 6.6.15-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.6.15-2kali1 (2024-05-17) x86_64 GNU/Linux
```

2. `cat /proc/version`

Shows the Linux kernel version and other details.

`cat /proc/version`

```
(kali㉿kali)-[~/Desktop]
$ cat /proc/version
Linux version 6.6.15-amd64 (devel@kali.org) (gcc-13 (Debian 13.2.0-24) 13.2.0, GNU ld (GNU Binutils for Debian) 2.42) #1 SMP PREEMPT_DYNAMIC Kali 6.6.15-2kali1 (2024-05-17)
```

3. `df -h`

Displays disk space usage for all mounted filesystems in a human-readable format.

`df -h`

```
(kali㉿kali)-[~/Desktop]
$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	948M	0	948M	0%	/dev
tmpfs	198M	988K	197M	1%	/run
/dev/sda1	79G	17G	58G	23%	/
tmpfs	989M	0	989M	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock

4. `free -m`

Displays memory usage in megabytes, including total, used, free, and available memory.

`free -m`

```
(kali㉿kali)-[~/Desktop]
$ free -m
```

	total	used	free	shared	buff/cache	available
Mem:	1976	924	330	34	901	1052
Swap:	1023	0	1023			

5. id

Shows the current user's UID (User ID), GID (Group ID), and group memberships.

Id

```
(kali㉿kali)-[~/Desktop]
$ id
uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(nv),106(bluetooth),113(scanner),136(wireshark),137(kaboxer),138(vboxsf)
```

6. whoami

Displays the username of the currently logged-in user.

Whoami

```
(kali㉿kali)-[~/Desktop]
$ whoami
kali
```

7. passwd

Changes the password for the current user or another user (requires sudo for other users).

Passwd

```
(kali㉿kali)-[~/Desktop]
$ passwd
Changing password for kali.
Current password:
New password:
```

8. useradd

Adds a new user to the system.

sudo useradd -m new_user

9. usermod -aG <group> <user>

Adds a user to a group (e.g., add a user to the sudo group for administrative access).

sudo usermod -aG sudo new_user

10. groupadd <group>

Creates a new user group.

sudo groupadd new_group

11. who

Displays a list of users currently logged in.

Who

```
(kali㉿kali)-[~/Desktop]
$ who
```

12. groups

Displays the groups that the current user or a specified user belongs to.

groups

```
(kali㉿kali)-[~/Desktop]
$ groups

kali adm dialout cdrom floppy sudo audio dip video plugdev users netdev bluetooth scanner wireshark kaboxer vboxsf
```

13. su - <username>

Switches to another user account (enter exit to return to the previous user).

su - new_user

14. top

Displays a real-time list of running processes, along with system usage stats like CPU and memory.

Top

```
(kali㉿kali)-[~/Desktop]
$ top

top - 07:15:23 up 2:04, 1 user, load average: 0.11, 0.08, 0.04
Tasks: 162 total, 1 running, 161 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.7 us, 1.4 sy, 0.0 ni, 96.5 id, 0.0 wa, 0.0 hi, 0.3 si, 0.0 st
MiB Mem : 1976.7 total, 331.1 free, 923.5 used, 902.2 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used. 1053.3 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
  813 root        20   0 522644 191368 83460 S   2.7   9.5   2:08.95 Xorg
 1377 kali        20   0 459156 104476 88252 S   0.7   5.2   0:13.58 qterminal
  986 kali        20   0 340316 28172 17664 S   0.3   1.4   0:01.07 xfce4-session
 1045 kali        20   0 215336 3840 3456 S   0.3   0.2   0:00.25 VBoxClient
 1173 kali        20   0 635832 99500 52244 S   0.3   4.9   0:05.78 xfdesktop
 1181 kali        20   0 363316 54324 22692 S   0.3   2.7   0:16.09 panel-13-cpugra
 1183 kali        20   0 340272 30104 20852 S   0.3   1.5   0:14.14 panel-15-genmon
    1 root        20   0 22556 13176 9976 S   0.0   0.7   0:01.12 systemd
    2 root        20   0 0 0 0 S   0.0   0.0   0:00.02 kthreadd
    3 root        20   0 0 0 0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
    5 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
    6 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-slub_
    7 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-netns
   11 root        20   0 0 0 0 I   0.0   0.0   0:00.00 kworker/u4:0-ext4-rsv-conversion
```

15. ps aux

Shows detailed information about all running processes.

ps aux

```
kali@kali:~$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.5	0.6	22560	13028	?	Ss	11:49	0:01	/sbin/init splash
root	2	0.0	0.0	0	0	?	S	11:49	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	S	11:49	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	11:49	0:00	[kworker/R-rcu_g]
root	5	0.0	0.0	0	0	?	I<	11:49	0:00	[kworker/R-rcu_p]
root	6	0.0	0.0	0	0	?	I<	11:49	0:00	[kworker/R-slub_]
root	7	0.0	0.0	0	0	?	I<	11:49	0:00	[kworker/R-netns]
root	8	0.0	0.0	0	0	?	I	11:49	0:00	[kworker/0:0-events]
root	9	0.0	0.0	0	0	?	I	11:49	0:00	[kworker/0:1-events]
root	10	0.0	0.0	0	0	?	I<	11:49	0:00	[kworker/0:0H-kblockd]
root	11	0.0	0.0	0	0	?	I	11:49	0:00	[kworker/u4:0-ext4-rsv-conversion]
root	12	0.0	0.0	0	0	?	I<	11:49	0:00	[kworker/R-mm_pe]
root	13	0.0	0.0	0	0	?	I	11:49	0:00	[rcu_tasks_kthread]
root	14	0.0	0.0	0	0	?	I	11:49	0:00	[rcu_tasks_rude_kthread]
root	15	0.0	0.0	0	0	?	I	11:49	0:00	[rcu_tasks_trace_kthread]
root	16	0.0	0.0	0	0	?	S	11:49	0:00	[ksoftirqd/0]
root	17	0.1	0.0	0	0	?	I	11:49	0:00	[rcu_preempt]
root	18	0.0	0.0	0	0	?	S	11:49	0:00	[migration/0]
root	19	0.0	0.0	0	0	?	S	11:49	0:00	[idle_inject/0]
root	20	0.0	0.0	0	0	?	S	11:49	0:00	[cpuhp/0]
root	21	0.0	0.0	0	0	?	S	11:49	0:00	[cpuhp/1]
root	22	0.0	0.0	0	0	?	S	11:49	0:00	[idle_inject/1]
root	23	0.0	0.0	0	0	?	S	11:49	0:00	[migration/1]
root	24	0.1	0.0	0	0	?	S	11:49	0:00	[ksoftirqd/1]
root	25	0.0	0.0	0	0	?	I	11:49	0:00	[kworker/1:0-cgroup_destroy]
root	26	0.0	0.0	0	0	?	I<	11:49	0:00	[kworker/1:0H-events_highpri]

16. sudo shutdown -h now

Shuts down the system immediately. Replace -h with -r to restart.

sudo shutdown -h now

17. ifconfig

Displays the network configuration of all network interfaces.

ifconfig

```
(kali@kali)~$ ifconfig
```

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
inet6 fe80::a00:27ff:fed2:2679 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:d2:26:79 txqueuelen 1000 (Ethernet)
RX packets 2 bytes 1180 (1.1 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 10 bytes 1340 (1.3 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 12 bytes 680 (680.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 12 bytes 680 (680.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

18. ping <hostname/IP>

Checks the connectivity between the current machine and another machine (e.g., ping google.com).

ping google.com

```
(kali㉿kali)-[~]
└─$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=232 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=153 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=177 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=112 time=200 ms
^C
— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 152.917/190.714/232.415/29.282 ms
```

19. chmod <permissions> <file>

Changes file or directory permissions. Example: chmod 755 script.sh.

chmod 755 file.txt

```
(kali㉿kali)-[~/Desktop/snp]
└─$ ls -alh
total 64K
drwxrwxr-x 7 kali kali 4.0K Oct 6 07:05 .
drwxr-xr-x 22 kali kali 4.0K Oct 6 07:07 ..
-rwxrwxr-x 1 kali kali 16K Jul 28 07:05 a.out
drwxr-xr-x 2 kali kali 4.0K Jul 28 06:50 chkpermission
-rw-rw-r-- 1 kali kali 55 Jul 28 03:17 hemlo.txt
drwxrwxr-x 2 kali kali 4.0K Jul 28 06:46 IT23363120
drwxrwxr-x 2 kali kali 4.0K Jul 28 03:21 snp2
-rw-rw-r-- 1 kali kali 0 Oct 6 07:04 snp_aasingment.txt
drwxrwxr-x 2 kali kali 4.0K Oct 6 07:02 snp_assingment
drwxrwxr-x 2 kali kali 4.0K Jul 28 06:40 student
-rw-rw-r-- 1 kali kali 0 Jul 28 06:40 test.txt
-rw-rw-r-- 1 kali kali 355 Jul 28 07:05 time.c
-rw-r--r-- 1 kali kali 12K Jul 28 07:05 .time.c.swp

(kali㉿kali)-[~/Desktop/snp]
└─$ chmod 777 hemlo.txt

(kali㉿kali)-[~/Desktop/snp]
└─$ ls -alh
total 64
drwxrwxr-x 7 kali kali 4096 Oct 6 07:05 .
drwxr-xr-x 22 kali kali 4096 Oct 6 07:07 ..
-rwxrwxr-x 1 kali kali 16088 Jul 28 07:05 a.out
drwxr-xr-x 2 kali kali 4096 Jul 28 06:50 chkpermission
-rwxrwxrwx 1 kali kali 55 Jul 28 03:17 hemlo.txt
drwxrwxr-x 2 kali kali 4096 Jul 28 06:46 IT23363120
drwxrwxr-x 2 kali kali 4096 Jul 28 03:21 snp2
-rw-rw-r-- 1 kali kali 0 Oct 6 07:04 snp_aasingment.txt
drwxrwxr-x 2 kali kali 4096 Oct 6 07:02 snp_assingment
drwxrwxr-x 2 kali kali 4096 Jul 28 06:40 student
-rw-rw-r-- 1 kali kali 0 Jul 28 06:40 test.txt
-rw-rw-r-- 1 kali kali 355 Jul 28 07:05 time.c
-rw-r--r-- 1 kali kali 12288 Jul 28 07:05 .time.c.swp
```

```
File Actions Edit View Help
drwxrwxr-x 2 kali kali 4.0K Jul 28 06:46 IT23363120
drwxrwxr-x 2 kali kali 4.0K Jul 28 03:21 snp2
-rw-rw-r-- 1 kali kali 0 Oct 6 07:04 snp_aasingment.txt
drwxrwxr-x 2 kali kali 4.0K Oct 6 07:02 snp_assingment
drwxrwxr-x 2 kali kali 4.0K Jul 28 06:40 student
-rw-rw-r-- 1 kali kali 0 Jul 28 06:40 test.txt
-rw-rw-r-- 1 kali kali 355 Jul 28 07:05 time.c
-rw-r--r-- 1 kali kali 12K Jul 28 07:05 .time.c.swp

(kali㉿kali)-[~/Desktop/snp]
└─$ chmod 777 hemlo.txt

(kali㉿kali)-[~/Desktop/snp]
└─$ ls -alh
total 64
drwxrwxr-x 7 kali kali 4096 Oct 6 07:05 .
drwxr-xr-x 22 kali kali 4096 Oct 6 07:07 ..
-rwxrwxr-x 1 kali kali 16088 Jul 28 07:05 a.out
drwxr-xr-x 2 kali kali 4096 Jul 28 06:50 chkpermission
-rwxrwxrwx 1 kali kali 55 Jul 28 03:17 hemlo.txt
drwxrwxr-x 2 kali kali 4096 Jul 28 06:46 IT23363120
drwxrwxr-x 2 kali kali 4096 Jul 28 03:21 snp2
-rw-rw-r-- 1 kali kali 0 Oct 6 07:04 snp_aasingment.txt
drwxrwxr-x 2 kali kali 4096 Oct 6 07:02 snp_assingment
drwxrwxr-x 2 kali kali 4096 Jul 28 06:40 student
-rw-rw-r-- 1 kali kali 0 Jul 28 06:40 test.txt
-rw-rw-r-- 1 kali kali 355 Jul 28 07:05 time.c
-rw-r--r-- 1 kali kali 12288 Jul 28 07:05 .time.c.swp
```

20. chown <user>:<group> <file>

Changes the owner and group of a file or directory.

sudo chown user:group file.t

4. Configuring DHCP server

DHCP (Dynamic Host Configuration Protocol) is a network service that automatically assigns IP addresses to devices (like computers, phones, or printers) when they join a network. It ensures each device gets a unique address, allowing it to communicate with other devices on the network.

Without DHCP, we'd have to manually assign an IP address (Local ip address) to every device, which can be time-consuming and prone to errors. DHCP makes this process automatic, ensuring that devices can connect to the network easily and function properly without manual configuration.

1. Role of DHCP: It automatically assigns IP addresses and other network configurations (like subnet mask, default gateway, and DNS servers) to devices in a network. This process is called "DHCP leasing."
2. DHCP Process: When a device connects to the network, it sends out a request for an IP address. The DHCP server responds by offering an IP address, which the device then accepts, making it active on the network.
3. IP Address Leases: The IP addresses assigned by the DHCP server are temporary and have a lease time. Once the lease expires, the device must renew it to keep the same IP address or get a new one.
4. Scope and Pools: The DHCP server has a range of IP addresses (a pool) that it can assign to devices. This range is defined within a particular subnet of the network.

So isc-dhcp server is open dhcp server where we can customize as we want and create our own private network

Installing isc-dhcp server. Using the command “**sudo apt install isc-dhcp-server**”

```
kavindu_kali_linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kavindu@kali: ~
File Actions Edit View Help
(kavindu@kali)~$ sudo apt install isc-dhcp server
Error: Unable to locate package isc-dhcp server
Error: Unable to locate package server

(kavindu@kali)~$ sudo apt install isc-dhcp-server
Installing:
isc-dhcp-server

Installing dependencies:
policycoreutils selinux-utils

Suggested packages:
isc-dhcp-server-ldap

Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 431
Download size: 1750 kB
Space needed: 7845 kB / 9683 MB available

Continue? [Y/n] Y
Get:2 http://http.kali.org/kali kali-rolling/main amd64 selinux-utils amd64 3.5-2+b2 [127 kB]
Get:1 http://mirror.primelink.net.id/kali kali-rolling/main amd64 isc-dhcp-server amd64 4.4.3-P1-5 [1479 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 policycoreutils amd64 3.5-2.1 [143 kB]
Fetched 1750 kB in 2s (1153 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database ... 395169 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-5_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-5) ...
Selecting previously unselected package selinux-utils.
Preparing to unpack .../selinux-utils_3.5-2+b2_amd64.deb ...
Unpacking selinux-utils (3.5-2+b2) ...
Selecting previously unselected package policycoreutils.
Preparing to unpack .../policycoreutils_3.5-2.1_amd64.deb ...
Unpacking policycoreutils (3.5-2.1) ...
Setting up policycoreutils (3.5-2.1) ...
Setting up selinux-utils (3.5-2+b2) ...
Setting up isc-dhcp-server (4.4.3-P1-5) ...
Generating /etc/default/isc-dhcp-server...
update-rc.d: We have no instructions for the selinux-autorelabel init script.
update-rc.d: It looks like a non-network service, we enable it.
Setting up isc-dhcp-server (4.4.3-P1-5) ...
Processing triggers for kali-menu (2024.3.1) ...
Processing triggers for man-db (2.12.1-2) ...

(kavindu@kali)~$
```

```
kavindu_kali_linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kavindu@kali: ~
File Actions Edit View Help
(kavindu@kali)~$ sudo apt install isc-dhcp-server
Installing:
isc-dhcp-server

Installing dependencies:
policycoreutils selinux-utils

Suggested packages:
isc-dhcp-server-ldap

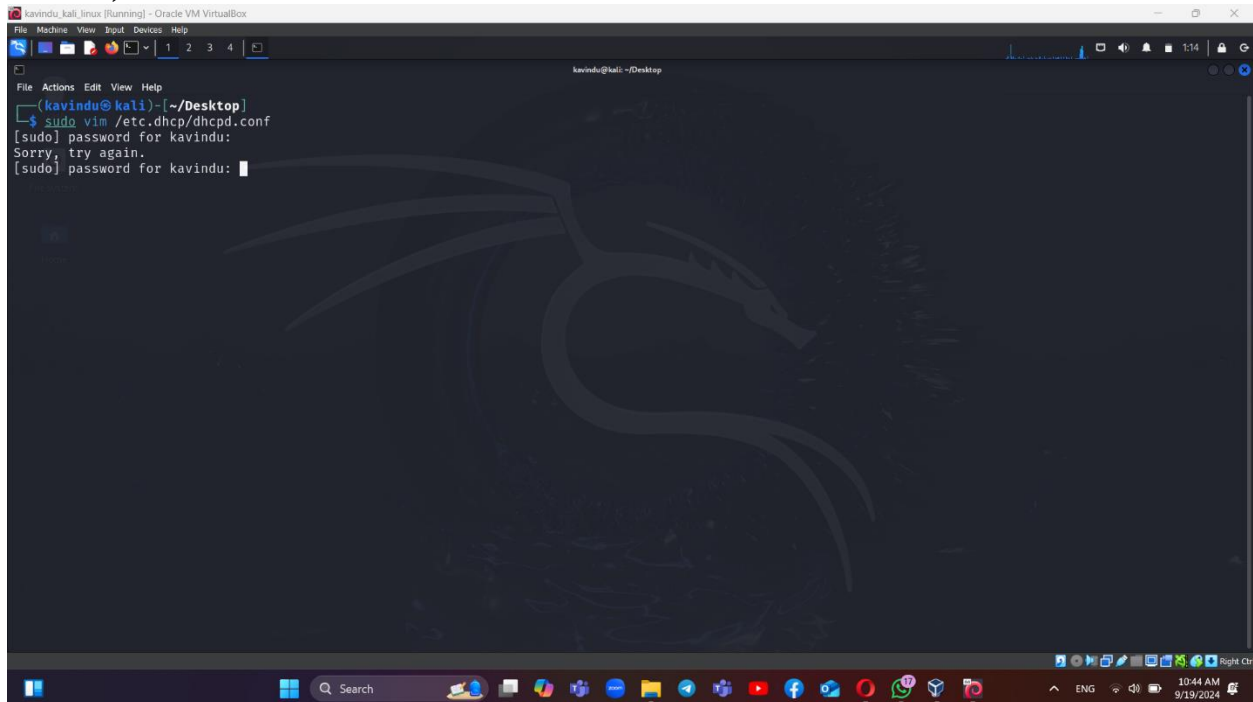
Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 431
Download size: 1750 kB
Space needed: 7845 kB / 9683 MB available

Continue? [Y/n] Y
Get:2 http://http.kali.org/kali kali-rolling/main amd64 selinux-utils amd64 3.5-2+b2 [127 kB]
Get:1 http://mirror.primelink.net.id/kali kali-rolling/main amd64 isc-dhcp-server amd64 4.4.3-P1-5 [1479 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 policycoreutils amd64 3.5-2.1 [143 kB]
Fetched 1750 kB in 2s (1153 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database ... 395169 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-5_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-5) ...
Selecting previously unselected package selinux-utils.
Preparing to unpack .../selinux-utils_3.5-2+b2_amd64.deb ...
Unpacking selinux-utils (3.5-2+b2) ...
Selecting previously unselected package policycoreutils.
Preparing to unpack .../policycoreutils_3.5-2.1_amd64.deb ...
Unpacking policycoreutils (3.5-2.1) ...
Setting up policycoreutils (3.5-2.1) ...
Setting up selinux-utils (3.5-2+b2) ...
Setting up isc-dhcp-server (4.4.3-P1-5) ...
Generating /etc/default/isc-dhcp-server...
update-rc.d: We have no instructions for the selinux-autorelabel init script.
update-rc.d: It looks like a non-network service, we enable it.
Setting up isc-dhcp-server (4.4.3-P1-5) ...
Processing triggers for kali-menu (2024.3.1) ...
Processing triggers for man-db (2.12.1-2) ...

(kavindu@kali)~$
```

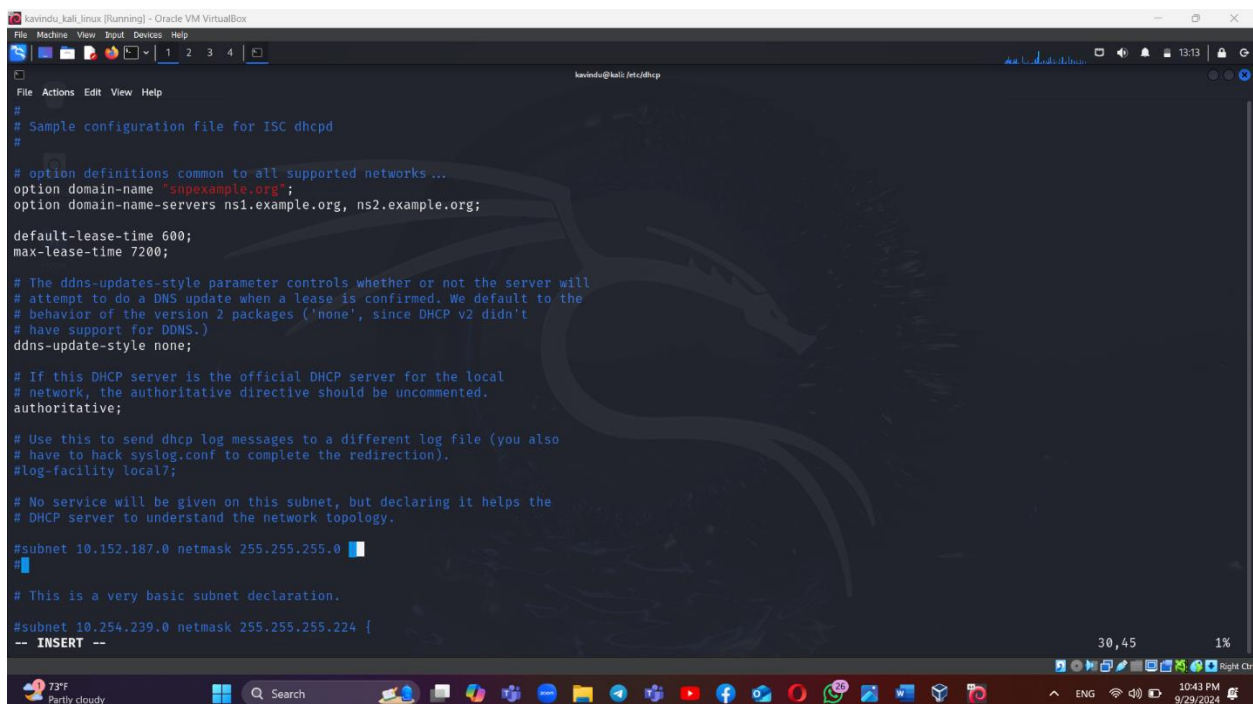
Once it's installed , lets configure the dhcp-server as we wanted. To do that we need to open the configuration file . It is named as dhcpd.conf in etc/dhcp directory , using the command “**sudo vim /etc/dhcp/dhcpd.conf**”

(I personally preferred to use vim text editor over nano text editor .There fore I used vim instead of nano)



```
kavindu@kali: ~/Desktop
File Actions Edit View Help
┌─(kavindu@kali)─[~/Desktop]
└─$ sudo vim /etc/dhcp/dhcpd.conf
[sudo] password for kavindu:
Sorry, try again.
[sudo] password for kavindu: 
```

once opened I made the my configuring domain name as “**snpexample.com**”.



```
kavindu@kali: /etc/dhcp
File Actions Edit View Help
#
# Sample configuration file for ISC dhcpd
#
# option definitions common to all supported networks ...
option domain-name 'snpexample.org';
option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;
max-lease-time 7200;

# The ddns-updates-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
# have support for DDNS.)
ddns-update-style none;

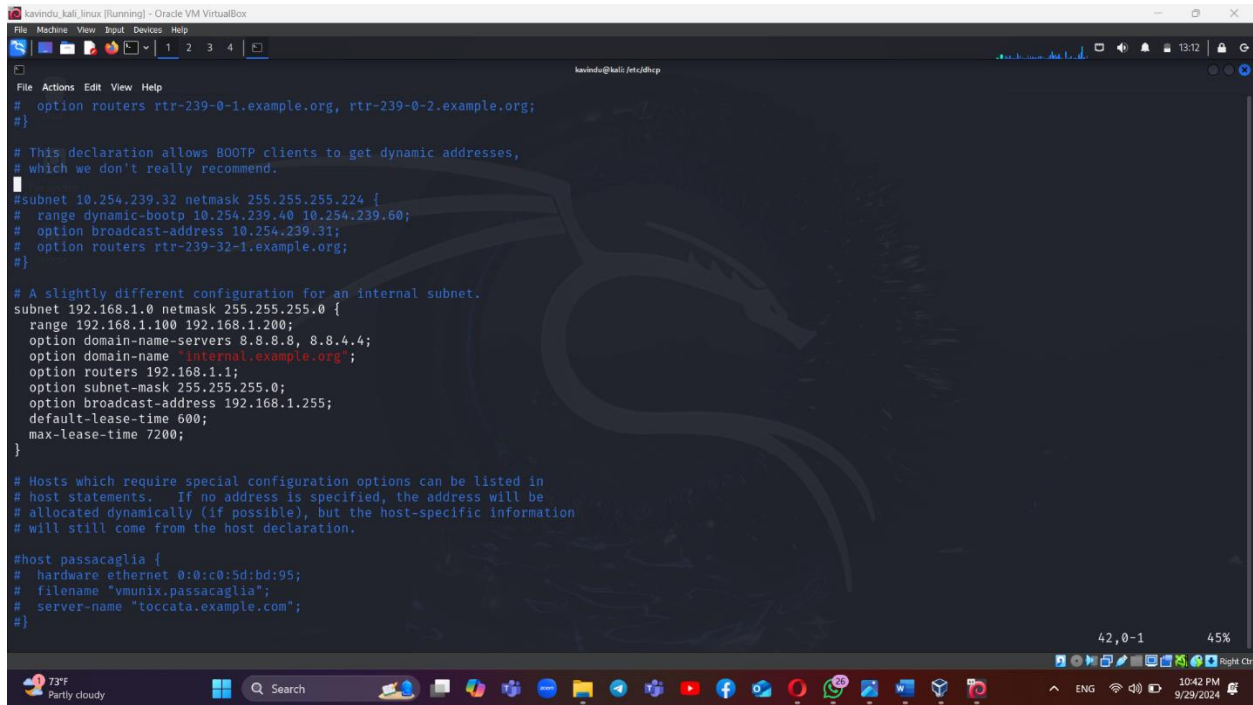
# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
#log-facility local7;

# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

#subnet 10.152.187.0 netmask 255.255.255.0
#
# This is a very basic subnet declaration.

#subnet 10.254.239.0 netmask 255.255.255.224 {
-- INSERT --
```

A screenshot of a terminal window running inside an Oracle VM VirtualBox. The terminal shows a DHCP configuration file being edited. The configuration includes a subnet declaration for 10.254.239.32, a range of dynamic-bootp addresses, and a separate subnet declaration for 192.168.1.0. The user is currently editing the 192.168.1.0 subnet configuration, adding domain-name-servers, domain-name, routers, subnet-mask, broadcast-address, default-lease-time, and max-lease-time. The terminal window has a dark background with a dragon logo in the background. The bottom of the window shows the Windows taskbar with various icons and the system clock.

```
kavindu_kali_linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kavindu@kali: /etc/dhcp

# option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
#}

# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.
#}

#subnet 10.254.239.32 netmask 255.255.255.0 {
#  range dynamic-bootp 10.254.239.40 10.254.239.60;
#  option broadcast-address 10.254.239.31;
#  option routers rtr-239-0-1.example.org;
#}

# A slightly different configuration for an internal subnet.
subnet 192.168.1.0 netmask 255.255.255.0 {
  range 192.168.1.100 192.168.1.200;
  option domain-name-servers 8.8.8.8, 8.8.4.4;
  option domain-name "internal.example.org";
  option routers 192.168.1.1;
  option subnet-mask 255.255.255.0;
  option broadcast-address 192.168.1.255;
  default-lease-time 600;
  max-lease-time 7200;
}

# Hosts which require special configuration options can be listed in
# host statements.  If no address is specified, the address will be
# allocated dynamically (if possible), but the host-specific information
# will still come from the host declaration.

#host passacaglia {
#  hardware ethernet 0:0:c0:5d:bd:95;
#  filename "vmunix.passacaglia";
#  server-name "toccata.example.com";
#}

42,0-1 45%
```

I added following lines to the file,

option domain-name "snpexample.org";

option domain-name-servers ns1.example.org, ns2.example.org;

default-lease-time 600;

max-lease-time 7200;

ddns-update-style none;

authoritative;

subnet 192.168.1.0 netmask 255.255.255.0 {

range 192.168.1.100 192.168.1.200;

option domain-name-servers 8.8.8.8, 8.8.4.4;

option domain-name "internal.example.org";

option routers 192.168.1.1;

option subnet-mask 255.255.255.0;

option broadcast-address 192.168.1.255;

default-lease-time 600;

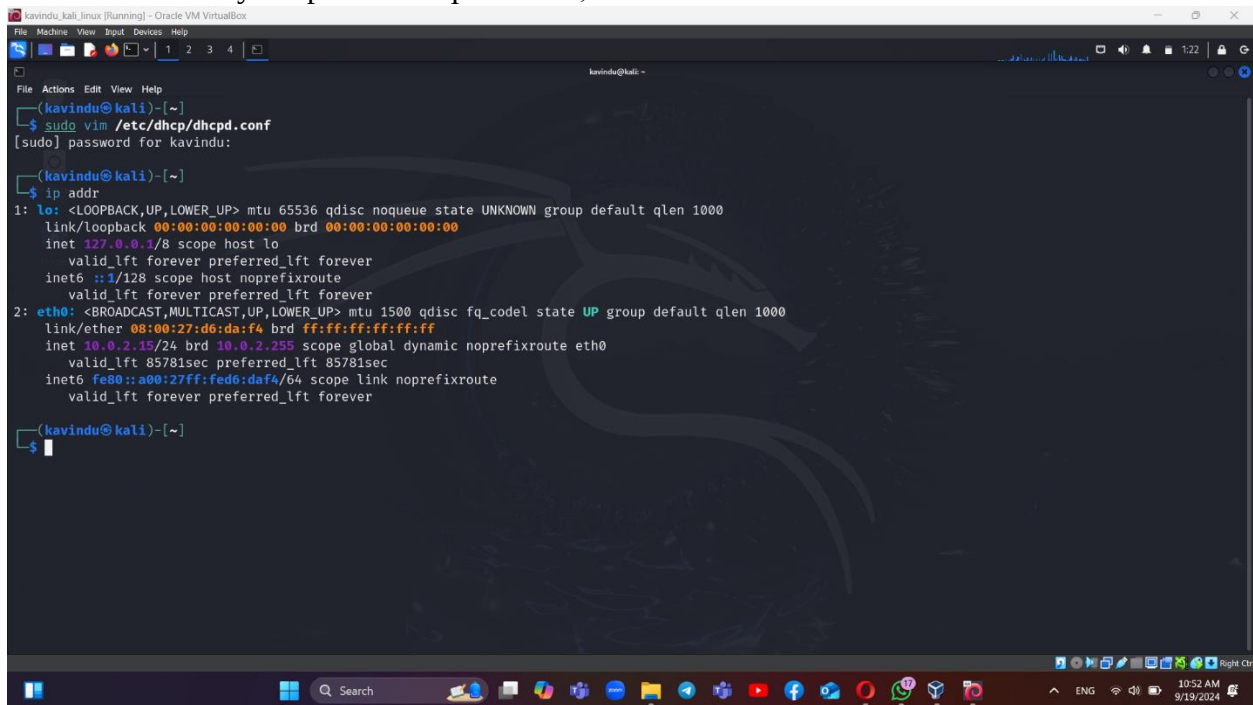
```
max-lease-time 7200;
```

```
}
```

Here,

- **option domain-name:** This specifies the domain name that DHCP clients that I'm going to use. In this case, the domain name is set to snpexample.org.
- **option domain-name-servers:** This defines the DNS servers that clients should use to resolve domain names. Here, the DNS servers ns1.example.org and ns2.example.org are specified.
- **default-lease-time:** This sets the default amount of time (in seconds) that a client can keep its IP address before needing to renew the lease. In this case, it's set to 600 seconds (10 minutes).
- **max-lease-time:** This defines the maximum amount of time (in seconds) a client can lease an IP address before it has to renew. In this case, it's set to 7200 seconds (2 hours)
- **ddns-update-style:** This option controls how the DHCP server interacts with Dynamic DNS. By setting it to none, it disables DDNS updates (meaning the server won't attempt to automatically update DNS records when a client gets a new lease).
- **authoritative:** This tells the DHCP server that it is the main DHCP server for the local network. If a client sends a request to the wrong server, the authoritative server will respond quickly and correct it. Uncommenting this line makes the server authoritative.
- **subnet 192.168.1.0 netmask 255.255.255.0:** This defines the subnet and subnet mask for the local network. Here, it's 192.168.1.0/24.
- **range 192.168.1.100 192.168.1.200:** This specifies the IP address range that the DHCP server will lease to clients. Here, it will lease IP addresses between 192.168.1.100 and 192.168.1.200.
- **option domain-name-servers 8.8.8.8, 8.8.4.4:** This provides DNS servers for the clients on this subnet. In this case, Google's public DNS servers (8.8.8.8 and 8.8.4.4) are used.
- **option domain-name "internal.example.org":** This specifies the domain name for this particular subnet (internal.example.org).
- **option routers 192.168.1.1:** This defines the default gateway (router) for the clients, which is set to 192.168.1.1.
- **option subnet-mask 255.255.255.0:** This tells the clients the subnet mask to use (255.255.255.0).
- **option broadcast-address 192.168.1.255:** This defines the broadcast address for the network (192.168.1.255).
- **default-lease-time 600 and max-lease-time 7200:** These are the lease times specifically for this subnet (same as mentioned earlier).

Now lets check my dhcp-server's ip address ,



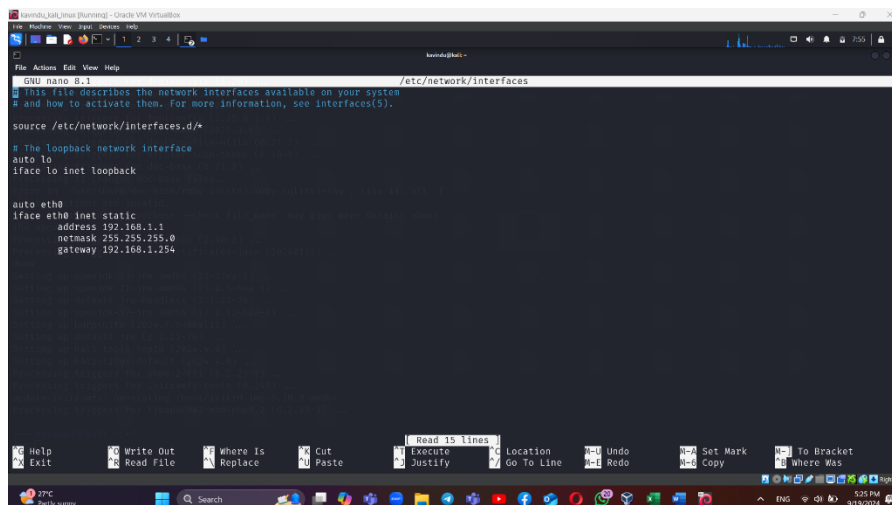
```
(kavindu@kali)~  
$ sudo vim /etc/dhcp/dhcpd.conf  
[sudo] password for kavindu:  
(kavindu@kali)~  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:da:f4:ff brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85781sec preferred_lft 85781sec  
    inet6 fe80::a00:27ff:fed0:daf4/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kavindu@kali)~  
$
```

Its showing as 10.0.2.15 which is okay, but lets change to a format like 192.168 , so to do that we have assign a static ip adrsss to eth0 port. To do that we have change it in network file, using the commad “**Sudo nano /etc/network/interfaces**”



```
(kavindu@kali)~  
$ sudo nano /etc/network/interfaces
```

And added the static ip address as following and set it **192.168.1. 1** and other details.



```
(kavindu@kali)~  
$ nano /etc/network/interfaces  
# This file describes the network interfaces available on your system  
# and how to activate them. For more information, see interfaces(5).  
  
source /etc/network/interfaces.d/*  
  
# The loopback network interface  
auto lo  
iface lo inet loopback  
  
auto eth0  
iface eth0 inet static  
    address 192.168.1.1  
    netmask 255.255.255.0  
    gateway 192.168.1.254  
(kavindu@kali)~
```

Now let's see our new IP address for eth0 after restarting the network. It is shown that it has changed from 10.0.2.15 to 192.168.1.1, indicating it worked.

```
kavindu_kali_linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kavindu@kali ~
File Actions Edit View Help
└─$ sudo cat /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto eth0
iface lo inet loopback
address 192.168.1.1
netmask 255.255.255.0
gateway 192.168.1.254

(kavindu@kali)-[~]
└─$ sudo nano /etc/network/interfaces
(kavindu@kali)-[~]
└─$ sudo systemctl restart networking

(kavindu@kali)-[~]
└─$ ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:d6:da:f4 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 84867sec preferred_lft 84867sec
    inet 192.168.1.1/24 brd 192.168.1.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed6:daf4/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kavindu@kali)-[~]
└─$
```

Now let's restart the dhcp server. Seems like it has some errors.

```
kavindu_kali_linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kavindu@kali ~
File Actions Edit View Help
# }
# pool {
#     allow members of "foo";
#     range 10.17.224.10 10.17.224.250;
# }
# pool {
#     deny members of "foo";
#     range 10.0.29.10 10.0.29.230;
# }
#}

(kavindu@kali)-[~]
└─$ sudo vim /etc/dhcp/dhcpd.conf

(kavindu@kali)-[~]
└─$ sudo systemctl restart isc-dhcp-server
sudo: systemctl: command not found

(kavindu@kali)-[~]
└─$ sudo systemctl restart isc-dhcp-server

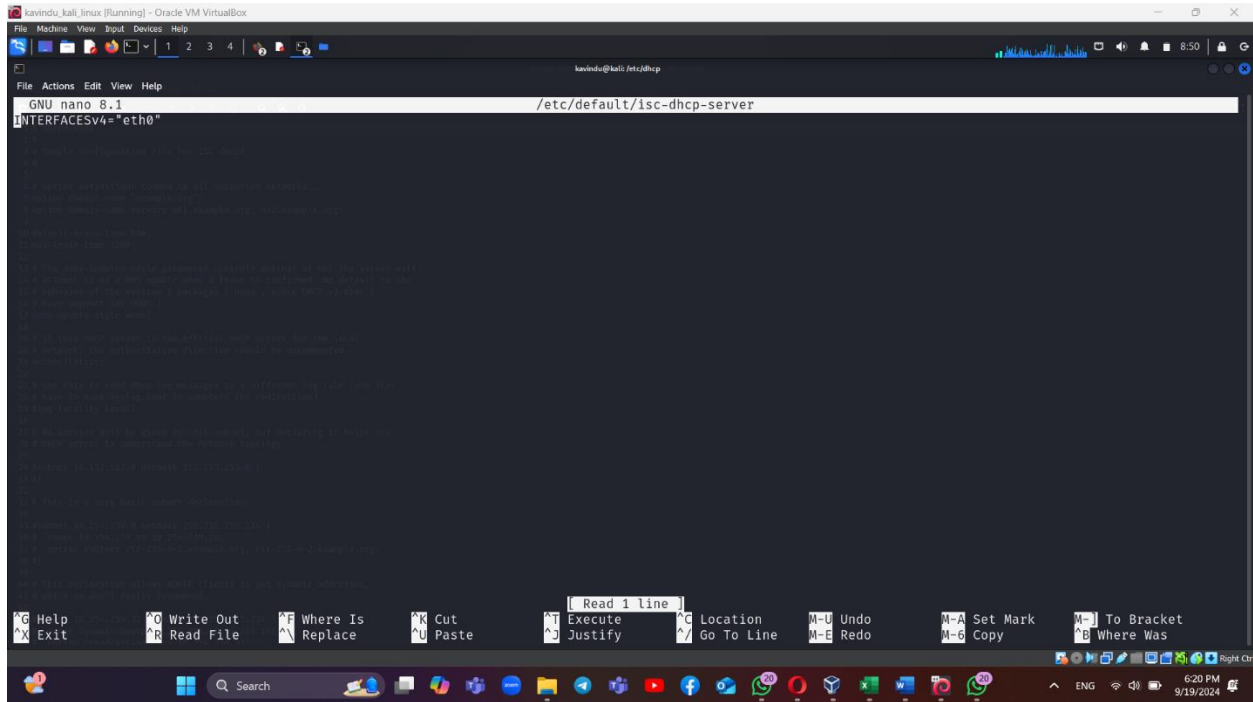
Job for isc-dhcp-server.service failed because the control process exited with error code.
See "systemctl status isc-dhcp-server.service" and "journalctl -xeu isc-dhcp-server.service" for details.

(kavindu@kali)-[~]
└─$ sudo systemctl restart isc-dhcp-server

Job for isc-dhcp-server.service failed because the control process exited with error code.
See "systemctl status isc-dhcp-server.service" and "journalctl -xeu isc-dhcp-server.service" for details.

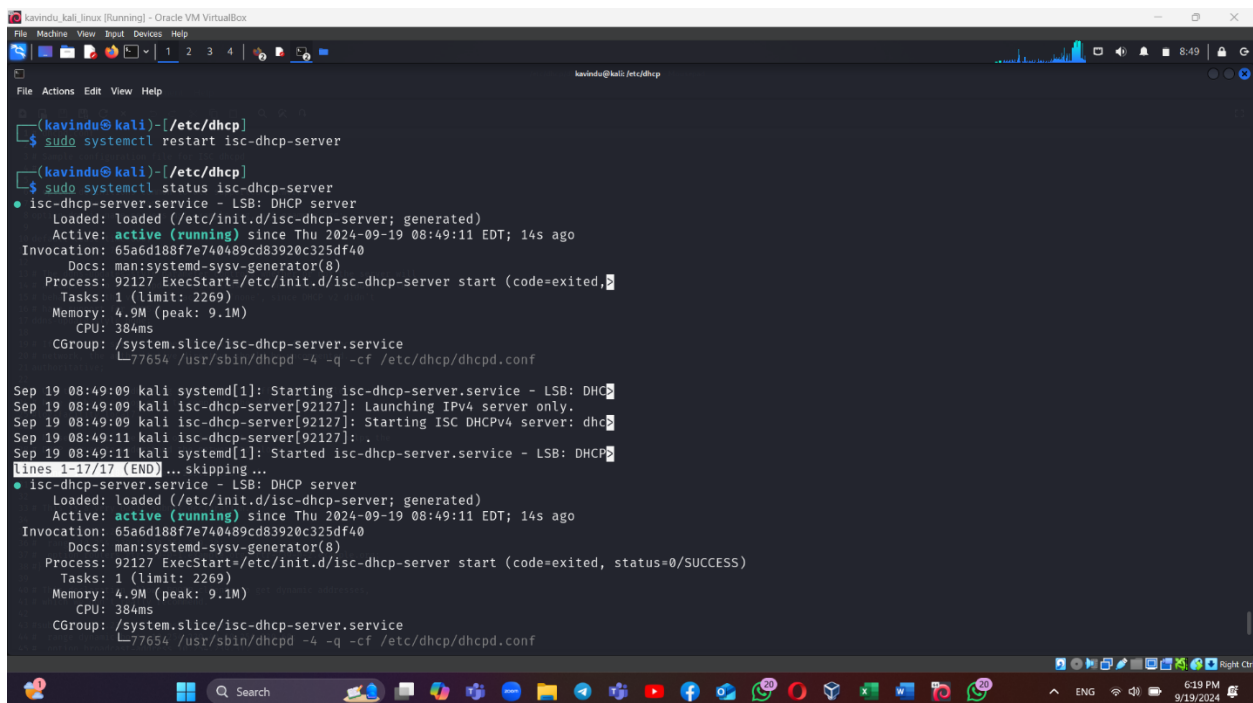
(kavindu@kali)-[~]
└─$
```

It may be caused by , since I haven't configured the network interface for dhcp server. So let's do that using command "**sudo nano /etc/default/isc-dhcp-server**" make the interface for ipv4 (since I'm using ip version 4 not 6) for the eth0 port (The port which I manually assigned a static ip previously)



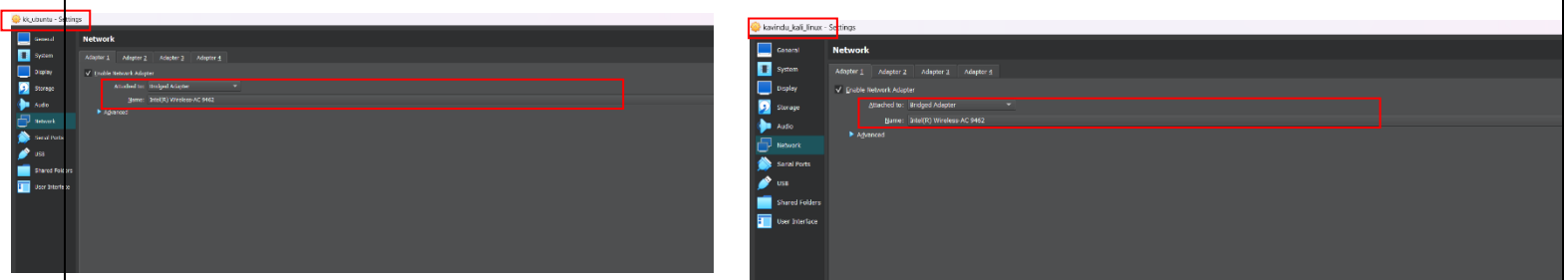
The screenshot shows a terminal window titled "kavindu_kali_linux [Running] - Oracle VM VirtualBox". The terminal is running the nano text editor on the file `/etc/default/isc-dhcp-server`. The first line of the file is `INTERFACESv4="eth0"`. The nano editor's status bar at the bottom shows "GNU nano 8.1" and various keyboard shortcuts like "Read 1 line", "Execute", "Location", "Undo", "Set Mark", "To Bracket", "Where Was", "Help", "Exit", "Write Out", "Read File", "Where Is", "Replace", "Cut", "Paste", "Justify", "Go To Line", "Redo", "Copy", and "Where Was". The system clock in the bottom right corner shows 6:20 PM on 9/19/2024.

Now let's restart and see the status .to do that we have to give system control command using "**sudo systemctl restart dhcp-server**" and to see the status of it , "**sudo systemctl status dhcp-server**"



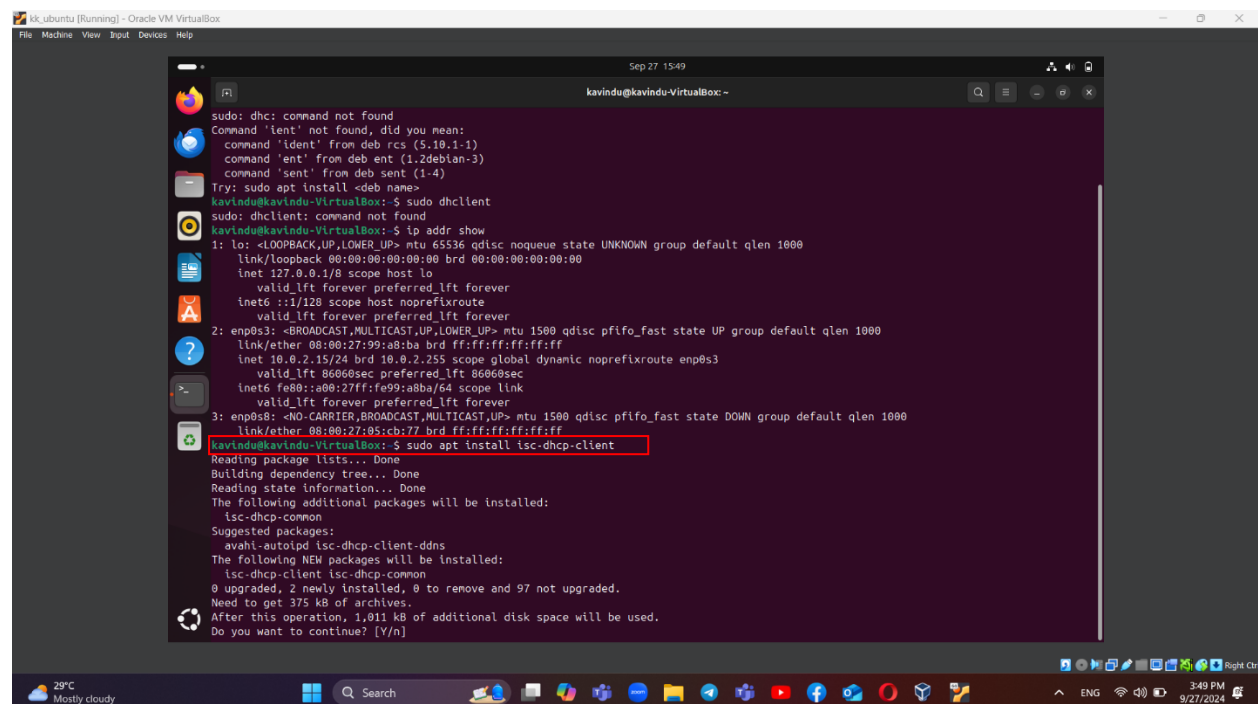
The screenshot shows a terminal window titled "kavindu_kali_linux [Running] - Oracle VM VirtualBox". The terminal is running the nano text editor on the file `/etc/dhcp`. The user has entered the command `sudo systemctl restart isc-dhcp-server` and then `sudo systemctl status isc-dhcp-server`. The output of the status command shows that the `isc-dhcp-server.service` is active (running) since Thu 2024-09-19 08:49:11 EDT; 14s ago. The output also shows the loaded path, invocation, docs, process, tasks, memory, CPU, and CGroup information. The system clock in the bottom right corner shows 6:19 PM on 9/19/2024.

It shows actively running (In green color). Before we check we whether our dhcp server is manually assigning the ip addresses to other devices, we need to ensure that both networks are in the same network. Here I have chosen bridge adapter network type, since it allows my 2 VM's to communicate and also to communicate with internet. NAT wont work cause it not allows to communicate between 2 or more VM's. Here I choose 2 ubuntu VM's to check



So lets check the ubuntu VM

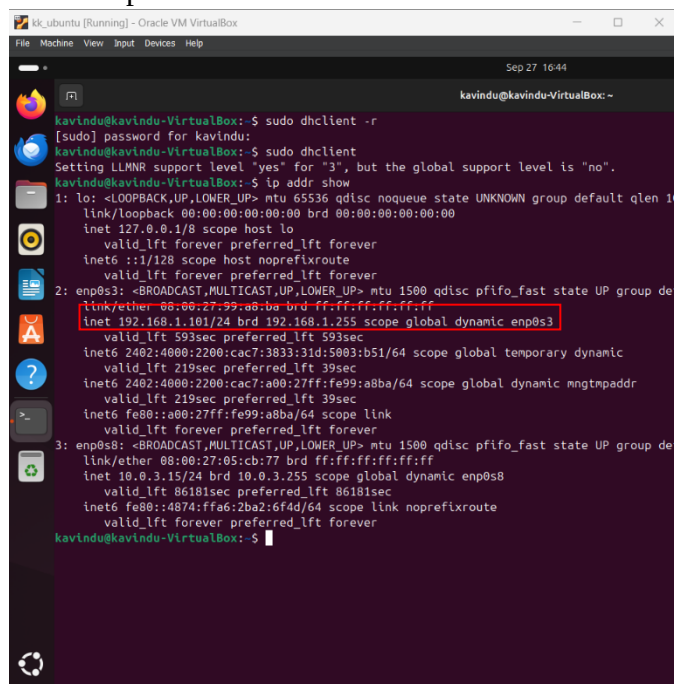
First of all, lets connect the dhcp server and assign an ip for the the ubuntu using dhclient command



Seems like initially I haven't installed dhclient. Therefore I have installed it using "Sudo apt install isc-dhcp-client". And lets request a dhclient now

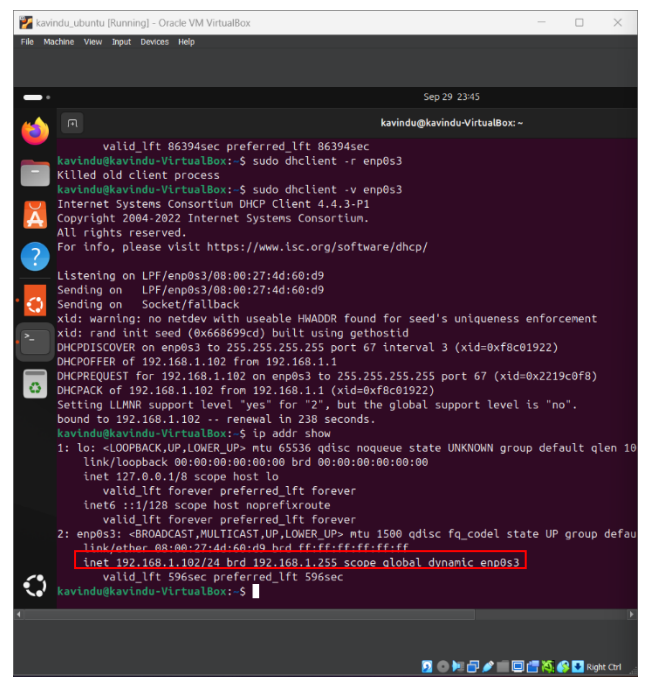
```
kavindu@kavindu-VirtualBox:~$ sudo dhclient -r
[sudo] password for kavindu:
kavindu@kavindu-VirtualBox:~$ sudo dhclient
Setting LLNMR support level "yes" for "3", but the global support level is "no".
kavindu@kavindu-VirtualBox:~$ ip addr show
```

Here using “**sudo dhclient -r**” I removed if any previously assigned dhcp client ip address and using “**sudo dhclient**” assigned a new ip address to my ubuntu VM. Using “**ip addr show**” lets see the new ip addresses.



```
kk_ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Sep 27 16:44
kavindu@kavindu-VirtualBox:~$ sudo dhclient -r
[sudo] password for kavindu:
kavindu@kavindu-VirtualBox:~$ sudo dhclient
Setting LLNMR support level "yes" for "3", but the global support level is "no".
kavindu@kavindu-VirtualBox:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:99:a8:ba-brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.101/24 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 593sec preferred_lft 593sec
    inet6 2402:4000:2200:cac7:3833:31d:5003:b51/64 scope global temporary dynamic
    valid_lft 219sec preferred_lft 39sec
    inet6 2402:4000:2200:cac7:a00:27ff:fe99:a8ba/64 scope global dynamic mngtppaddr
    valid_lft 219sec preferred_lft 39sec
    inet6 fe80::a00:27ff:fe99:a8ba/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:05:cb:77 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic enp0s8
        valid_lft 86181sec preferred_lft 86181sec
    inet6 fe80::4874:ffa6:2ba2:6f4d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kavindu@kavindu-VirtualBox:~$
```

Ubuntu VM 1 (kk_ubuntu)



```
kavindu_ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Sep 29 23:45
kavindu@kavindu-VirtualBox:~$ sudo dhclient -r enp0s3
killed old client process
kavindu@kavindu-VirtualBox:~$ sudo dhclient -v enp0s3
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/enp0s3/08:00:27:4d:60:d9
Sending on LPF/enp0s3/08:00:27:4d:60:d9
Sending on Socket/fallback
xid: warning: no netdev with useable HWADDR found for seed's uniqueness enforcement
xid: rand init seed (0x668699cd) built using gethostid
DHCPDISCOVER on enp0s3 to 255.255.255.255 port 67 interval 3 (xid=0xf8c01922)
DHCPOFFER of 192.168.1.102 from 192.168.1.1
DHCPREQUEST for 192.168.1.102 on enp0s3 to 255.255.255.255 port 67 (xid=0x2219c0f8)
DHCPACK of 192.168.1.102 from 192.168.1.1 (xid=0xf8c01922)
Setting LLNMR support level "yes" for "2", but the global support level is "no".
bound to 192.168.1.102 -- renewal in 238 seconds.
kavindu@kavindu-VirtualBox:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group defau
    link/ether 08:00:27:4d:60:d9-brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.102/24 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 596sec preferred_lft 596sec
kavindu@kavindu-VirtualBox:~$
```

Ubuntu VM 2 (kavindu_ubuntu)

VM1 got the ip address of 192.168.1.101 and VM2 got the ip address of 192.168.1.102 . Which seems to be accurate since I assigned a range to be given for new devices (the range was given in dhcpd.conf file , the range was range **192.168.1.100 - 192.168.1.200**). There for both ip addresses are in that range. There for using ping command lets check these VMS can actually connect with our dhcp server installed in kali VM.

```
(kavindu@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.1 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fed6:daf4 prefixlen 64 scopeid 0<link>
    ether 08:00:27:d6:da:f4 txqueuelen 1000 (Ethernet)
    RX packets 310 bytes 78163 (76.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 25 bytes 4183 (4.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kavindu@kali)-[~]
$ ping 192.168.1.101
PING 192.168.1.101 (192.168.1.101) 56(84) bytes of data:
 64 bytes from 192.168.1.101: icmp_seq=1 ttl=64 time=1.89 ms
 64 bytes from 192.168.1.101: icmp_seq=2 ttl=64 time=0.788 ms
 64 bytes from 192.168.1.101: icmp_seq=3 ttl=64 time=0.754 ms
 64 bytes from 192.168.1.101: icmp_seq=4 ttl=64 time=0.740 ms
 64 bytes from 192.168.1.101: icmp_seq=5 ttl=64 time=0.942 ms

kavindu@kavindu-VirtualBox:~$ sudo dhclient -r
[sudo] password for kavindu:
kavindu@kavindu-VirtualBox:~$ sudo dhclient
Setting LLNMR support level "yes" for "3", but the global support level is "no".
kavindu@kavindu-VirtualBox:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:99:a8:ba brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.101/24 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 593sec preferred_lft 593sec
    inet6 2402:4000:2200:cac7:3833:31d:5003:b51/64 scope global temporary dynamic
        valid_lft 219sec preferred_lft 39sec
    inet6 2402:4000:2200:cac7:a00:27ff:fe99:a8ba/64 scope global dynamic mngtnpaddr
        valid_lft 219sec preferred_lft 39sec
    inet6 fe80::a00:27ff:fe99:a8ba/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:05:cb:77 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic enp0s8
        valid_lft 86181sec preferred_lft 86181sec
    inet6 fe80::487a:ffa6:2ba2:6f4d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kavindu@kavindu-VirtualBox:~$
```

VM 1 (kk_ubuntu) . it seems like they can connect.

```
kavindu@kavindu-VirtualBox:~$ ssh 192.168.1.1
kavindu@kavindu-VirtualBox:~$ ssh 192.168.1.0
kavindu@kavindu-VirtualBox:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data:
 64 bytes from 192.168.1.1: icmp_seq=1 ttl=63 time=23.9 ms
 64 bytes from 192.168.1.1: icmp_seq=2 ttl=63 time=3.90 ms
 64 bytes from 192.168.1.1: icmp_seq=3 ttl=63 time=9.99 ms
 64 bytes from 192.168.1.1: icmp_seq=4 ttl=63 time=4.69 ms
 64 bytes from 192.168.1.1: icmp_seq=5 ttl=63 time=3.82 ms
 64 bytes from 192.168.1.1: icmp_seq=6 ttl=63 time=5.53 ms
 64 bytes from 192.168.1.1: icmp_seq=7 ttl=63 time=8.53 ms
 64 bytes from 192.168.1.1: icmp_seq=8 ttl=63 time=9.88 ms
 64 bytes from 192.168.1.1: icmp_seq=9 ttl=63 time=2.89 ms
 64 bytes from 192.168.1.1: icmp_seq=10 ttl=63 time=4.26 ms
 64 bytes from 192.168.1.1: icmp_seq=11 ttl=63 time=20.0 ms
 64 bytes from 192.168.1.1: icmp_seq=12 ttl=63 time=13.2 ms
 64 bytes from 192.168.1.1: icmp_seq=13 ttl=63 time=5.96 ms

kavindu@kavindu-VirtualBox:~$ sudo nano /etc/default/isc-dhcp-server
kavindu@kavindu-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    link/ether 08:00:27:d6:da:f4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.1/24 brd 192.168.1.255 scope global eth0
        valid_lft forever preferred_lft forever
kavindu@kavindu-VirtualBox:~$
```

VM1 kk_ubuntu connecting with kali . it also working. (I added this just make sure).

Following screnn shots further elaborate that as 0% packet loss

```
kavindu@kali:~$ ping 192.168.1.101
64 bytes from 192.168.1.101: icmp_seq=16 ttl=64 time=0.927 ms
64 bytes from 192.168.1.101: icmp_seq=17 ttl=64 time=1.27 ms
64 bytes from 192.168.1.101: icmp_seq=18 ttl=64 time=0.992 ms
64 bytes from 192.168.1.101: icmp_seq=19 ttl=64 time=1.04 ms
64 bytes from 192.168.1.101: icmp_seq=20 ttl=64 time=0.920 ms
64 bytes from 192.168.1.101: icmp_seq=21 ttl=64 time=0.980 ms
64 bytes from 192.168.1.101: icmp_seq=22 ttl=64 time=1.04 ms
64 bytes from 192.168.1.101: icmp_seq=23 ttl=64 time=0.826 ms
64 bytes from 192.168.1.101: icmp_seq=24 ttl=64 time=1.06 ms
64 bytes from 192.168.1.101: icmp_seq=25 ttl=64 time=1.39 ms
64 bytes from 192.168.1.101: icmp_seq=26 ttl=64 time=0.726 ms
64 bytes from 192.168.1.101: icmp_seq=27 ttl=64 time=1.06 ms
64 bytes from 192.168.1.101: icmp_seq=28 ttl=64 time=0.982 ms
64 bytes from 192.168.1.101: icmp_seq=29 ttl=64 time=1.15 ms
64 bytes from 192.168.1.101: icmp_seq=30 ttl=64 time=1.09 ms
64 bytes from 192.168.1.101: icmp_seq=31 ttl=64 time=0.945 ms
64 bytes from 192.168.1.101: icmp_seq=32 ttl=64 time=13.6 ms
64 bytes from 192.168.1.101: icmp_seq=33 ttl=64 time=1.02 ms
64 bytes from 192.168.1.101: icmp_seq=34 ttl=64 time=1.14 ms
64 bytes from 192.168.1.101: icmp_seq=35 ttl=64 time=0.730 ms
64 bytes from 192.168.1.101: icmp_seq=36 ttl=64 time=0.755 ms
64 bytes from 192.168.1.101: icmp_seq=37 ttl=64 time=0.824 ms
64 bytes from 192.168.1.101: icmp_seq=38 ttl=64 time=1.19 ms
64 bytes from 192.168.1.101: icmp_seq=39 ttl=64 time=0.741 ms
^C
--- 192.168.1.101 ping statistics ---
39 packets transmitted, 39 received, 0% packet loss, time 38173ms
rtt min/avg/max/mdev = 0.629/1.315/13.586/2.003 ms

(kavindu@kali)-[~]
$
```

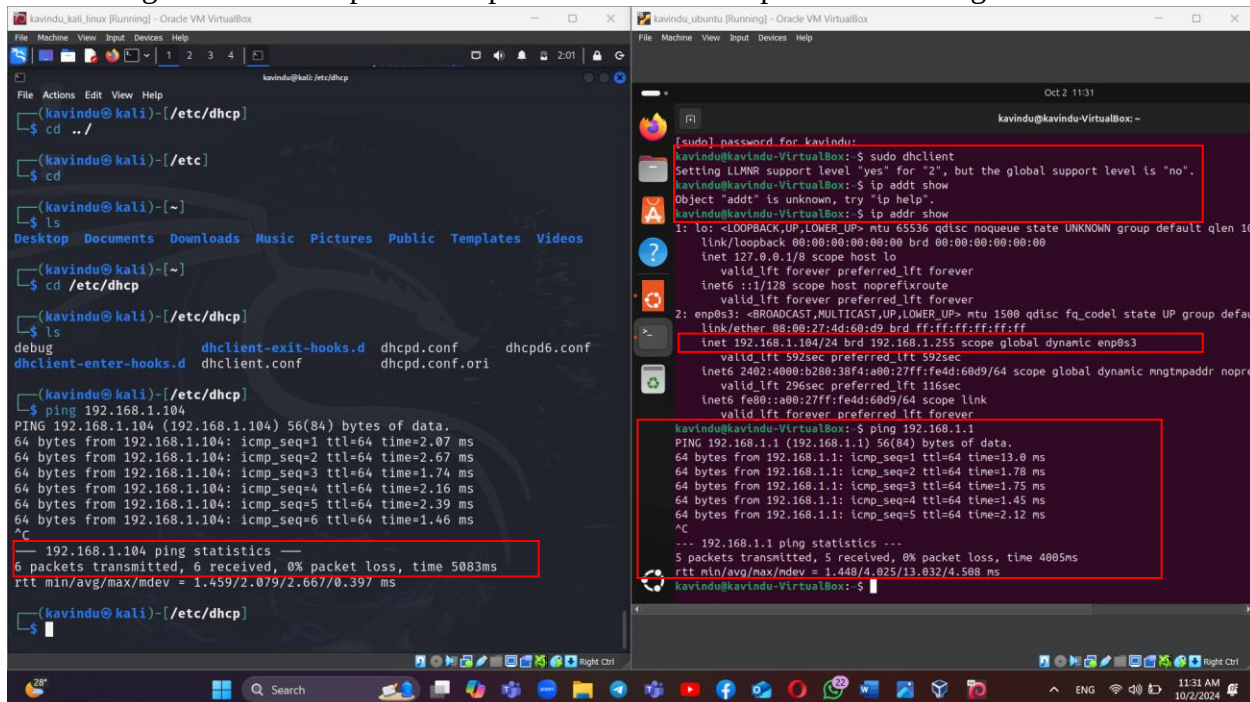
```
kavindu@kavindu-VirtualBox:~$ sudo dhclient -r
[sudo] password for kavindu:
kavindu@kavindu-VirtualBox:~$ sudo dhclient
Setting LLNMR support level 'yes' for '3', but the global support level is "no".
kavindu@kavindu-VirtualBox:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp8s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:99:a8:ba brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.101/24 brd 192.168.1.255 scope global dynamic enp8s3
        valid_lft 593sec preferred_lft 593sec
    inet6 2402:4000:2200:cac7:3833:31d:5003:b51/64 scope global temporary dynamic
        valid_lft 219sec preferred_lft 39sec
    inet6 2402:4000:2200:cac7:a00:27ff:fe99:a8ba/64 scope global dynamic mngt mpaddr
        valid_lft 219sec preferred_lft 39sec
    inet6 fe80::a00:27ff:fe99:a8ba/64 scope link
        valid_lft forever preferred_lft forever
3: enp8s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:05:cb:77 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic enp8s8
        valid_lft 86181sec preferred_lft 86181sec
    inet6 fe80::4874:ffa6:2ba2:6f4d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kavindu@kavindu-VirtualBox:~$
```

For VM2 similarly .

```
kavindu@kali:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp8s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:99:a8:ba brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.101/24 brd 192.168.1.255 scope global dynamic enp8s3
        valid_lft 593sec preferred_lft 593sec
    inet6 2402:4000:2200:cac7:3833:31d:5003:b51/64 scope global temporary dynamic
        valid_lft 219sec preferred_lft 39sec
    inet6 2402:4000:2200:cac7:a00:27ff:fe99:a8ba/64 scope global dynamic mngt mpaddr
        valid_lft 219sec preferred_lft 39sec
    inet6 fe80::a00:27ff:fe99:a8ba/64 scope link
        valid_lft forever preferred_lft forever
3: enp8s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group def
    link/ether 08:00:27:05:cb:77 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic enp8s8
        valid_lft 86181sec preferred_lft 86181sec
    inet6 fe80::4874:ffa6:2ba2:6f4d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kavindu@kali:~$ ping 192.168.1.102
PING 192.168.1.102 (192.168.1.102) 56(84) bytes of data.
64 bytes from 192.168.1.102: icmp_seq=1 ttl=64 time=1.40 ms
64 bytes from 192.168.1.102: icmp_seq=2 ttl=64 time=0.708 ms
64 bytes from 192.168.1.102: icmp_seq=3 ttl=64 time=0.524 ms
^C
--- 192.168.1.102 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2037ms
rtt min/avg/max/mdev = 0.524/0.878/1.404/0.378 ms

(kavindu@kali)-[~]
$
```

I have assigned a new dhcp address Ip for VM2 to further prove its working

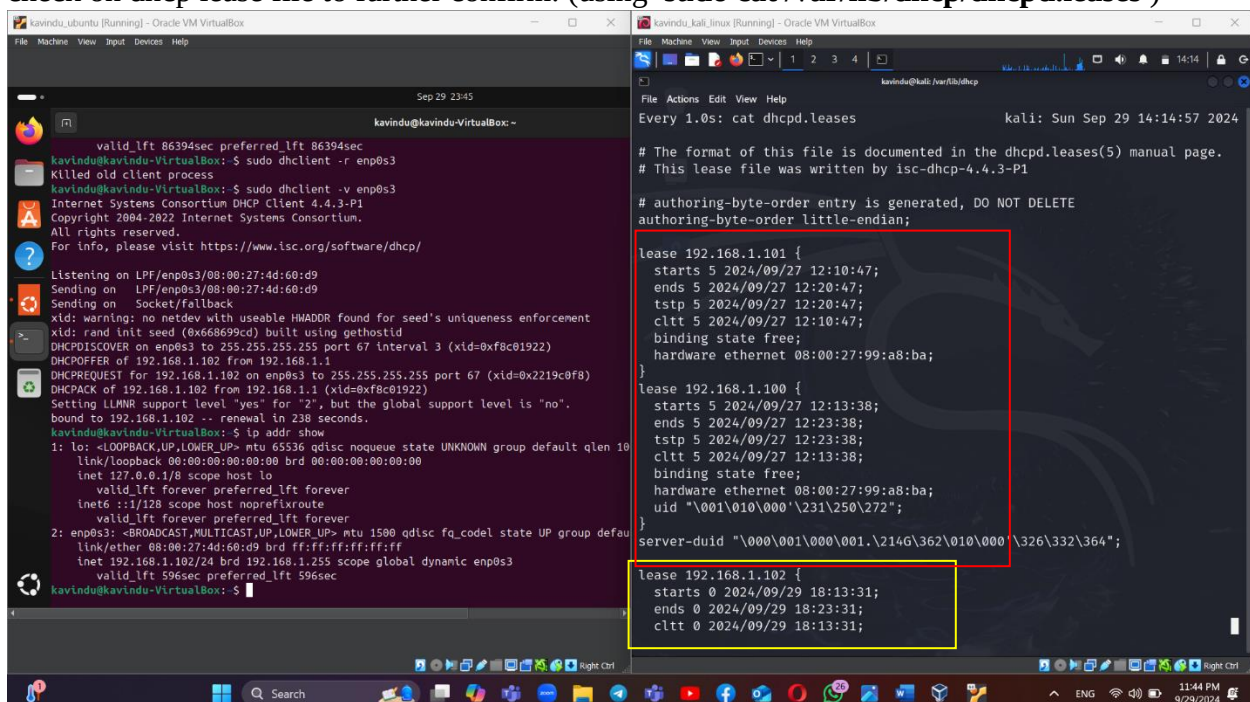


The image shows two terminal windows. The left window is a Kali Linux terminal where the user navigates to `/etc/dhcp`, lists files, and pings `192.168.1.104`. The output of the ping shows successful connections with 0% packet loss. The right window is an Ubuntu terminal where the user runs `sudo dhclient`, which assigns the IP `192.168.1.104` to the interface `enp0s3`. The user then pings `192.168.1.1` and shows the statistics.

```
(kavindu@kali)-[/etc/dhcp]
$ cd ../
(kavindu@kali)-[/etc]
$ cd
(kavindu@kali)-[~]
$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
(kavindu@kali)-[~]
$ cd /etc/dhcp
(kavindu@kali)-[/etc/dhcp]
$ ls
debug dhclient-enter-hooks.d dhclient-exit-hooks.d dhcpd.conf dhcpd6.conf
(kavindu@kali)-[/etc/dhcp]
$ ping 192.168.1.104
PING 192.168.1.104 (192.168.1.104) 56(84) bytes of data:
64 bytes from 192.168.1.104: icmp_seq=1 ttl=64 time=2.07 ms
64 bytes from 192.168.1.104: icmp_seq=2 ttl=64 time=2.67 ms
64 bytes from 192.168.1.104: icmp_seq=3 ttl=64 time=1.74 ms
64 bytes from 192.168.1.104: icmp_seq=4 ttl=64 time=2.16 ms
64 bytes from 192.168.1.104: icmp_seq=5 ttl=64 time=2.39 ms
64 bytes from 192.168.1.104: icmp_seq=6 ttl=64 time=1.46 ms
^C
--- 192.168.1.104 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5083ms
rtt min/avg/max/mdev = 1.459/2.079/2.667/0.397 ms
(kavindu@kali)-[/etc/dhcp]
$

[sudo] password for kavindu:
kavindu@kavindu-VirtualBox:~$ sudo dhclient
Setting LLNMR support level "yes" for "2", but the global support level is "no".
kavindu@kavindu-VirtualBox:~$ ip addt show
Object "addt" is unknown, try "ip help".
kavindu@kavindu-VirtualBox:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:4d:60:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.104/24 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 592sec preferred_lft 592sec
    inet6 2402:4000:b280:38f4:a00:27ff:fe4d:60d9/64 scope global dynamic mngtppaddr nopre
        valid_lft 296sec preferred_lft 116sec
    inet6 fe80::a00:27ff:fe4d:60d9/64 scope link
        valid_lft forever preferred_lft forever
kavindu@kavindu-VirtualBox:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data:
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=13.0 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=1.78 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=1.75 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=1.45 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=64 time=2.12 ms
^C
--- 192.168.1.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4805ms
rtt min/avg/max/mdev = 1.448/4.025/13.032/4.588 ms
kavindu@kavindu-VirtualBox:~$
```

It has assigned a new ip as 192.168.1.104 as ping it in kali and shows it connects. Now for lets check on dhcp lease file to further confirm. (using `sudo cat /var/lib/dhcp/dhcpd.leases`)



The image shows two terminal windows. The left window is a Kali Linux terminal where the user runs `sudo dhclient -r enp0s3` to release the IP, and then `sudo dhclient -v enp0s3` to request a new IP. The output shows the DHCP process and the assignment of `192.168.1.102`. The right window is an Ubuntu terminal where the user runs `cat /var/lib/dhcp/dhcpd.leases` to view the lease file. The output shows the lease file content, with the lease for `192.168.1.101` highlighted in red and the lease for `192.168.1.102` highlighted in yellow.

```
kavindu@kavindu-VirtualBox:~$ sudo dhclient -r enp0s3
Killed old client process.
kavindu@kavindu-VirtualBox:~$ sudo dhclient -v enp0s3
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/enp0s3/08:00:27:4d:60:d9
Sending on   LPF/enp0s3/08:00:27:4d:60:d9
Sending on   Socket/Fallback
xid: warning: no netdev with useable HWADDR found for seed's uniqueness enforcement
xid: rand init seed (0x668699cd) built using gethostid
DHCPDISCOVER on enp0s3 to 255.255.255.255 port 67 interval 3 (xid=0xf8c01922)
DHCPOFFER of 192.168.1.102 from 192.168.1.1
DHCPREQUEST for 192.168.1.102 on enp0s3 to 255.255.255.255 port 67 (xid=0x2219c0f8)
DHCPACK of 192.168.1.102 from 192.168.1.1 (xid=0xf8c01922)
Setting LLNMR support level "yes" for "2", but the global support level is "no".
bound to 192.168.1.102 -- renewal in 238 seconds.
kavindu@kavindu-VirtualBox:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:4d:60:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.102/24 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 596sec preferred_lft 596sec
kavindu@kavindu-VirtualBox:~$

# The format of this file is documented in the dhcpd.leases(5) manual page.
# This lease file was written by isc-dhcp-4.4.3-P1

# authoring-byte-order entry is generated, DO NOT DELETE
authoring-byte-order little-endian;

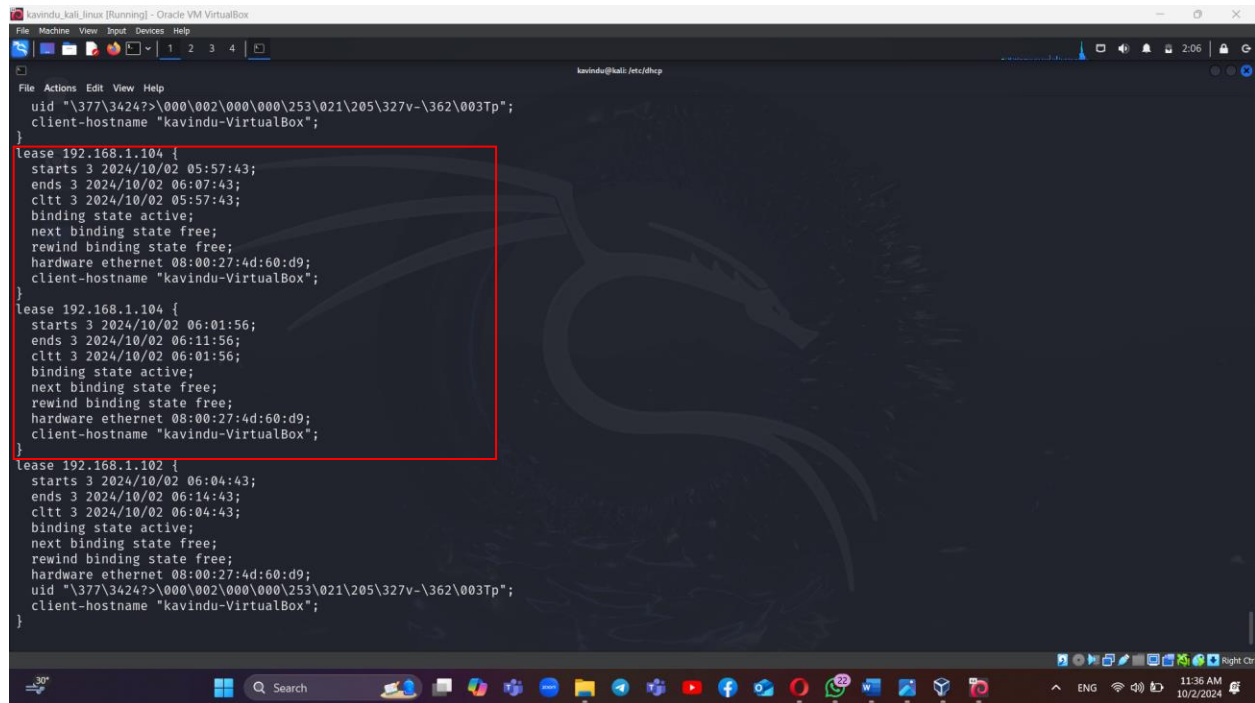
lease 192.168.1.101 {
    starts 5 2024/09/27 12:10:47;
    ends 5 2024/09/27 12:20:47;
    tstp 5 2024/09/27 12:20:47;
    cltt 5 2024/09/27 12:10:47;
    binding state free;
    hardware ethernet 08:00:27:99:a8:ba;
}

lease 192.168.1.100 {
    starts 5 2024/09/27 12:13:38;
    ends 5 2024/09/27 12:23:38;
    tstp 5 2024/09/27 12:23:38;
    cltt 5 2024/09/27 12:13:38;
    binding state free;
    hardware ethernet 08:00:27:99:a8:ba;
    uid "\001\010\000\231\250\272";
}

server-duid "\000\001\000\001\214g\362\010\000\326\332\364";

lease 192.168.1.102 {
    starts 0 2024/09/29 18:13:31;
    ends 0 2024/09/29 18:23:31;
    cltt 0 2024/09/29 18:13:31;
}
```

Red color shows the lease was assigned to my VM1 and yellow color shows the lease was assigned to my VM2.



```
kavindu@kali: /etc/dhcp
uid "\377\34247>\000\002\000\000\253\021\205\327v-\362\003Tp";
client-hostname "kavindu-VirtualBox";
}
lease 192.168.1.104 {
starts 3 2024/10/02 05:57:43;
ends 3 2024/10/02 06:07:43;
cltt 3 2024/10/02 05:57:43;
binding state active;
next binding state free;
rewind binding state free;
hardware ethernet 08:00:27:4d:60:d9;
client-hostname "kavindu-VirtualBox";
}
lease 192.168.1.104 {
starts 3 2024/10/02 06:01:56;
ends 3 2024/10/02 06:11:56;
cltt 3 2024/10/02 06:01:56;
binding state active;
next binding state free;
rewind binding state free;
hardware ethernet 08:00:27:4d:60:d9;
client-hostname "kavindu-VirtualBox";
}
lease 192.168.1.102 {
starts 3 2024/10/02 06:04:43;
ends 3 2024/10/02 06:14:43;
cltt 3 2024/10/02 06:04:43;
binding state active;
next binding state free;
rewind binding state free;
hardware ethernet 08:00:27:4d:60:d9;
uid "\377\34247>\000\002\000\000\253\021\205\327v-\362\003Tp";
client-hostname "kavindu-VirtualBox";
}
}
```

This shows the newly assigned ip for VM2 also has been recorded (.104).

5.DNS (Domain Name System)

The **DNS (Domain Name System)** is a critical part of the internet infrastructure. As in example perspective It acts like a phonebook, mapping human-readable domain names (like example.com) to IP addresses (like 192.0.2.1), which computers use to identify each other on the network.

When you enter a domain name in your browser, the DNS system resolves it into the corresponding IP address, allowing your computer to connect to the target website.

Configuring bind DNS server

Bind also known as Berkeley Internet Name Domain is one of the most widely used DNS server software .

So to install we must update and upgrade our repository to make sure our installing version is the latest by executing recsective commands in order.

Sudo apt update

Sudo apt upgrade

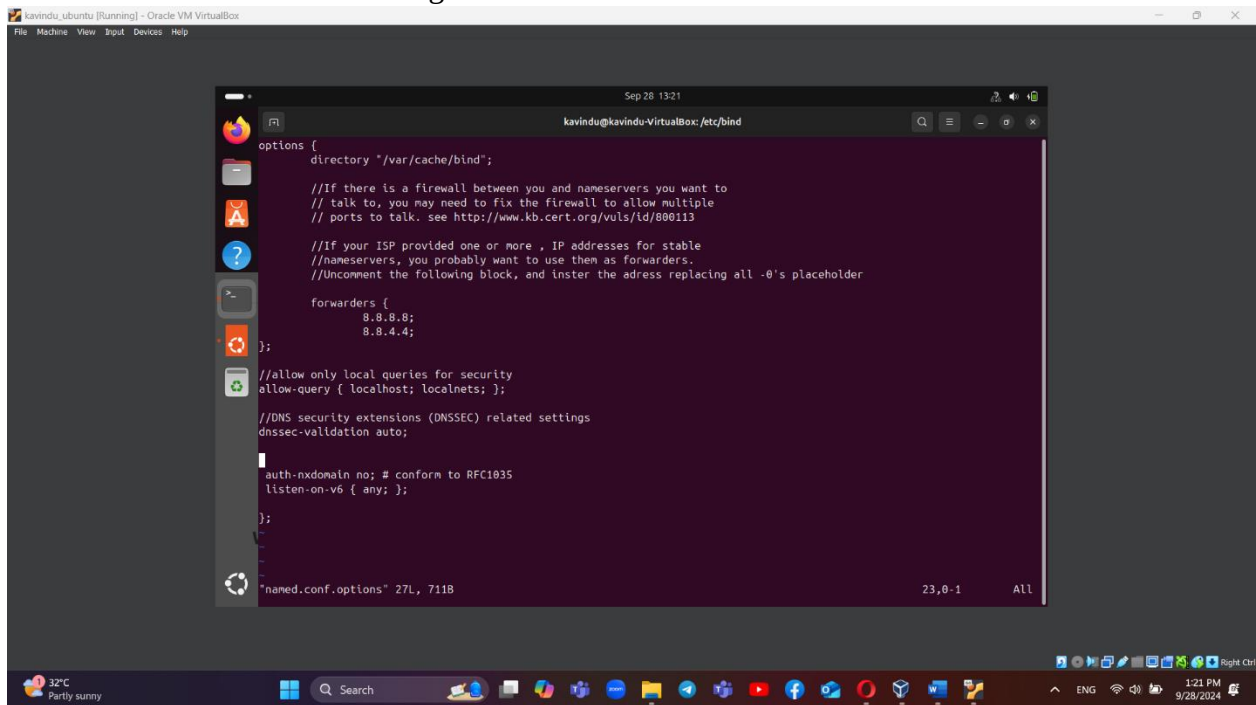
So after upgrading our repository , lets install the bind DNS server using command
sudo apt install bind9 bind9utils bind9-doc.

```
kavindu@kavindu-VirtualBox:/etc/bind$ sudo apt install bind9 bind9utils bind9-doc
```

after installing , lets start configuring the bind dns server. So lets configure option file which named as **named.conf.options**. I used vim editor to edit the file

```
kavindu@kavindu-VirtualBox:/etc/bind$ sudo vim named.conf.options
```

Under that file included following code.



```
kavindu@kavindu-VirtualBox: /etc/bind
options {
    directory "/var/cache/bind";

    //If there is a firewall between you and nameservers you want to
    // talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    //If your ISP provided one or more , IP addresses for stable
    //nameservers, you probably want to use them as forwarders.
    //Uncomment the following block, and inster the adress replacing all -0's placeholder

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

    //allow only local queries for security
    allow-query { localhost; localnets; };

    //DNS security extensions (DNSSEC) related settings
    dnssec-validation auto;

    auth-nxdomain no; # conform to RFC1035
    listen-on-v6 { any; };
};

#named.conf.options" 27L, 711B
23,0-1 ALL
```

```
options {
    directory "/var/cache/bind";

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

    allow-query { localnets; localhost; };

    dnssec-validation auto;

    auth-nxdomain no; # conform to RFC1035

    listen-on-v6 { any; };

};
```

dnssec-validation auto: It checks if DNS responses are secure by validating them with DNSSEC (a security feature for DNS). **In simple terms**, it adds extra security to ensure the website info you get is real and trustworthy.

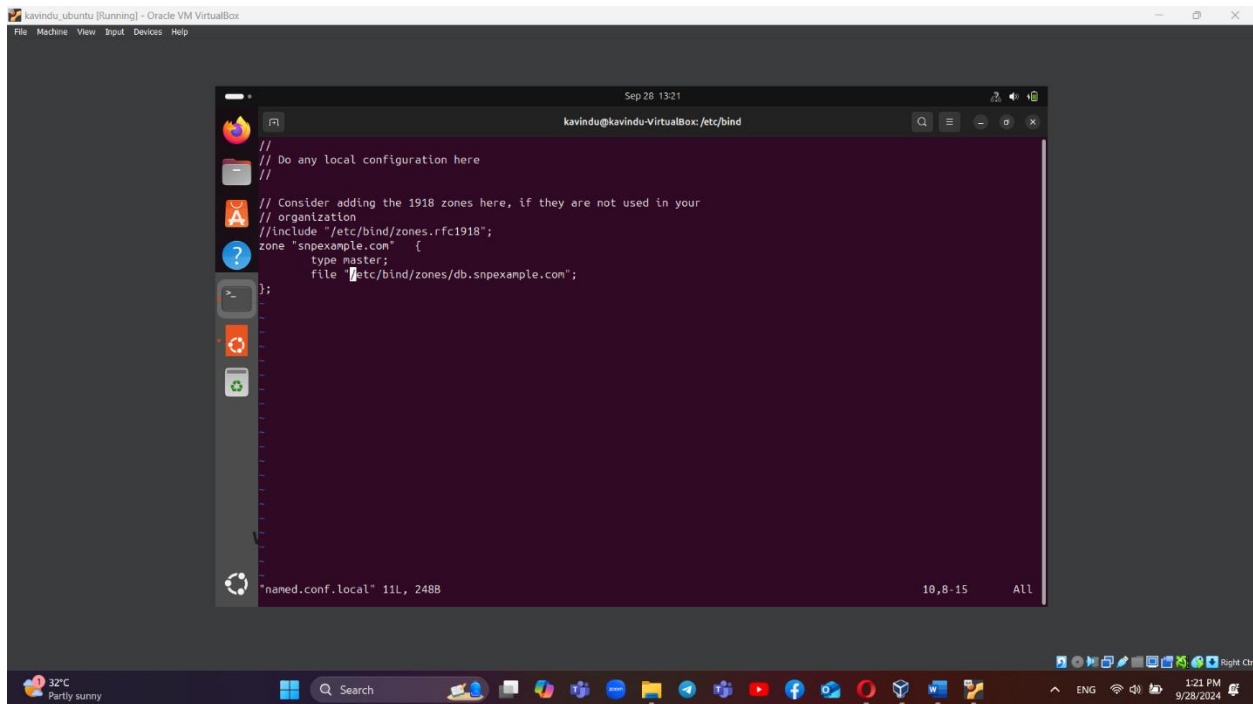
2. auth-nxdomain no: It makes the DNS server say "I don't know this domain" instead of pretending it's in charge of it. **In simple terms** the server doesn't falsely claim to know when a domain doesn't exist.

3. listen-on-v6 { any; }:: The server listens for DNS queries on any available IPv6 address. **In simple terms** the DNS server is open to requests on all IPv6 internet addresses.

4. forwarders: It tells your DNS server to ask other DNS servers (like Google's 8.8.8.8) for help when it doesn't know the answer. Your DNS server can ask other servers (like Google's) for help if it doesn't know an answer.

Now let's configure our zone files. They will specify which domains in the DNS server will handle. To edit I have used

sudo nano /etc/bind/named.conf.local command



```
kavindu@kavindu-VirtualBox: /etc/bind
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";
zone "snpexample.com" {
    type master;
    file "/etc/bind/zones/db.snpexample.com";
};

"named.conf.local" 11L, 248B
```

```
zone "snpexample.com" {
    type master;
    file "/etc/bind/zones/db.snpexample.com";
};
```

Zone “**snpexample.com**” defines the domain name or zone for which DNS is authoritative. As the domain name I have chosen “snpexample.com”

Type master :- indicates that the dns server is the primary or master server for this zone. It holds the original and authoritative records for the zone. It can distribute these records to secondary (slave) servers.

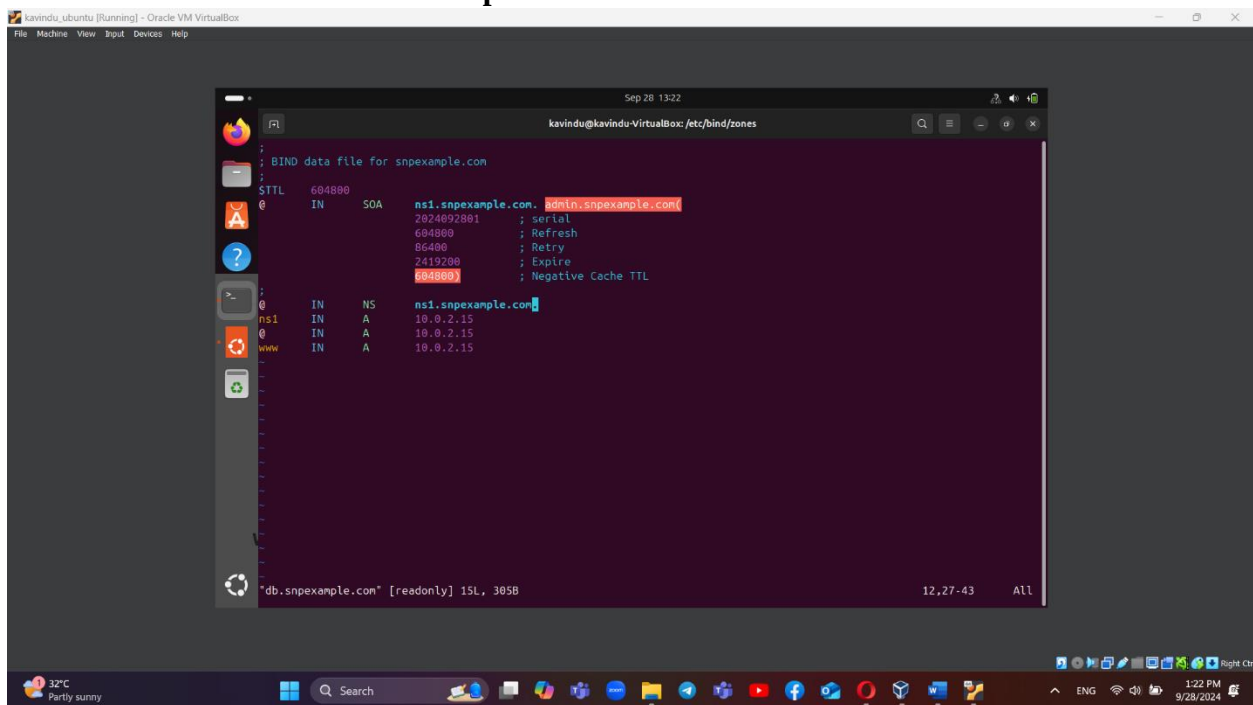
file “**/etc/bind/zones/db.snpexample.com**” ; : This specifies the location of the zone file for the domain.

So simply , this code sets up the snpexample.com domain on your DNS server, indicating that your server holds the master copy of DNS records for this domain. The actual DNS data like Ip addresses is stored in the file “**/etc/bind/zones/db.snpexample.com**”

so lets create that zone file to store the dns data. 1st create the directory zones then create the db.snpexample.com

Using **sudo mkdir /etc/bind/zones**

sudo nano /etc/bind/zones/db.example.com



The screenshot shows a terminal window titled "kavindu@kavindu-VirtualBox: /etc/bind/zones". The terminal displays the contents of the BIND data file for snpexample.com. The file starts with a comment: "BIND data file for snpexample.com". It then defines the zone's SOA (Start of Authority) record with the following parameters: STTL 604800, IN, SOA, ns1.snpexample.com. admin.snpexample.com, 2024092801, 604800, 86400, 2419200, and 604800. The SOA record is followed by a list of nameservers (NS) and their IP addresses (A records). The nameservers are ns1.snpexample.com, ns1, and www, all pointing to the IP address 10.0.2.15. The terminal also shows the status of the zone file: "db.snpexample.com" [readonly] 15L, 305B. The bottom of the terminal shows the system status: 12.27-43 All.

```

$ cat /etc/bind/zones/db.snpexample.com
; BIND data file for snpexample.com
$TTL 604800
@ IN SOA ns1.snpexample.com. admin.snpexample.com.
    2024092801      ; serial
    604800         ; Refresh
    86400          ; Retry
    2419200        ; Expire
    604800         ; Negative Cache TTL
;
@ IN NS ns1.snpexample.com
ns1 IN A 10.0.2.15
@ IN A 10.0.2.15
www IN A 10.0.2.15

```

Command used,

;

; BIND data file for example.com

;

\$TTL 604800

@ IN SOA ns1.snpexample.com. admin.example.com. (

2 ; Serial

604800 ; Refresh

86400 ; Retry

2419200 ; Expire

604800) ; Negative Cache TTL

;

@ IN NS ns1.snpexample.com.

ns1 IN A 10.0.2.15

@ IN A 10.0.2.15

www IN A 10.0.2.15

\$TTL 604800 : \$TTL stands for Time to Live ,and 604800 seconds is equivalent to 7 days.This means that DNS records in this zone can be cached for 7 days before the DNS resolver must query the authoritative server for fresh information.

@ IN SOA ns1.example.com. admin.example.com. (

2 ; Serial

604800 ; Refresh

86400 ; Retry

2419200 ; Expire

604800) ; Negative Cache TTL

@: Refers to the **current domain** (example.com in this case).

IN: Internet class (this is the standard class for DNS).

SOA: Stands for **Start of Authority**. This defines the primary name server for the domain and contact info.

ns1.example.com.: The **primary name server** for example.com.

admin.example.com.: The **email address** of the administrator, written in a format where the "@" symbol is replaced by a period (.). In reality, this would be admin@example.com.

2: Serial number. This should increment whenever changes are made to the zone file, so that secondary (slave) DNS servers know that they need to update their records.

604800: Refresh (in seconds) – how often secondary servers should check for updates from the master server. Here it's set to **7 days**.

86400: Retry (in seconds) – how long secondary servers should wait before retrying a failed zone transfer. Here it's **24 hours**.

2419200: Expire (in seconds) – the time after which secondary servers should stop serving the zone if they cannot reach the master. Here it is **28 days**.

604800: Negative Cache TTL – how long other DNS servers should cache a negative response (when a record isn't found). Here it's **7 days**.

@ IN NS ns1.snpexample.com.

- This defines a **Name Server (NS)** record.
- **@:** Refers to the example.com domain.
- **NS ns1.snpexample.com.:** Specifies that ns1.example.com is the name server for this domain.

ns1 IN A 10.0.2.15

- **ns1:** Refers to the **ns1.snpexample.com** subdomain.
- **A:** Address record – it maps a domain name to an **IPv4 address**.
- **10.0.2.15:** The **IPv4 address** for ns1.example.com.

@ IN A 10.0.2.15

- **@:** Refers to the example.com domain.
- **A:** Address record – it maps the domain name example.com to the **IPv4 address** 10.0.2.15

www IN A 10.0.2.15

- **www**: Refers to the **www.snpexample.com** subdomain.
- **A**: Address record – it maps **www.example.com** to the **IPv4 address 10.0.2.15**

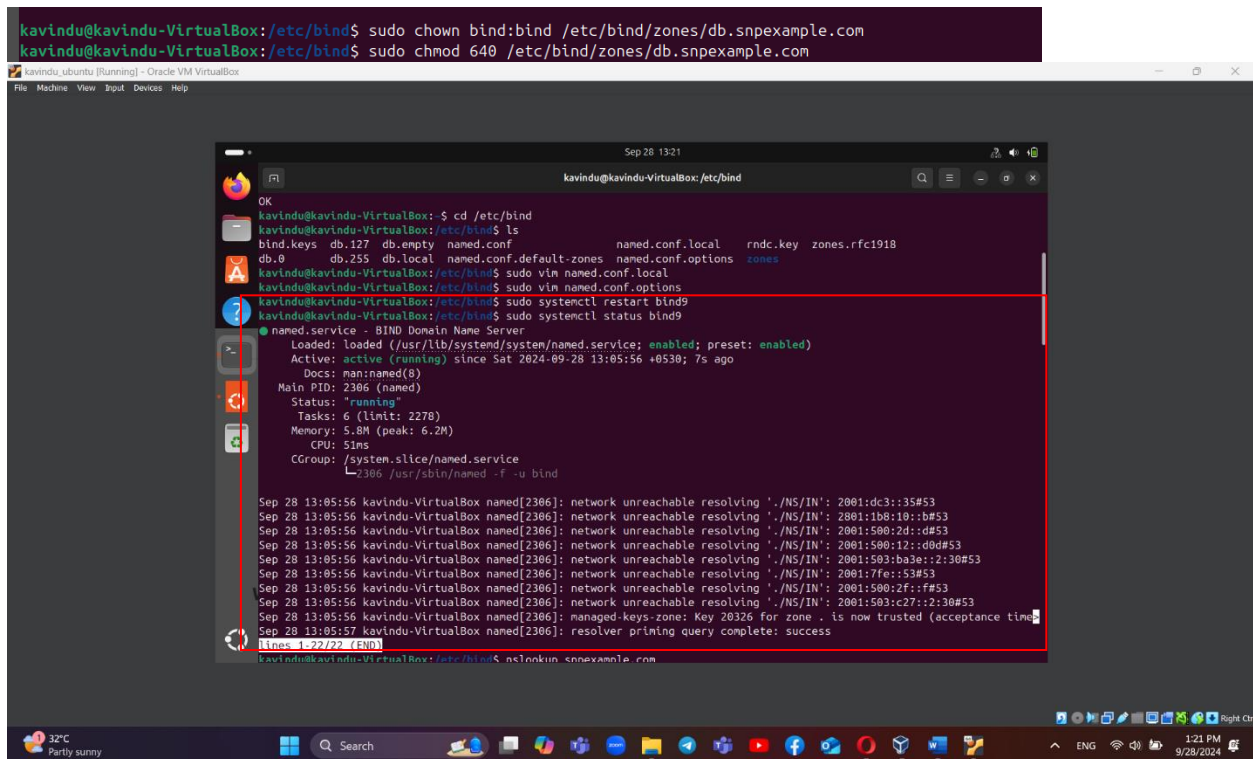
So in summary ,this file configures DNS for **example.com**, with **ns1.example.com** as its name server.The domain and its subdomains (**ns1**, **www**) are all mapped to the IP address **10.0.2.15**.The SOA record tells secondary DNS servers how often to check for updates and what to do if they cannot reach the primary server.

Lets restart the bind9 server and check the status using command ,

sudo systemctl restart bind9 ,

sudo systemctl status bind9

```
kavindu@kavindu-VirtualBox:/etc/bind$ sudo chown bind:bind /etc/bind/zones/db.snpexample.com
kavindu@kavindu-VirtualBox:/etc/bind$ sudo chmod 640 /etc/bind/zones/db.snpexample.com
```



```
File Machine View Input Devices Help
kavindu@kavindu-VirtualBox: /etc/bind
Sep 28 13:21
kavindu@kavindu-VirtualBox: /etc/bind
kavindu@kavindu-VirtualBox: $ cd /etc/bind
kavindu@kavindu-VirtualBox:/etc/bind$ ls
bind.keys  db.127  db.empty  named.conf  named.conf.local  rndc.key  zones.rfc1918
db.0      db.255  db.local  named.conf.default-zones  named.conf.options  zones
kavindu@kavindu-VirtualBox:/etc/bind$ sudo vim named.conf.local
kavindu@kavindu-VirtualBox:/etc/bind$ sudo vim named.conf.options
kavindu@kavindu-VirtualBox:/etc/bind$ sudo systemctl restart bind9
kavindu@kavindu-VirtualBox:/etc/bind$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sat 2024-09-28 13:05:56 +0530; 7s ago
     Docs: man:named(8)
    Main PID: 2306 (named)
      Status: "running"
        Tasks: 6 (limit: 2270)
      Memory: 5.8M (peak: 6.2M)
         CPU: 51ms
       CGroup: /system.slice/named.service
              └─2306 /usr/sbin/named -f -u bind

Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:dc3::35#53
Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:1b8:10::b#53
Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:500:2d::d#53
Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:500:12::d#53
Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:503:b3e::2:30#53
Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:770::53#53
Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:500:2f::f#53
Sep 28 13:05:56 kavindu-VirtualBox named[2306]: network unreachable resolving './NS/IN': 2001:503:c27::2:30#53
Sep 28 13:05:57 kavindu-VirtualBox named[2306]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance time
Sep 28 13:05:57 kavindu-VirtualBox named[2306]: resolver priming query complete: success
lines 1-22/22 (END)
kavindu@kavindu-VirtualBox:/etc/bind$ nslookup snpexample.com
```

To make sure lets change the file permissions and change firewalls allowing to accept and send requests using udp and tcp ports

```
kavindu@kavindu-VirtualBox:/etc/bind$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
kavindu@kavindu-VirtualBox:/etc/bind$ sudo iptables -A INPUT -p tcp --dport 53 -j ACCEPT
kavindu@kavindu-VirtualBox:/etc/bind$ sudo iptables -A INPUT -p udp --dport 53 -j ACCEPT
kavindu@kavindu-VirtualBox:/etc/bind$ sudo iptables-save | save tee /etc/iptables/rules.v4
Command 'save' not found, but can be installed with:
sudo apt install atfs
Command 'sudp' not found, did you mean:
  command 'ssdp' from snap ssdp (0.0.1)
  command 'sudp' from deb sudo (1.9.14p2-1ubuntu1)
  command 'sudp' from deb sudo:ldap (1.9.14p2-1ubuntu1)
  command 'sup' from deb sup (20100519-3)
  command 'sfdp' from deb graphviz (2.42.2-7build3)
See 'snap info <snapname>' for additional versions.
kavindu@kavindu-VirtualBox:/etc/bind$ nslookup snpexample.com 127.0.0.1
Server:
  127.0.0.1
Address:
  127.0.0.1#53

Name:   snpexample.com
Address: 10.0.2.15

kavindu@kavindu-VirtualBox:/etc/bind$ dig @127.0.0.1 snpexample.com
; <<>> DiG 9.18.24-0ubuntu5-Ubuntu <<>> @127.0.0.1 snpexample.com
; (1 server found)
;; global options: +cmd
```

To make sure the syntax of edited files are valid , I did the following commands.

```
Sep 28 13:05:56 kavindu-VirtualBox named[2386]: network unreachable resolving '././NS/IN': 2001:503:c27::2:30#53
Sep 28 13:05:57 kavindu-VirtualBox named[2386]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance time
Sep 28 13:05:57 kavindu-VirtualBox named[2386]: resolver pruning query complete: success
kavindu@kavindu-VirtualBox:/etc/bind$ nslookup snpexample.com
Server:
  127.0.0.53
Address:
  127.0.0.53#53

** server can't find snpexample.com: NXDOMAIN

kavindu@kavindu-VirtualBox:/etc/bind$ sudo vim named.conf.local
kavindu@kavindu-VirtualBox:/etc/bind$ sudo systemctl restart bind9
kavindu@kavindu-VirtualBox:/etc/bind$ nslookup snpexample.com
Server:
  127.0.0.53
Address:
  127.0.0.53#53

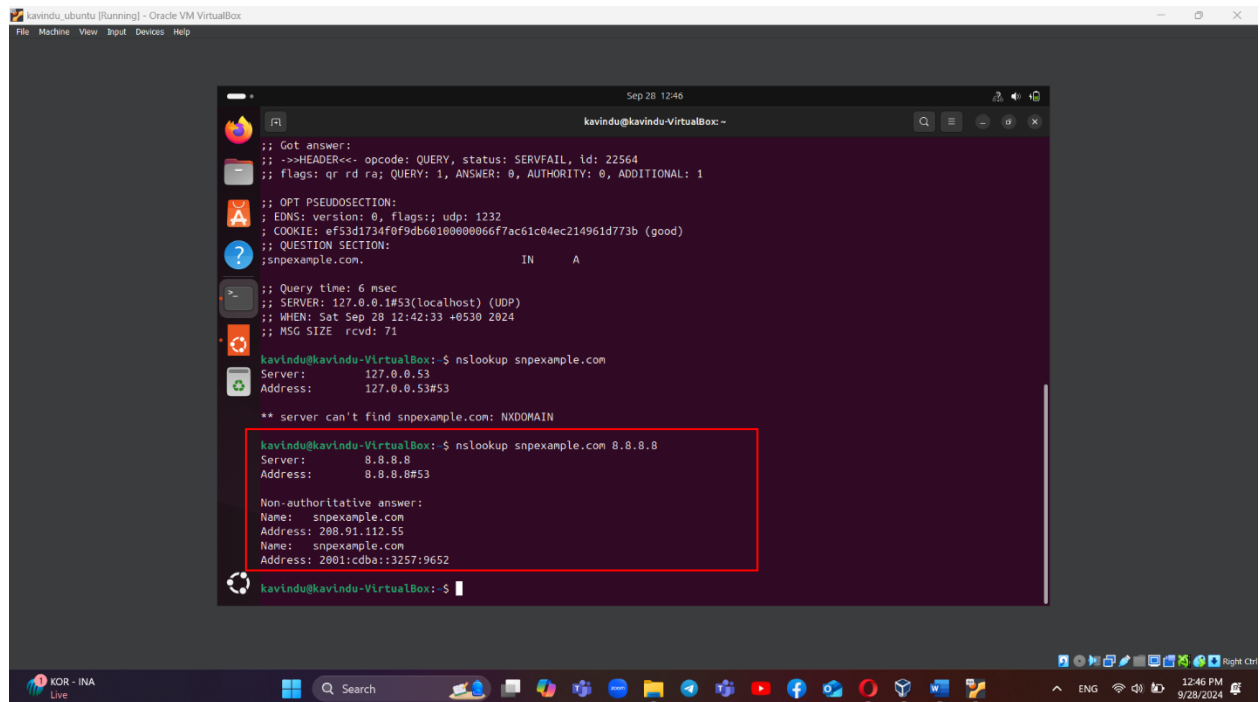
** server can't find snpexample.com: NXDOMAIN

kavindu@kavindu-VirtualBox:/etc/bind$ sudo chown bind:bind /etc/bind/zones/db.snpexample.com
kavindu@kavindu-VirtualBox:/etc/bind$ sudo chmod 640 /etc/bind/zones/db.snpexample.com
kavindu@kavindu-VirtualBox:/etc/bind$ sudo named-checkconf
kavindu@kavindu-VirtualBox:/etc/bind$ sudo named-checkzone snpexample.com /etc/bind/zones/db.snpexample.com
zone snpexample.com/IN: loaded serial 2024092801
OK
kavindu@kavindu-VirtualBox:/etc/bind$ sudo systemctl restart bind9
kavindu@kavindu-VirtualBox:/etc/bind$ sudo systemd-resolve --flush-caches
sudo: systemd-resolve: command not found
kavindu@kavindu-VirtualBox:/etc/bind$ nslookup snpexample.com
Server:
  127.0.0.53
Address:
  127.0.0.53#53

** server can't find snpexample.com: NXDOMAIN
```

Here I have mistakenly tried to check its working within the same server , due to that I got the error as no server is to be found.(yellow color)

There fore lets check our dns server is working with an external server. I used google's server 8.8.8.8



```
kavindu@kavindu-VirtualBox:~$ nslookup snpexample.com
Server:      127.0.0.53
Address:     127.0.0.53#53

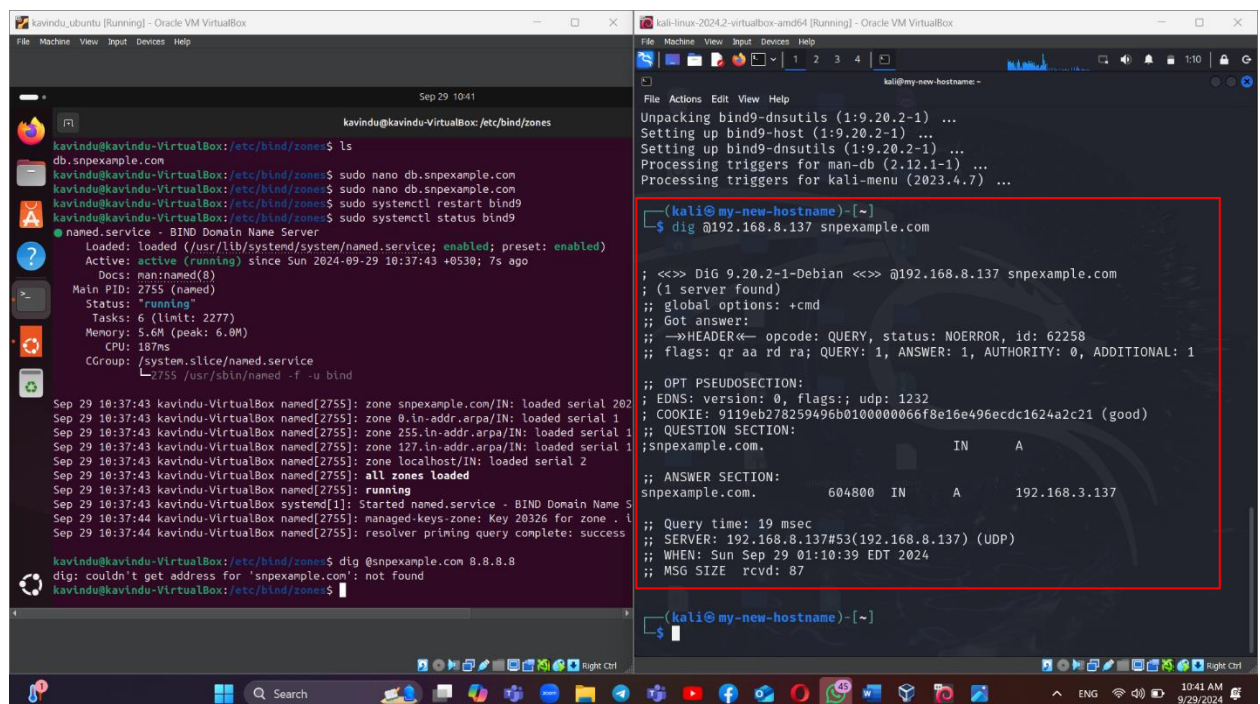
** server can't find snpexample.com: NXDOMAIN

kavindu@kavindu-VirtualBox:~$ nslookup snpexample.com 8.8.8.8
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   snpexample.com
Address: 2001:cdba::3257:9652
Name:   snpexample.com
Address: 2001:cdba::3257:9652

kavindu@kavindu-VirtualBox:~$
```

Now with another VM I used kali. (In the later in process I changed the VM's ip to an static ip address)(Note , Both VMs must have Bridgeadapter network type (which I prefere , but definitely not NAT))



```
kavindu@kavindu-VirtualBox:~$ ls /etc/bind/zones
db.snpexample.com
db.snpexample.com
kavindu@kavindu-VirtualBox:~$ sudo nano db.snpexample.com
kavindu@kavindu-VirtualBox:~$ sudo nano db.snpexample.com
kavindu@kavindu-VirtualBox:~$ sudo systemctl restart bind9
kavindu@kavindu-VirtualBox:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sun 2024-09-29 10:37:43 +0530; 7s ago
     Main PID: 2755 (named)
       Status: "running"
         Tasks: 6 (limit: 2277)
        Memory: 5.6M (peak: 6.0M)
           CPU: 187ms
      CGroup: /system.slice/named.service
              └─2755 /usr/sbin/named -f -u bind

Sep 29 10:37:43 kavindu-VirtualBox named[2755]: zone snpexample.com/IN: loaded serial 202
Sep 29 10:37:43 kavindu-VirtualBox named[2755]: zone 0.in-addr.arpa/IN: loaded serial 1
Sep 29 10:37:43 kavindu-VirtualBox named[2755]: zone 255.in-addr.arpa/IN: loaded serial 1
Sep 29 10:37:43 kavindu-VirtualBox named[2755]: zone 127.in-addr.arpa/IN: loaded serial 1
Sep 29 10:37:43 kavindu-VirtualBox named[2755]: zone localhost/IN: loaded serial 2
Sep 29 10:37:43 kavindu-VirtualBox named[2755]: all zones loaded
Sep 29 10:37:43 kavindu-VirtualBox named[2755]: running
Sep 29 10:37:43 kavindu-VirtualBox system[1]: Started named.service - BIND Domain Name S
Sep 29 10:37:44 kavindu-VirtualBox named[2755]: managed-keys-zone: Key 20326 for zone . i
Sep 29 10:37:44 kavindu-VirtualBox named[2755]: resolver priming query complete: success

kavindu@kavindu-VirtualBox:~$ dig @snpexample.com 8.8.8.8
dig: couldn't get address for "snpexample.com": not found
kavindu@kavindu-VirtualBox:~$
```

```
kali@my-new-hostname:~$ dig @192.168.8.137 snpexample.com

; <<>> DiG 9.20.2-1-Debian <<>> @192.168.8.137 snpexample.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 62258
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags: udp: 1232
;; COOKIE: 9119eb278259496b0100000066f8e16e496ecd1624a2c21 (good)
;; QUESTION SECTION:
;; snpexample.com.                IN      A
;; ANSWER SECTION:
snpexample.com.                604800  IN      A      192.168.3.137

;; Query time: 19 msec
;; SERVER: 192.168.8.137#53(192.168.8.137) (UDP)
;; WHEN: Sun Sep 29 01:10:39 EDT 2024
;; MSG SIZE rcvd: 87

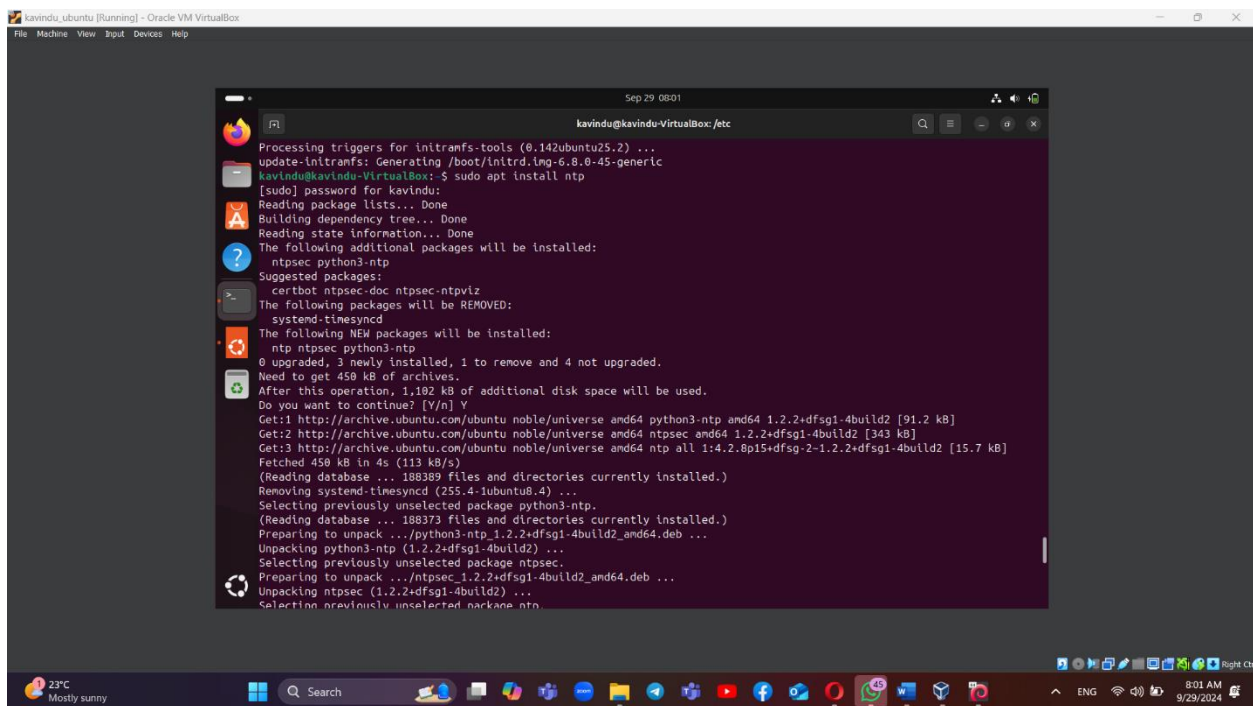
kali@my-new-hostname:~$
```

6.NTP(Network Transfer Protocol)

Accurate system time is crucial for various reasons, such as:

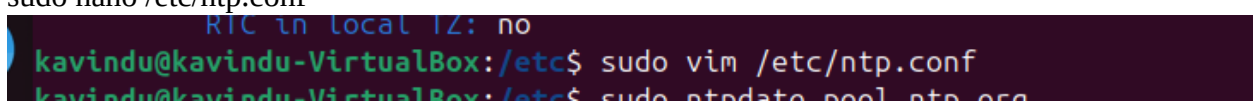
- **Security:** Time discrepancies can affect timestamp-based security mechanisms like log analysis, SSL/TLS certificate validation, and authentication processes.
- **Logging and Auditing:** Coordinating timestamps across systems ensures accurate event tracking and debugging.
- **Network Coordination:** Time-sensitive applications (e.g., financial transactions, database operations) require synchronized time to avoid issues like data inconsistency.
- **File System Integrity:** Ensures correct file timestamps, which helps in backups and file synchronization.

1st of all , lets install ntp



```
kavindu@kavindu-VirtualBox: /etc$ sudo apt install ntp
[sudo] password for kavindu:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ntpsec python3-ntp
Suggested packages:
  certbot ntpsec-doc ntpsec-ntpvlz
The following packages will be REMOVED:
  systemd-timesyncd
The following NEW packages will be installed:
  ntp ntpsec python3-ntp
0 upgraded, 3 newly installed, 1 to remove and 4 not upgraded.
Need to get 450 kB of archives.
After this operation, 1,102 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-ntp amd64 1.2.2+dfsg1-4build2 [91.2 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 ntpsec amd64 1.2.2+dfsg1-4build2 [343 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 ntp all 1:4.2.8p15+dfsg-2-1.2.2+dfsg1-4build2 [15.7 kB]
Fetched 450 kB in 4s (113 kB/s)
(Reading database ... 188389 files and directories currently installed.)
Removing systemd-timesyncd (255.4-1ubuntu8.4) ...
Selecting previously unselected package python3-ntp.
(Reading database ... 188373 files and directories currently installed.)
Preparing to unpack .../python3-ntp_1.2.2+dfsg1-4build2_amd64.deb ...
Unpacking python3-ntp (1.2.2+dfsg1-4build2) ...
Selecting previously unselected package ntpsec.
Preparing to unpack .../ntpsec_1.2.2+dfsg1-4build2_amd64.deb ...
Unpacking ntpsec (1.2.2+dfsg1-4build2) ...
Selecting previously unselected package ntp.
Preparing to unpack .../ntp_1:4.2.8p15+dfsg-2-1.2.2+dfsg1-4build2_all.deb ...
Unpacking ntp (1:4.2.8p15+dfsg-2-1.2.2+dfsg1-4build2) ...
Setting up python3-ntp (1.2.2+dfsg1-4build2) ...
Setting up ntpsec (1.2.2+dfsg1-4build2) ...
Setting up ntp (1:4.2.8p15+dfsg-2-1.2.2+dfsg1-4build2) ...
```

Now lets start configuring. configuration file is located at /etc/ntp.conf. You can edit this file to specify which NTP server(s) to synchronize with. Therefore to access it I used,
`sudo nano /etc/ntp.conf`



```
kavindu@kavindu-VirtualBox: /etc$ sudo vim /etc/ntp.conf
```

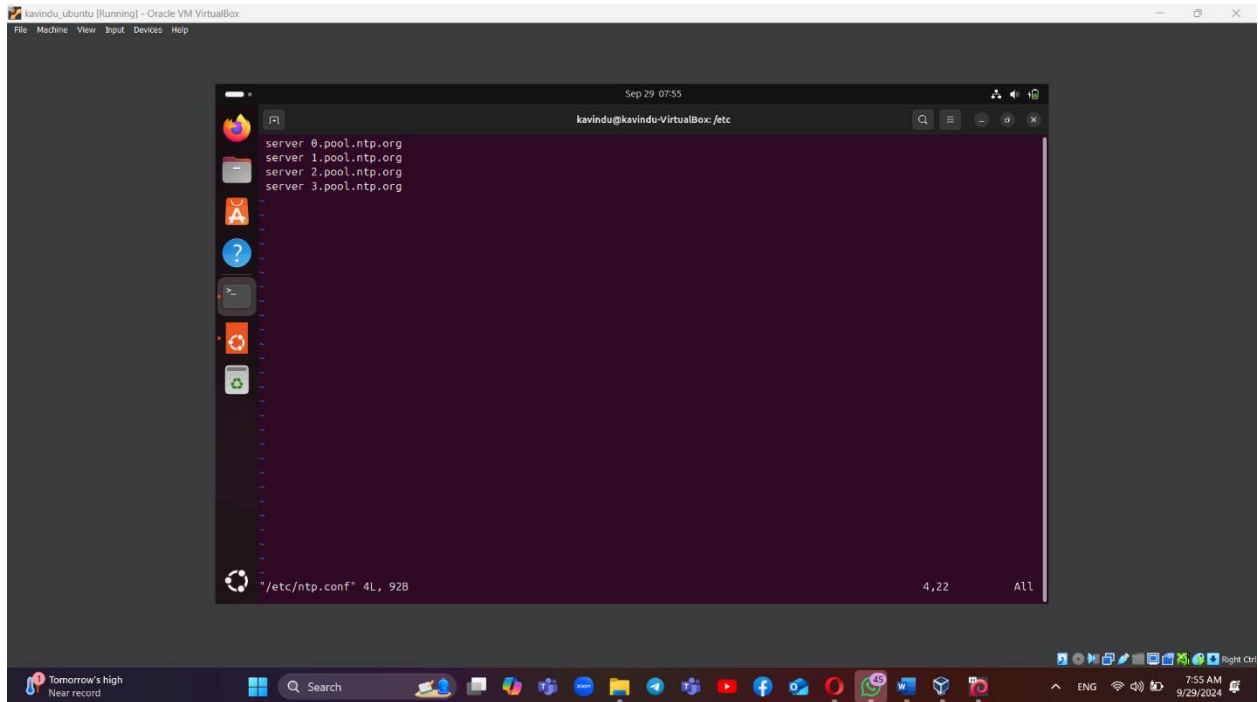
In that file added following lines,

server 0.pool.ntp.org

server 1.pool.ntp.org

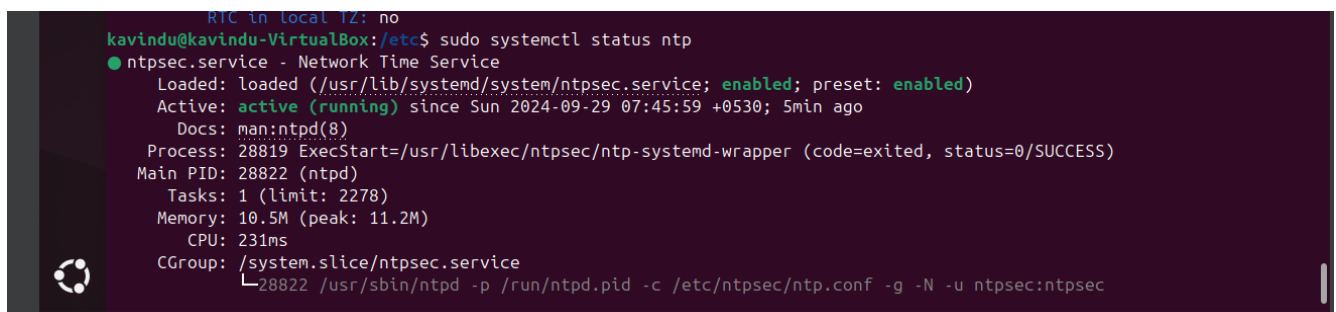
server 2.pool.ntp.org

server 3.pool.ntp.org



now lets start the ntp server using sudo systemctl start ntp command

```
kavindu@kavindu-VirtualBox:/etc$ sudo systemctl start ntp
```



Now lets check its working or not , using **ntpq-p** command

```
kavindu@kavindu-VirtualBox: /etc$ ntpq -p
remote          refid      st t when poll reach  delay  offset  jitter
=====
0.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
1.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
2.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
3.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
prod-ntp-5.ntp4.ps5.canonical.com 201.68.88.106 2 u 39 64 7 185.9218 8.7807 8.6262
+time.cloudflare.com 10.4.8.56 3 u 59 64 1 45.5840 15.1821 7.6942
+time.cloudflare.com 10.4.8.56 3 u 59 64 1 34.2666 8.9893 4.6870
time.cloudflare.com .STEP.    16 u - 64  0 0.0000 0.0000 0.0001
time.cloudflare.com .STEP.    16 u - 64  0 0.0000 0.0000 0.0001

kavindu@kavindu-VirtualBox: /etc$ timedatectl
Local time: Sun 2024-09-29 07:53:04 +0530
Universal time: Sun 2024-09-29 02:23:04 UTC
RTC time: Sun 2024-09-29 02:23:14
Time zone: Asia/Colombo (+0530, +0530)
System clock synchronized: no
NTP service: n/a
RTC in local TZ: no
kavindu@kavindu-VirtualBox: /etc$ sudo ntpdate pool.ntp.org
sudo: ntpdate: command not found
kavindu@kavindu-VirtualBox: /etc$ sudo apt install ntpdate
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
```

Seems like its not properly working and further clarified using **timedatectl** command.so to update the pool servers I had to install it again and then check again(yellow)

```
kavindu@kavindu-VirtualBox: /etc$ timedatectl
Local time: Sun 2024-09-29 07:54:44 +0530
Universal time: Sun 2024-09-29 02:24:44 UTC
RTC time: Sun 2024-09-29 02:24:53
Time zone: Asia/Colombo (+0530, +0530)
System clock synchronized: no
NTP service: n/a
RTC in local TZ: no
kavindu@kavindu-VirtualBox: /etc$ sudo vim /etc/ntp.conf
kavindu@kavindu-VirtualBox: /etc$ sudo ntpdate pool.ntp.org
2024-09-29 07:57:07.832249 (+0530) -0.010684 +/- 0.019846 pool.ntp.org 162.159.200.1 s3 no-leap
kavindu@kavindu-VirtualBox: /etc$ ntpq -p
remote          refid      st t when poll reach  delay  offset  jitter
=====
0.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
1.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
2.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
3.ubuntu.pool.ntp.org .POOL.    16 p - 256  0 0.0000 0.0000 0.0001
*prod-ntp-5.ntp4.ps5.canonical.com 201.68.88.106 2 u 36 64 177 171.5450 1.1323 5.4996
+time.cloudflare.com 10.4.8.56 3 u 56 64 37 19.7558 -2.4306 13.0528
+time.cloudflare.com 10.4.8.56 3 u 51 64 37 21.9184 2.7448 3.6925
time.cloudflare.com .STEP.    16 u - 64  0 0.0000 0.0000 0.0001
time.cloudflare.com .STEP.    16 u - 128 0 0.0000 0.0000 0.0001

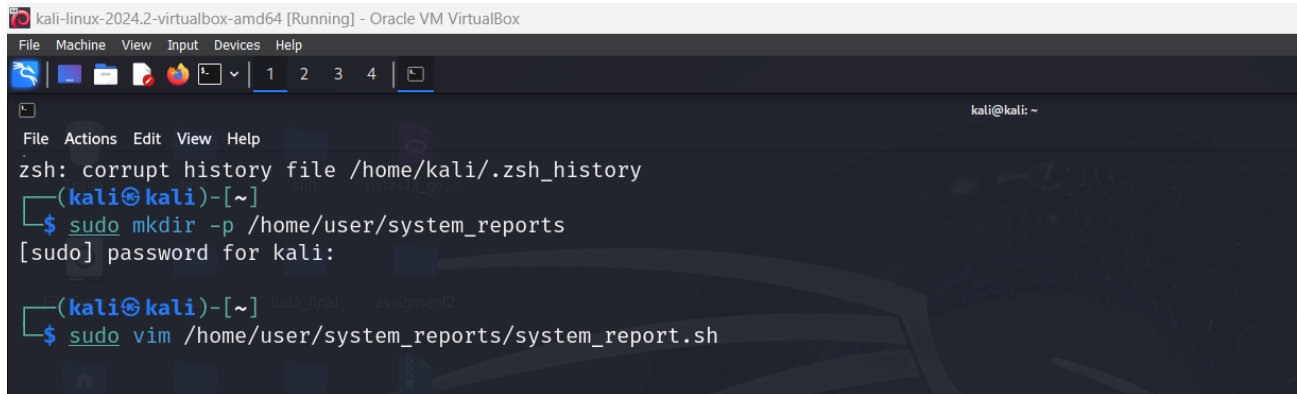
kavindu@kavindu-VirtualBox: /etc$ timedatectl
Local time: Sun 2024-09-29 07:57:44 +0530
Universal time: Sun 2024-09-29 02:27:44 UTC
RTC time: Sun 2024-09-29 02:27:44
Time zone: Asia/Colombo (+0530, +0530)
System clock synchronized: yes
NTP service: n/a
RTC in local TZ: no
kavindu@kavindu-VirtualBox: /etc$
```

Now its working .

7.Shell scripting and security

Task 01

Let's make the directory and the file where the task is mentioned.

A screenshot of a Kali Linux terminal window. The window title is "kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox". The terminal shows a zsh prompt where the user enters "corrupt history file /home/kali/.zsh_history". Then, the user enters "\$ sudo mkdir -p /home/user/system_reports", and the terminal shows "[sudo] password for kali:". Finally, the user enters "\$ sudo vim /home/user/system_reports/system_report.sh".

```
kali@kali: ~  
File Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)~  
$ sudo mkdir -p /home/user/system_reports  
[sudo] password for kali:  
(kali@kali)~  
$ sudo vim /home/user/system_reports/system_report.sh
```

I used vim editor to configure the script file.

#!/bin/bash

Defining the destination i wanted

destinationdir="/home/user/system_reports"

Create the destination directory if it does not exist (just in case)

mkdir -p "\$DEST_DIR"

Get current date as the output

date=\$(date +"%Y-%m-%d %H:%M:%S")

Get system uptime

uptime=\$(uptime -p)

Get free memory available

free_memory=\$(free -h | grep Mem | awk '{print \$4}')

```
kali@kali:~$ cat /dev/null > /tmp/.Xauthority
```

```
#!/bin/bash
```

```
# Defining the destination i wanted  
destinationdir="/home/user/system_reports"
```

```
# Create the destination directory if it does not exist (just in case)  
mkdir -p "$DEST_DIR"
```

```
# Get current date as the output  
date=$(date +"%Y-%m-%d %H:%M:%S")
```

```
# Get system uptime  
uptime=$(uptime -p)
```

```
# Get free memory available  
free_memory=$(free -h | grep Mem | awk '{print $4}')
```

```
# Get disk usage  
diskusage=$(df -h | grep '^/dev/' | awk '{print $1, $3, $2, $5}')
```

```
# Create the report filename as  
report="$destinationdir/system_report_$(date +%Y%m%d).log"
```

```
# Write the report to the file (the output we collected)  
{  
    echo "Report Date: $date"  
    echo "Uptime: $uptime"  
    echo "Free Memory: $free_memory"  
    echo "Disk Usage:"  
    echo "$diskusage"  
}  
> "$report"
```

```
echo "Report generated at $report"
```

```
-- INSERT --
```

code is pretty much explain it self, but available free memory command is quite complicated. Lets break it down

```
DISK_USAGE=$(df -h | grep '^/dev' | awk '{print $1, $3, $2, $5}')
```

df -h : used to display the amount of disk space used and available on linux file system.

-h : to humar-readble format.

grep '^/dev' :grep filters the output to show only lines that start with /dev, which usually correspond to disk partitions (e.g., /dev/sda1).The caret ^ is a regular expression that matches the beginning of a line.

awk '{print \$1, \$3, \$2, \$5}' :awk is a text-processing command that allows you to select and format specific fields from the output.

In this case, it's printing the first, third, second, and fifth fields from each line:

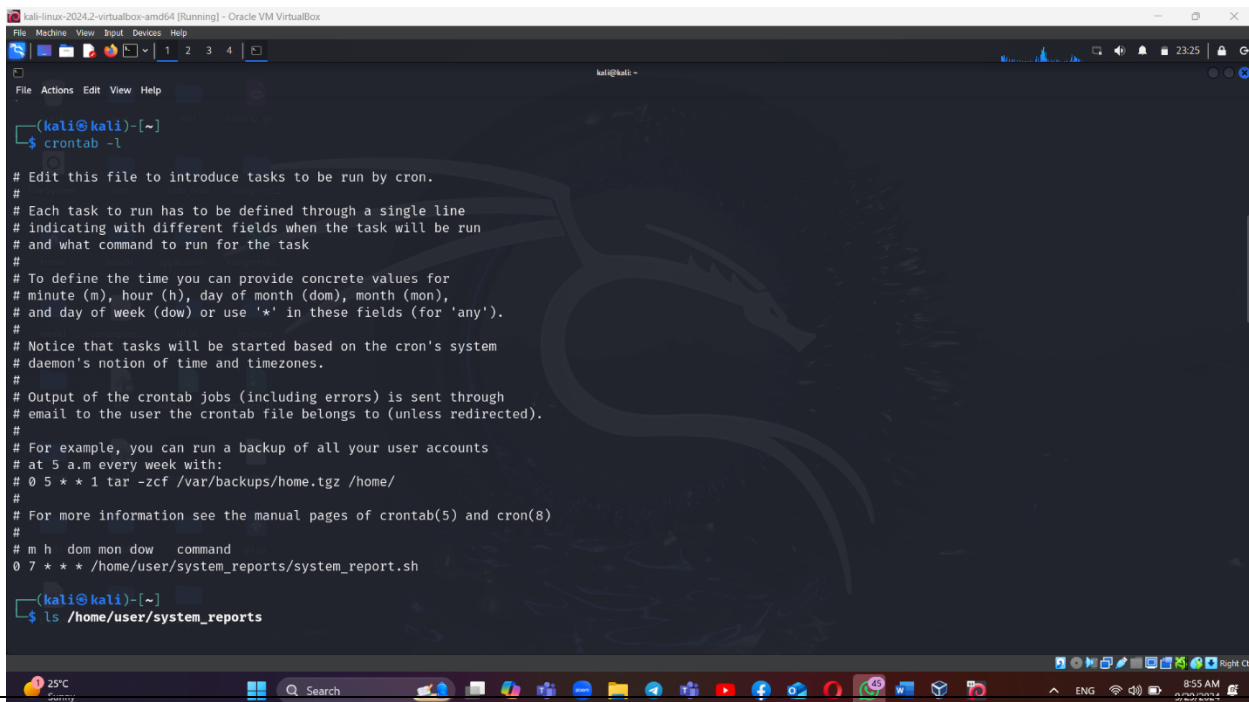
- \$1: The first field is the device name (e.g., /dev/sda1).
- \$3: The third field is the amount of space used (e.g., 25G).
- \$2: The second field is the total disk size (e.g., 50G).
- \$5: The fifth field is the percentage of disk space used (e.g., 50%).

The output would be:

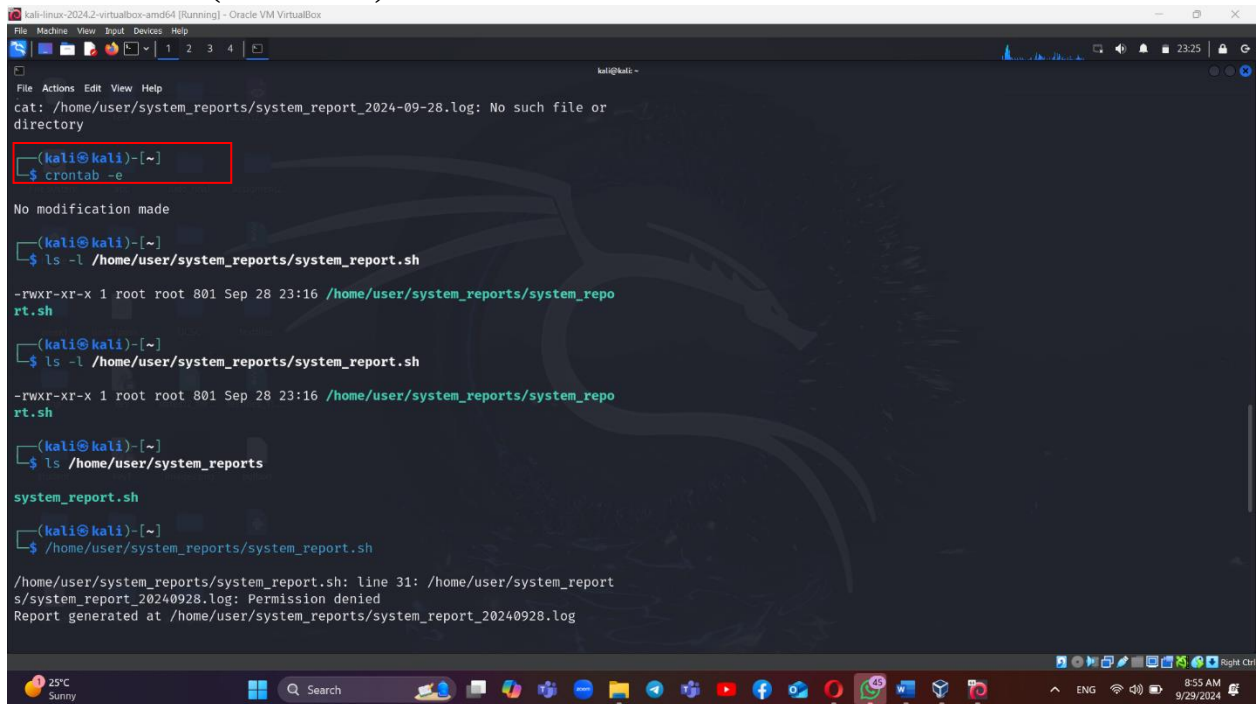
```
/dev/sda1 25G 50G 50%
```

Assigning to DISK_USAGE : The final result of this df | grep | awk pipeline is assigned to theDISK_USAGE variable.

Now lets open the crontab and configure it to give this task to occur .

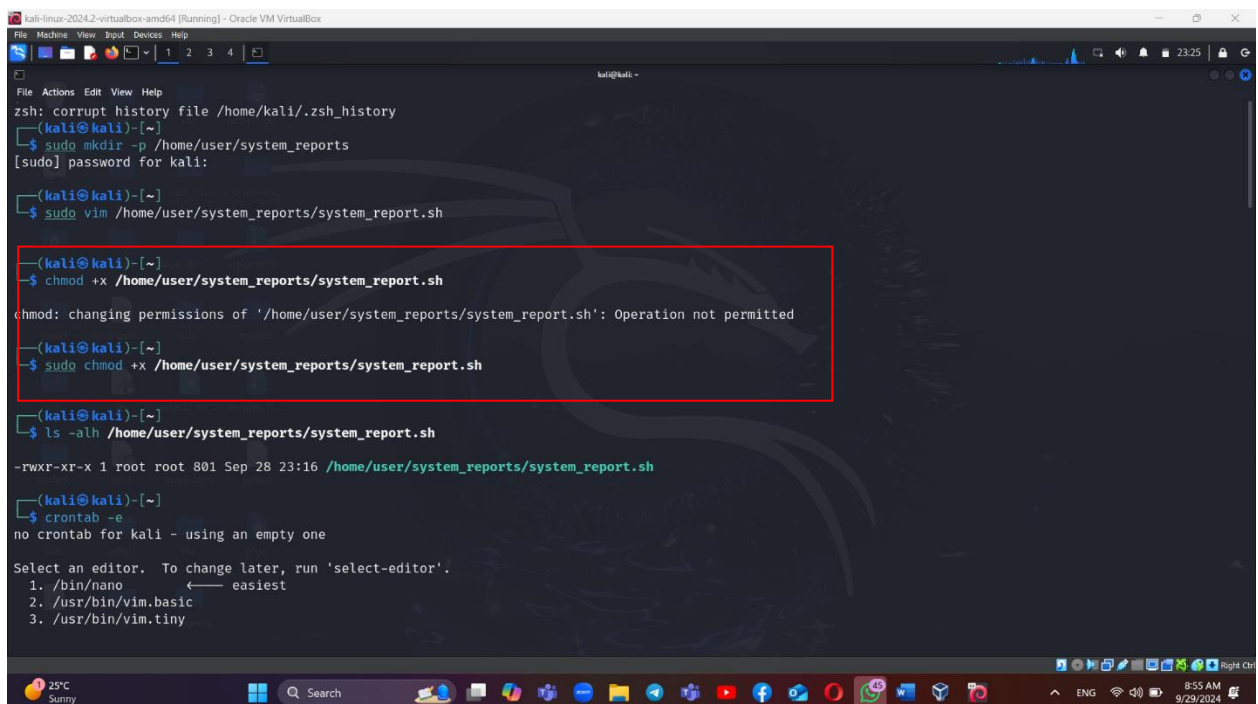
A screenshot of a Kali Linux terminal window. The terminal shows the command 'crontab -l' being executed, which displays the current crontab configuration. The output includes instructions on how to define tasks, time schedules, and commands. At the bottom, the command 'ls /home/user/system_reports' is entered, and the output shows the directory listing of the specified path. The terminal window has a dark background with a Kali Linux logo watermark. The top of the window shows the title bar 'kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox' and the bottom shows the system tray with various icons and the date/time '8:55 AM 2024-04-24'.

As we can see I have initially set it execute everyday 7 am. But to check this at the right moment without waiting for 7am , we can change that line into all *. It will look like this,
* * * * * /home/user/system_reports/system_report.sh . This is will run every minute.To edit I used crontab -e (editor mode).



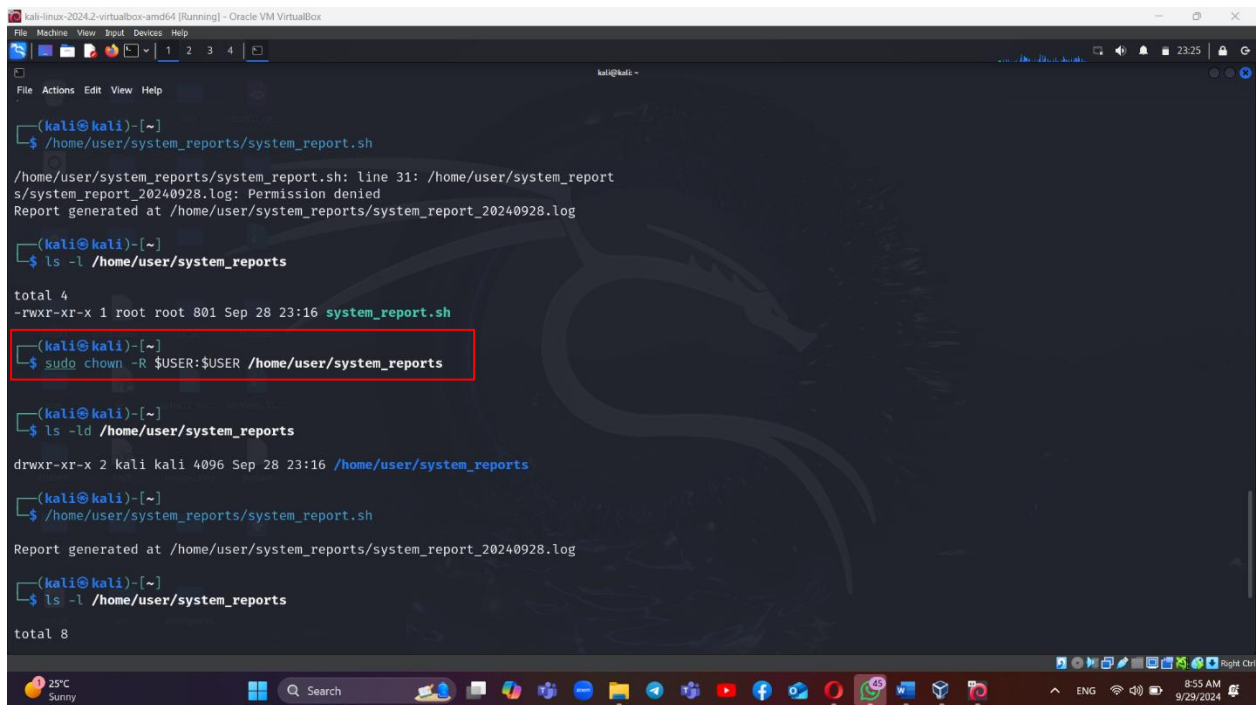
```
kali@kali: ~  
$ crontab -e  
No modification made  
kali@kali: ~  
$ ls -l /home/user/system_reports/system_report.sh  
-rwxr-xr-x 1 root root 801 Sep 28 23:16 /home/user/system_reports/system_report.sh  
kali@kali: ~  
$ ls -l /home/user/system_reports/system_report.sh  
-rwxr-xr-x 1 root root 801 Sep 28 23:16 /home/user/system_reports/system_report.sh  
kali@kali: ~  
$ ls /home/user/system_reports  
system_report.sh  
kali@kali: ~  
$ /home/user/system_reports/system_report.sh  
/home/user/system_reports/system_report.sh: line 31: /home/user/system_reports/system_report_20240928.log: Permission denied  
Report generated at /home/user/system_reports/system_report_20240928.log
```

Now lets change scripts permission to be execute. Using chmod command



```
kali@kali: ~  
$ sudo mkdir -p /home/user/system_reports  
[sudo] password for kali:  
kali@kali: ~  
$ sudo vim /home/user/system_reports/system_report.sh  
kali@kali: ~  
$ chmod +x /home/user/system_reports/system_report.sh  
chmod: changing permissions of '/home/user/system_reports/system_report.sh': Operation not permitted  
kali@kali: ~  
$ sudo chmod +x /home/user/system_reports/system_report.sh  
kali@kali: ~  
$ ls -alh /home/user/system_reports/system_report.sh  
-rwxr-xr-x 1 root root 801 Sep 28 23:16 /home/user/system_reports/system_report.sh  
kali@kali: ~  
$ crontab -e  
no crontab for kali - using an empty one  
Select an editor. To change later, run 'select-editor'.  
1. /bin/nano ← easiest  
2. /usr/bin/vim.basic  
3. /usr/bin/vim.tiny
```

Changing user group of the file (ownership) , for further making sure to execute.



```
kali@kali:~$ /home/user/system_reports/system_report.sh
/home/user/system_reports/system_report.sh: line 31: /home/user/system_report
s/system_report_20240928.log: Permission denied
Report generated at /home/user/system_reports/system_report_20240928.log

kali@kali:~$ ls -l /home/user/system_reports
total 4
-rwxr-xr-x 1 root root 801 Sep 28 23:16 system_report.sh

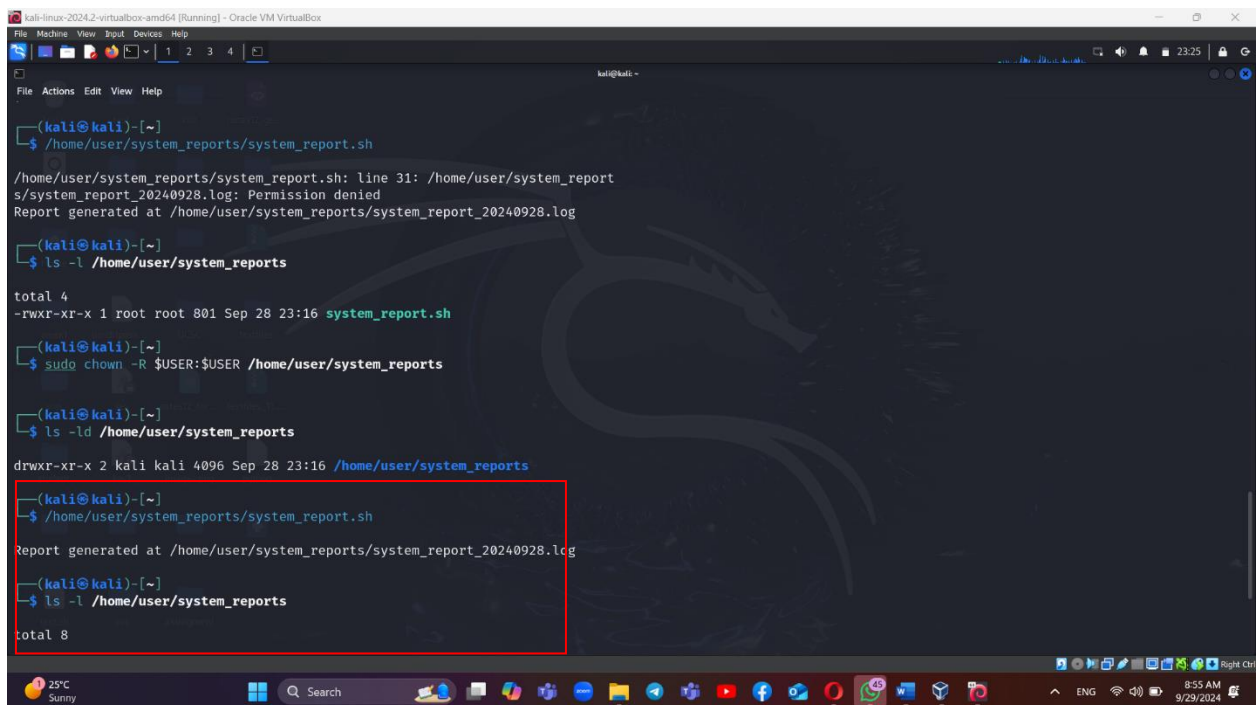
kali@kali:~$ sudo chown -R $USER:$USER /home/user/system_reports

kali@kali:~$ ls -ld /home/user/system_reports
drwxr-xr-x 2 kali kali 4096 Sep 28 23:16 /home/user/system_reports

kali@kali:~$ /home/user/system_reports/system_report.sh
Report generated at /home/user/system_reports/system_report_20240928.log

kali@kali:~$ ls -l /home/user/system_reports
total 8
```

Now lets execute the command.



```
kali@kali:~$ /home/user/system_reports/system_report.sh
/home/user/system_reports/system_report.sh: line 31: /home/user/system_report
s/system_report_20240928.log: Permission denied
Report generated at /home/user/system_reports/system_report_20240928.log

kali@kali:~$ ls -l /home/user/system_reports
total 4
-rwxr-xr-x 1 root root 801 Sep 28 23:16 system_report.sh

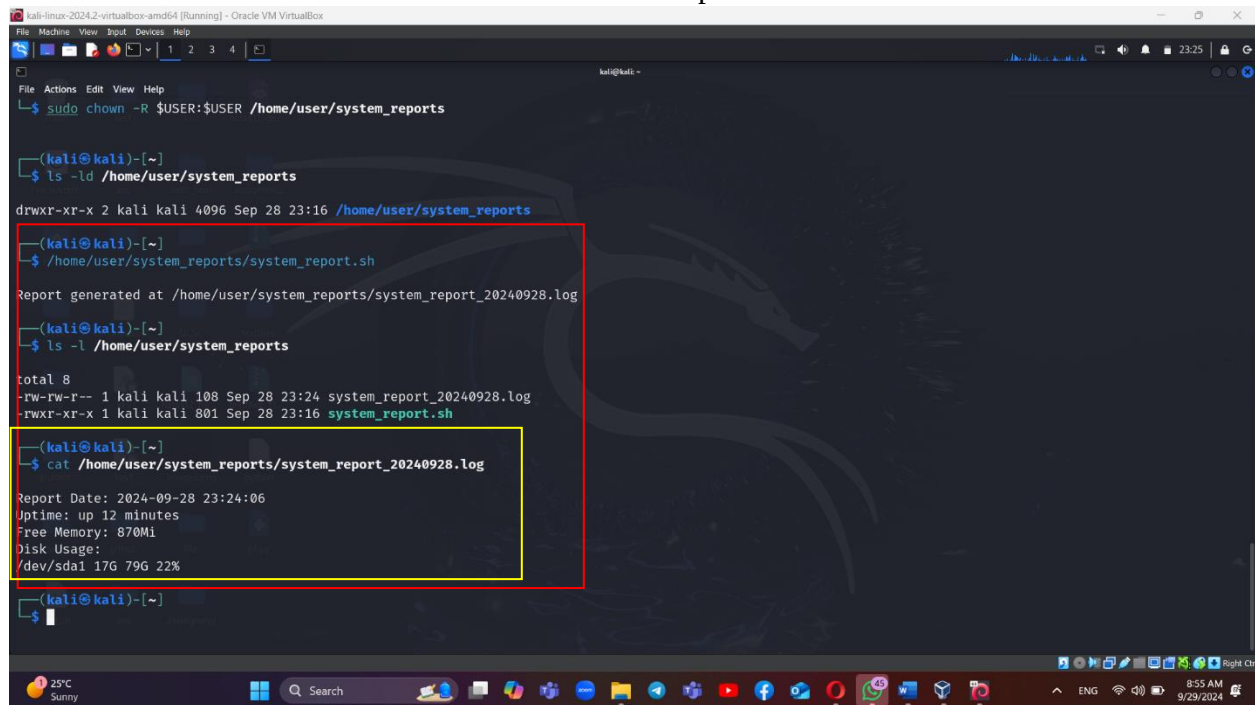
kali@kali:~$ sudo chown -R $USER:$USER /home/user/system_reports

kali@kali:~$ ls -ld /home/user/system_reports
drwxr-xr-x 2 kali kali 4096 Sep 28 23:16 /home/user/system_reports

kali@kali:~$ /home/user/system_reports/system_report.sh
Report generated at /home/user/system_reports/system_report_20240928.log

kali@kali:~$ ls -l /home/user/system_reports
total 8
```

As it shown. It has executed and file has created as expected. Now let's see what is created.



```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help
$ sudo chown -R $USER:$USER /home/user/system_reports

(kali@kali)-[~]
$ ls -ld /home/user/system_reports
drwxr-xr-x 2 kali kali 4096 Sep 28 23:16 /home/user/system_reports

(kali@kali)-[~]
$ /home/user/system_reports/system_report.sh
Report generated at /home/user/system_reports/system_report_20240928.log

(kali@kali)-[~]
$ ls -l /home/user/system_reports
total 8
-rw-rw-r-- 1 kali kali 108 Sep 28 23:24 system_report_20240928.log
-rwxr-xr-x 1 kali kali 801 Sep 28 23:16 system_report.sh

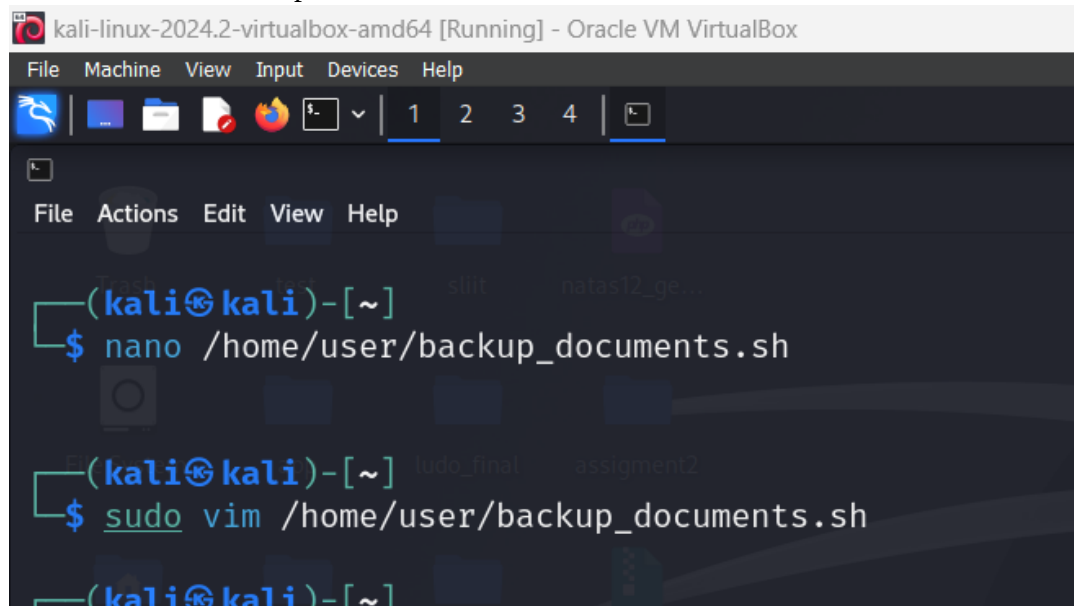
(kali@kali)-[~]
$ cat /home/user/system_reports/system_report_20240928.log
Report Date: 2024-09-28 23:24:06
Uptime: up 12 minutes
Free Memory: 870Mi
Disk Usage:
/dev/sda1 17G 79G 22%

(kali@kali)-[~]
$
```

Task 2

Let's create the script file as instructed. I used

`vim /home/user/backup_documents.sh` command. The editor chosen was vim.



```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help
(kali@kali)-[~]
$ nano /home/user/backup_documents.sh

(kali@kali)-[~]
$ sudo vim /home/user/backup_documents.sh

(kali@kali)-[~]
```

Now lets add following commands in the file

```
#!/bin/bash
```

```
# Define source and destination directories which are going to be used in the code
```

```
source_directory="/home/user/documents"
```

```
destination_directory="/home/user/backup/documents"
```

```
# Create the destination directory if it doesn't exist already
```

```
mkdir -p "$destination_directory"
```

```
# Check if the source directory exists or not . If not printing a error msg in the terminal
```

```
if [ ! -d "$source_directory" ]; then
```

```
    echo "Error: Source directory $source_directory does not exist."
```

```
    exit 1
```

```
fi
```

```
# Get the current date for future use in naming the file
```

```
current_date=$(date +%Y-%m-%d)
```

```
# Define the backup file name
```

```
backupfile="$destination_directory/documents_backup_${current_date}.tar.gz"
```

```
# Creating a compressed backup of source directory
```

```
tar -czf "$backupfile" -C "$source_directory" .
```

```
# Printing a message indicating the backup which we was creating is successful created
```

```
echo "Backup of $source_directory completed successfully!"
```

```
echo "Backup file created: $backupfile"
```

code is pretty much straight forward it self with comments.

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali: ~/Desktop
# /bin/bash

# Define source and destination directories which are going to be used in the code
source_directory="/home/user/documents"
destination_directory="/home/user/backup/documents"

# Create the destination directory if it doesn't exist already
mkdir -p "$destination_directory"

# Check if the source directory exists or not . If not printing a error msg in the terminal
if [ ! -d "$source_directory" ]; then
    echo "Error: Source directory $source_directory does not exist."
    exit 1
fi

# Get the current date for future use in naming the file
current_date=$(date +%Y-%m-%d)

# Define the backup file name
backupfile="$destination_directory/documents_backup_$current_date.tar.gz"

# Creating a compressed backup of source directory
tar -czf "$backupfile" -C "$source_directory" .

# Printing a message indicating the backup which we was creating is successful created
echo "Backup of $source_directory completed successfully!"
echo "Backup file created: $backupfile"
```

Lets make this script executable by changing mode

```
(kali@kali)-[~]
$ sudo chmod +x /home/user/backup_documents.sh

(kali@kali)-[~]
$ sudo mkdir -p /home/user/backup/documents
```

And also creating a directory named as documents to store these, as instructed.

Lets execute it.

```
(kali@kali)-[~]
$ /home/user/backup_documents.sh
```

Ran into some issues. seems like I haven't created the backup directory as I mentioned previously. (mkdir command will be useful)

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali:~/home/user/backup
File Actions Edit View Help
tar: Error is not recoverable: exiting now
/home/user/backup/documents/documents_backup_2024-09-28.tar.gz: Cannot open:
Permission denied
tar (child): Error is not recoverable: exiting now
Backup of /home/user/documents completed successfully!
Backup file created at : /home/user/backup/documents/documents_backup_2024-09-28.tar.gz

(kali@kali)-[~]
$ ls -l /home/user/backup/documents

total 0

(kali@kali)-[~]
$ ls -l /home/user

total 12
drwxr-xr-x 3 root root 4096 Sep 28 23:29 backup
-rwxr-xr-x 1 root root 624 Sep 28 23:28 backup_documents.sh
drwxr-xr-x 2 kali kali 4096 Sep 28 23:24 system_reports

(kali@kali)-[~]
$ cd backup
cd: no such file or directory: backup

(kali@kali)-[~]
$ cd /home/user/backup

(kali@kali)-[/home/user/backup]
$ ls
documents
```

To check now, I added some custom files as file 1, file 2. Etc..

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali:~/home/user/backup
File Actions Edit View Help
Backup file created at : /home/user/backup/documents/documents_backup_2024-09-28.tar.gz

(kali@kali)-[/home/user/backup]
$ sudo vim /home/user/backup_documents.sh

(kali@kali)-[/home/user/backup]
$ /home/user/backup_documents.sh

Error: Source directory /home/user/documents does not exist.

(kali@kali)-[/home/user/backup]
$ ls -l /home/user

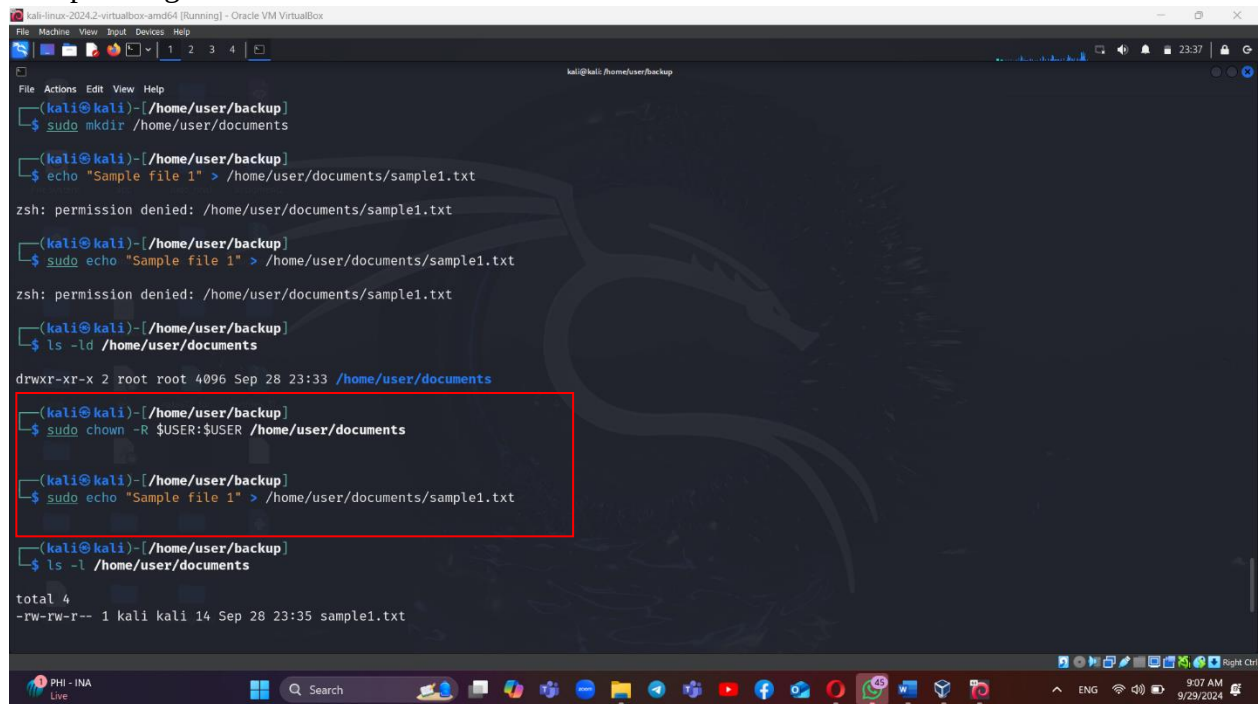
total 12
drwxr-xr-x 3 root root 4096 Sep 28 23:29 backup
-rwxr-xr-x 1 root root 728 Sep 28 23:32 backup_documents.sh
drwxr-xr-x 2 kali kali 4096 Sep 28 23:24 system_reports

(kali@kali)-[/home/user/backup]
$ mkdir /home/user/documents
mkdir: cannot create directory '/home/user/documents': Permission denied

(kali@kali)-[/home/user/backup]
$ sudo mkdir /home/user/documents

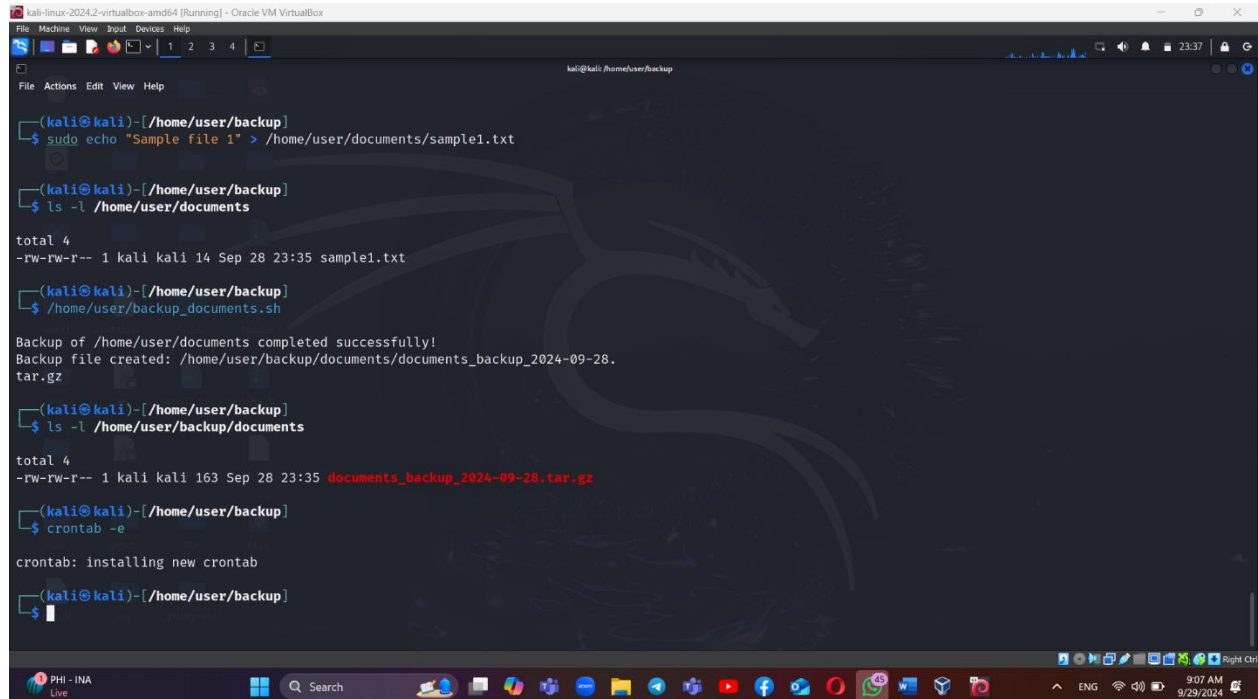
(kali@kali)-[/home/user/backup]
$ echo "Sample file 1" > /home/user/documents/sample1.txt
zsh: permission denied: /home/user/documents/sample1.txt
```

As you can see , I cannot add new files. There I used ownership command , and allwod super user prevelegos



```
kali@kali:~/home/user/backup
$ sudo mkdir /home/user/documents
$ echo "Sample file 1" > /home/user/documents/sample1.txt
zsh: permission denied: /home/user/documents/sample1.txt
$ sudo echo "Sample file 1" > /home/user/documents/sample1.txt
zsh: permission denied: /home/user/documents/sample1.txt
$ ls -ld /home/user/documents
drwxr-xr-x 2 root root 4096 Sep 28 23:33 /home/user/documents
$ sudo chown -R $USER:$USER /home/user/documents
$ sudo echo "Sample file 1" > /home/user/documents/sample1.txt
$ ls -l /home/user/documents
total 4
-rw-rw-r-- 1 kali kali 14 Sep 28 23:35 sample1.txt
```

Now if I used **ls -l** I can see that tar file has created. Which tells me it worked.



```
kali@kali:~/home/user/backup
$ sudo echo "Sample file 1" > /home/user/documents/sample1.txt
$ ls -l /home/user/documents
total 4
-rw-rw-r-- 1 kali kali 14 Sep 28 23:35 sample1.txt
$ /home/user/backup_documents.sh
Backup of /home/user/documents completed successfully!
Backup file created: /home/user/backup/documents/documents_backup_2024-09-28.tar.gz
$ ls -l /home/user/backup/documents
total 4
-rw-rw-r-- 1 kali kali 163 Sep 28 23:35 documents_backup_2024-09-28.tar.gz
$ crontab -e
crontab: installing new crontab
$
```

8.SSH(Secure Shell)

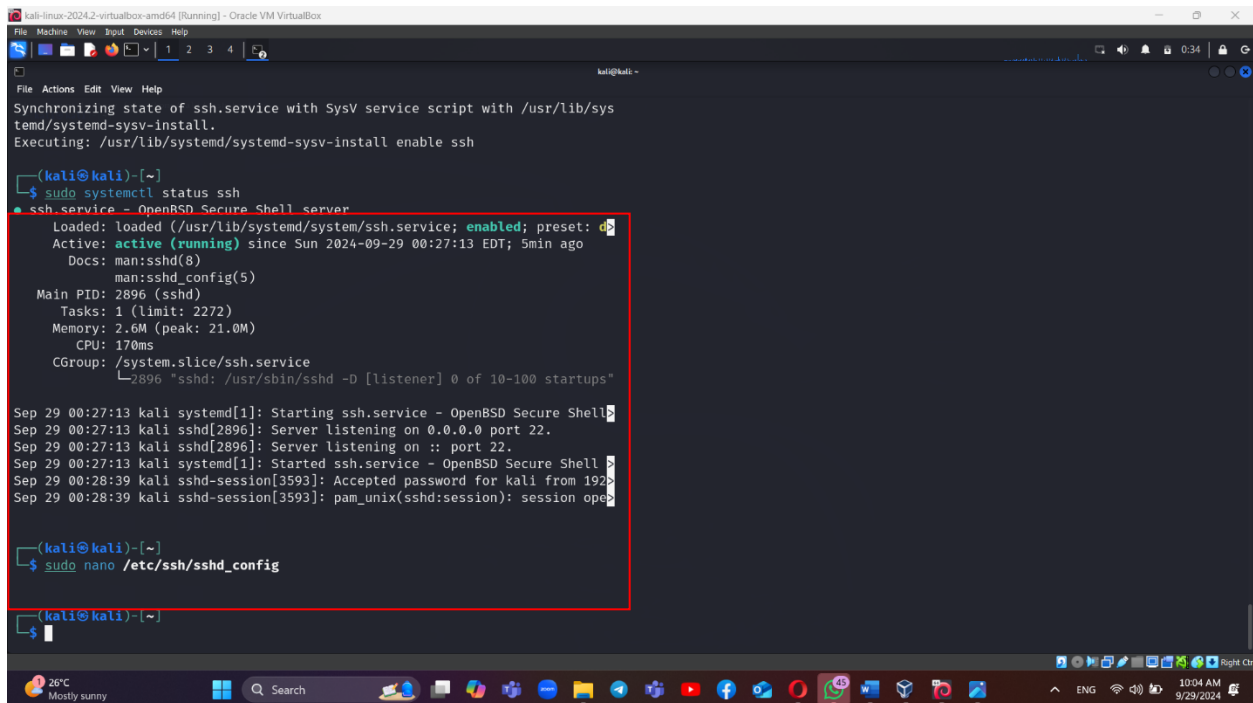
Ssh is used to securely access our linux or any other machines. So to configure it, let's install the open ssl. I used the command **sudo apt install openssh-server**.

Once install let's start the server.

```
(kali㉿kali)-[~]
$ sudo systemctl start ssh

(kali㉿kali)-[~]
$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/sys
temd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
```

And let's check our ssh server status. using command I checked it “**sudo systemctl status ssh**”



```
kali-linux-2024.2-virtualbox-amd64 (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali: ~
File Actions Edit View Help
Synchronizing state of ssh.service with SysV service script with /usr/lib/sys
temd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh

(kali㉿kali)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: d
   Active: active (running) since Sun 2024-09-29 00:27:13 EDT; 5min ago
   Docs: man:sshd(8)
        man:sshd_config(5)
   Main PID: 2896 (sshd)
     Tasks: 1 (limit: 2272)
    Memory: 2.6M (peak: 21.0M)
       CPU: 170ms
    CGroup: /system.slice/ssh.service
           └─2896 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

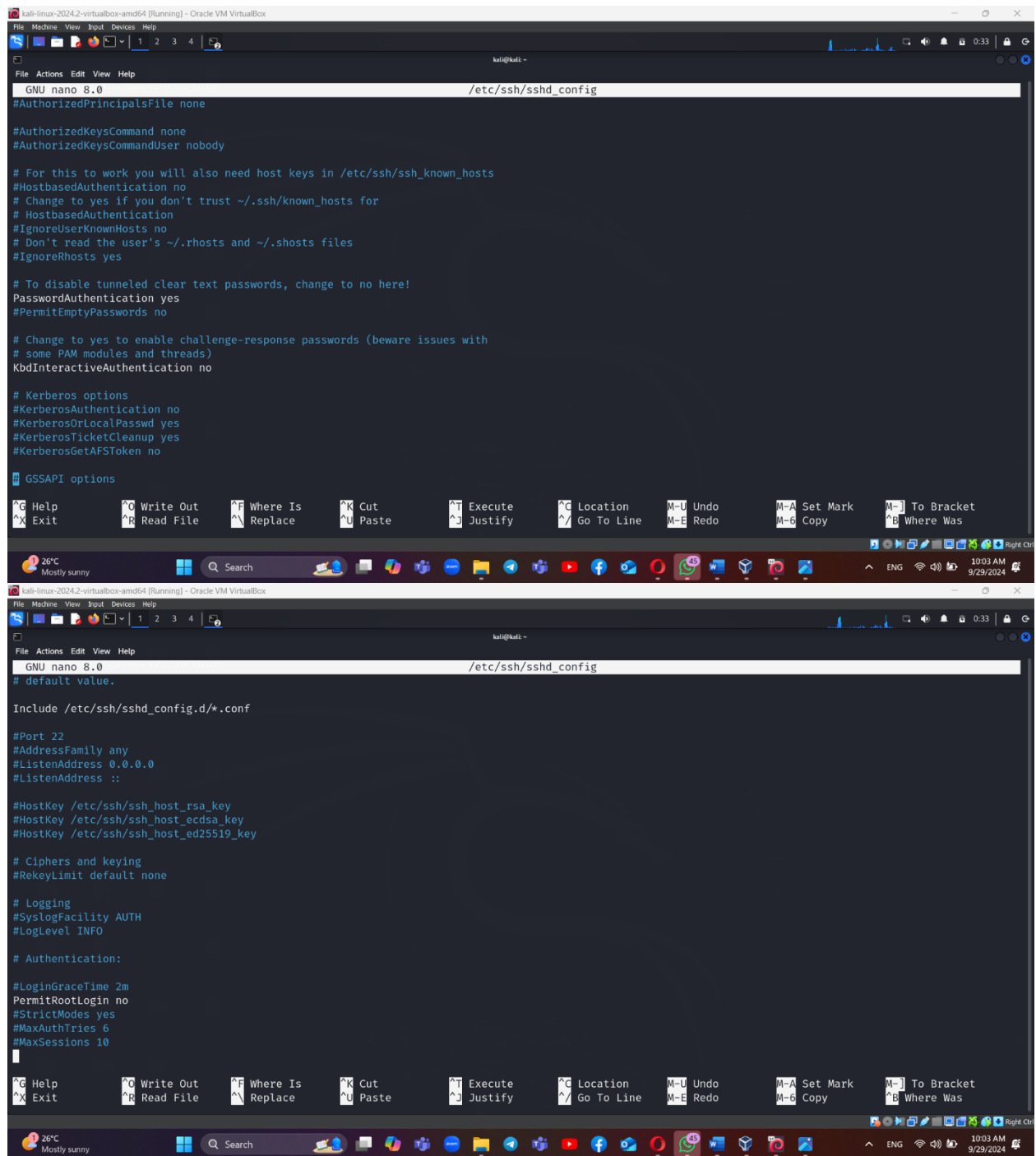
Sep 29 00:27:13 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell
Sep 29 00:27:13 kali sshd[2896]: Server listening on 0.0.0.0 port 22.
Sep 29 00:27:13 kali sshd[2896]: Server listening on :: port 22.
Sep 29 00:27:13 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell
Sep 29 00:28:39 kali sshd-session[3593]: Accepted password for kali from 192
Sep 29 00:28:39 kali sshd-session[3593]: pam_unix(sshd:session): session ope

(kali㉿kali)-[~]
$ sudo nano /etc/ssh/sshd_config

(kali㉿kali)-[~]
$
```

It says actively running, so seems fine up to now. Let's start configuring. To do that let's access to the configuration file. I used “**sudo nano /etc/ssh/sshd_config**”

```
(kali㉿kali)-[~]
$ sudo nano /etc/ssh/sshd_config
```



```
GNU nano 8.0 /etc/ssh/sshd_config
#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPISupport no

# X11
#X11Forwarding no
#X11UseLocalhost yes

# OpenSSH
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

I changed following lines **PermitRootLogin no**

PasswordAuthentication yes, which will ensure that root login is disabled for security and password-based authentication is allowed.

Now lets check our ip address , using the command **ip addr show**,

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali ~
File Actions Edit View Help

Sep 29 00:27:13 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell
Sep 29 00:27:13 kali sshd[2896]: Server listening on 0.0.0.0 port 22.
Sep 29 00:27:13 kali sshd[2896]: Server listening on :: port 22.
Sep 29 00:27:13 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell
Sep 29 00:28:39 kali sshd-session[3593]: Accepted password for kali from 192.
Sep 29 00:28:39 kali sshd-session[3593]: pam_unix(sshd:session): session opene

(kali@kali)-[~]
$ sudo nano /etc/ssh/sshd_config

(kali@kali)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d2:26:79 brd ff:ff:ff:ff:ff:ff
    inet 192.168.8.157/24 brd 192.168.8.255 scope global dynamic eth0
        valid_lft 85945sec preferred_lft 85945sec
    inet6 2402:4000:2300:c8a1:a00:27ff:fed2:2679/64 scope global dynamic mngtmpaddr proto kernel_ra
        valid_lft 295sec preferred_lft 115sec
    inet6 fe80::a00:27ff:fed2:2679/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$
```

To make sure my firewall wont intersect in connecting for ssh , I used following commands.

sudo ufw allow ssh

sudo ufw enable

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali ~
File Actions Edit View Help

Sep 29 00:28:39 kali sshd-session[3593]: Accepted password for kali from 192.
Sep 29 00:28:39 kali sshd-session[3593]: pam_unix(sshd:session): session opene

(kali@kali)-[~]
$ sudo nano /etc/ssh/sshd_config

(kali@kali)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d2:26:79 brd ff:ff:ff:ff:ff:ff
    inet 192.168.8.157/24 brd 192.168.8.255 scope global dynamic eth0
        valid_lft 85945sec preferred_lft 85945sec
    inet6 2402:4000:2300:c8a1:a00:27ff:fed2:2679/64 scope global dynamic mngtmpaddr proto kernel_ra
        valid_lft 295sec preferred_lft 115sec
    inet6 fe80::a00:27ff:fed2:2679/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)

(kali@kali)-[~]
$
```

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help
kali@kali: ~
$ sudo nano /etc/ssh/sshd_config

(kali@kali)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d2:26:79 brd ff:ff:ff:ff:ff:ff
    inet 192.168.8.157/24 brd 192.168.8.255 scope global dynamic eth0
        valid_lft 85945sec preferred_lft 85945sec
    inet6 2402:4000:2300:c8a1:a00:27ff:fed2:2679/64 scope global dynamic mngmtaddr proto kernel_r
        valid_lft 295sec preferred_lft 115sec
    inet6 fe80::a00:27ff:fed2:2679/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)

(kali@kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

(kali@kali)-[~]
$
```

Now lets check its working. Before using ssh I used ping command to check the communication between 2 VM im going to be using (Ubuntu (client) , kali vm (ssh server)).

```
kavindu.ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help
Sep 29 09:59
kali@kali: ~
^C--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/ndev = 73.328/81.144/88.961/7.816 ms
kavindu@kavindu-VirtualBox: $ ping 192.168.8.157
PING 192.168.8.157 (192.168.8.157) 56(84) bytes of data.
64 bytes from 192.168.8.157: icmp_seq=1 ttl=64 time=2.46 ms
64 bytes from 192.168.8.157: icmp_seq=2 ttl=64 time=1.19 ms
64 bytes from 192.168.8.157: icmp_seq=3 ttl=64 time=0.453 ms
^C
--- 192.168.8.157 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/ndev = 0.453/1.366/2.460/0.829 ms
kavindu@kavindu-VirtualBox: $ ssh kali@192.168.8.157
The authenticity of host '192.168.8.157 (192.168.8.157)' can't be established.
ED25519 key fingerprint is SHA256:19B/FNAsODHxrl8n0rDnTsgGouFDttatVoARNIrfl4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.8.157' (ED25519) to the list of known hosts.
kali@192.168.8.157's password:
Linux kali 6.6.15-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.6.15-2kali1 (2024-05-17) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 28 23:54:22 2024 from ::1
zsh: corrupt history file /home/kali/.zsh_history

(kali@kali)-[~]
$
```

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help
Active: active (running) since Sun 2024-09-29 00:27:13 EDT; 5s ago
Docs: man:sshd(8)
      man:sshd_config(5)
Process: 2894 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
Main PID: 2896 (sshd)
Tasks: 1 (limit: 2272)
Memory: 1.1M (peak: 1.4M)
CPU: 27ms
CGroup: /system.slice/ssh.service
└─2896 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

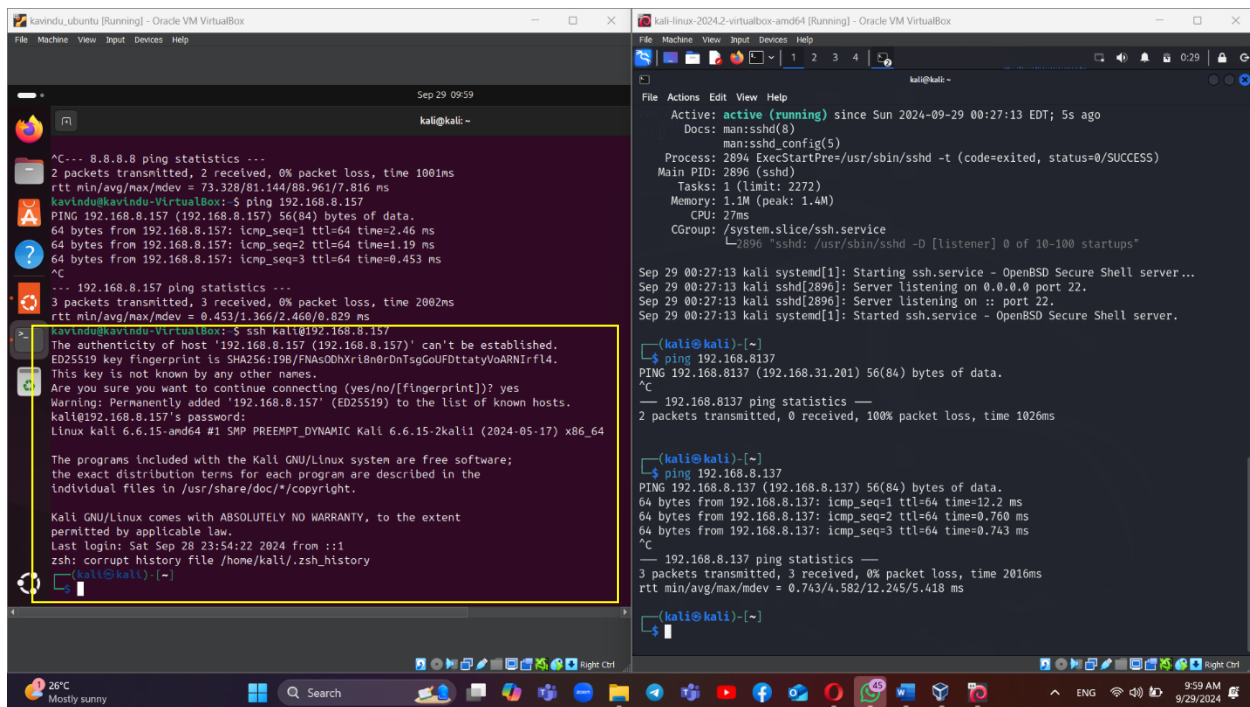
Sep 29 00:27:13 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 29 00:27:13 kali sshd[2896]: Server listening on 0.0.0.0 port 22.
Sep 29 00:27:13 kali sshd[2896]: Server listening on :: port 22.
Sep 29 00:27:13 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(kali@kali)-[~]
$ ping 192.168.8.137
PING 192.168.8.137 (192.168.8.137) 56(84) bytes of data.
^C
--- 192.168.8.137 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1026ms

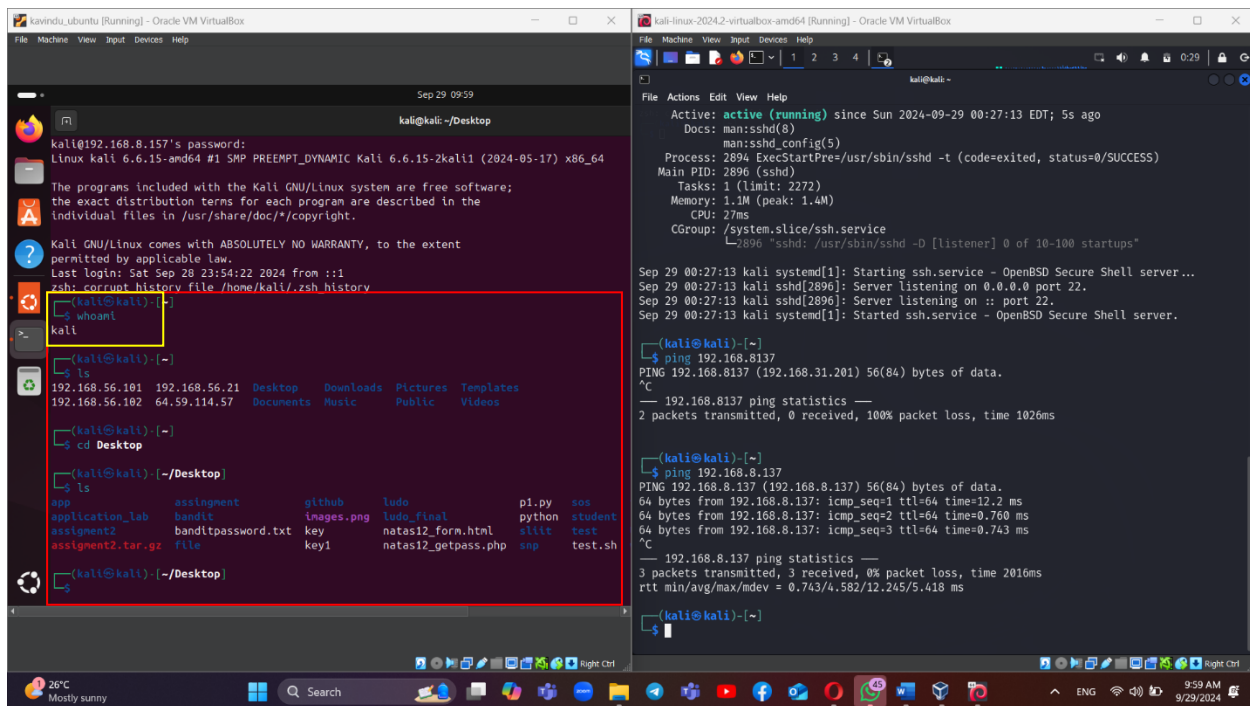
(kali@kali)-[~]
$ ping 192.168.8.137
PING 192.168.8.137 (192.168.8.137) 56(84) bytes of data.
64 bytes from 192.168.8.137: icmp_seq=1 ttl=64 time=12.2 ms
64 bytes from 192.168.8.137: icmp_seq=2 ttl=64 time=0.760 ms
64 bytes from 192.168.8.137: icmp_seq=3 ttl=64 time=0.743 ms
^C
--- 192.168.8.137 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2016ms
rtt min/avg/max/ndev = 0.743/4.582/12.245/5.418 ms

(kali@kali)-[~]
$
```

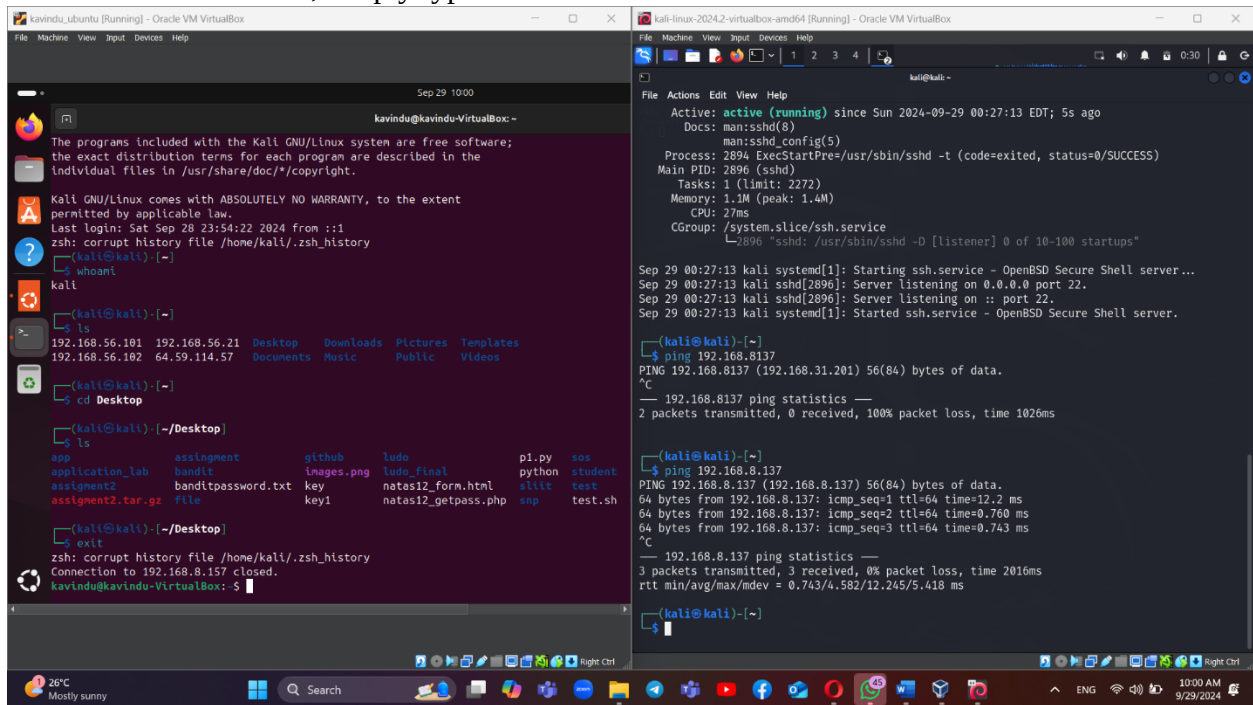
Since it seems to be fine lets use ssh command as ssh kali@192.168.8.157



In the end u can see I have kali environment. Lets further see it.



So to exit from this ssh , simply type exit .



```
kavindu@kavindu-VirtualBox: ~
$ ls
The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 28 23:54:22 2024 from ::1
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali) ~
$ whoami
kali
(kali@kali) ~
$ ls
192.168.56.101 192.168.56.21 Desktop Downloads Pictures Templates
192.168.56.102 64.59.114.57 Documents Music Public Videos
(kali@kali) ~
$ cd Desktop
(kali@kali) ~/Desktop
$ ls
app      assignment  github    ludo      p1.py    sos
application_lab  bandit    images.png ludo_final python  student
assignment2  banditpassword.txt key       nates12_form.html sliit  test
assignment2.tar.gz file      key1      nates12_getpass.php snp     test.sh
(kali@kali) ~/Desktop
$ exit
zsh: corrupt history file /home/kali/.zsh_history
Connection to 192.168.8.137 closed.
kavindu@kavindu-VirtualBox: ~
$
```

```
kali@kali: ~
File Actions Edit View Help
Active: active (running) since Sun 2024-09-29 00:27:13 EDT; 5s ago
Docs: man:sshd(8)
man:sshd_config(5)
Process: 2894 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
Main PID: 2896 (sshd)
Tasks: 1 (limit: 2272)
Memory: 1.1M (peak: 1.4M)
CPU: 27ms
CGroup: /system.slice/ssh.service
└─2896 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 29 00:27:13 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 29 00:27:13 kali sshd[2896]: Server listening on 0.0.0.0 port 22.
Sep 29 00:27:13 kali sshd[2896]: Server listening on :: port 22.
Sep 29 00:27:13 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(kali@kali) ~
$ ping 192.168.8137
PING 192.168.8137 (192.168.31.201) 56(84) bytes of data.
^C
--- 192.168.8137 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1026ms

(kali@kali) ~
$ ping 192.168.8.137
PING 192.168.8.137 (192.168.8.137) 56(84) bytes of data.
64 bytes from 192.168.8.137: icmp_seq=1 ttl=64 time=12.2 ms
64 bytes from 192.168.8.137: icmp_seq=2 ttl=64 time=0.760 ms
64 bytes from 192.168.8.137: icmp_seq=3 ttl=64 time=0.743 ms
^C
--- 192.168.8.137 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2016ms
rtt min/avg/max/mdev = 0.743/4.582/12.245/5.418 ms

(kali@kali) ~
$
```

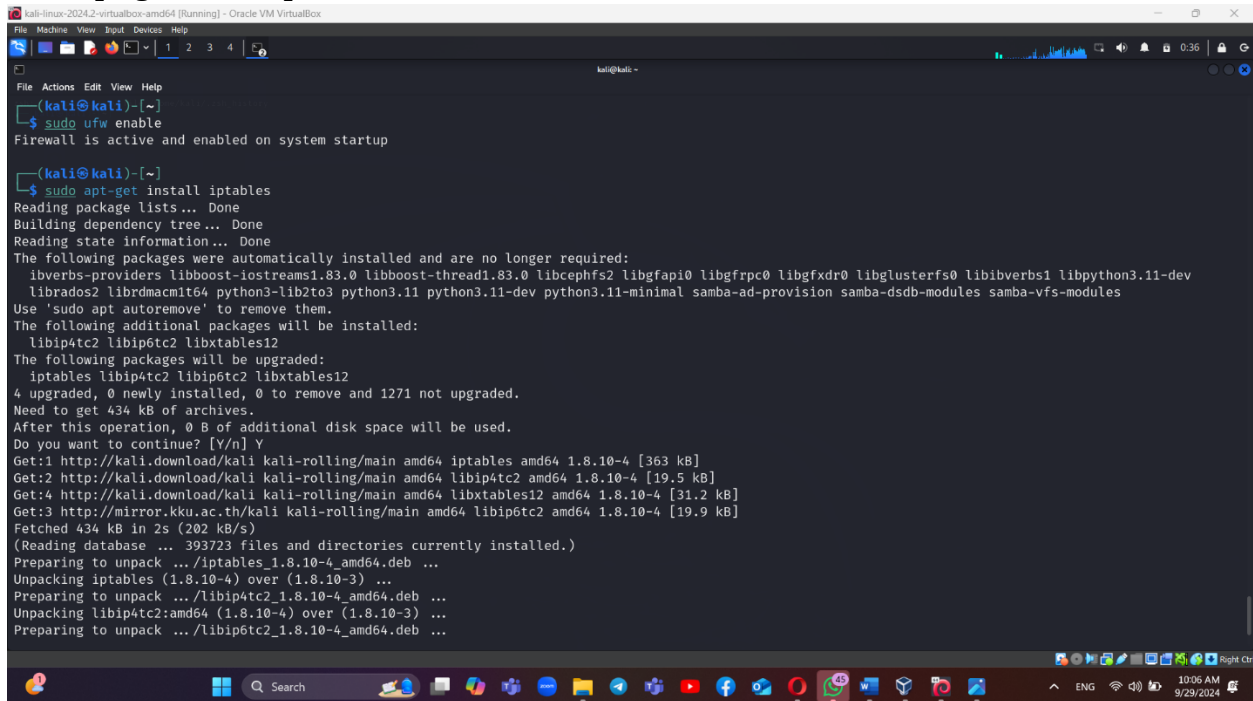
9.Iptables and ACLs

To manage network traffic and enhance security on our Linux machine, we can use iptables to create rules based on our requirements. Below are the basic iptables commands for each of the specified security measures.

So lets first upgrade our repository and install iptables .

sudo apt-get update

sudo apt-get install iptables



```
kali@kali:~$ sudo ufw enable
Firewall is active and enabled on system startup

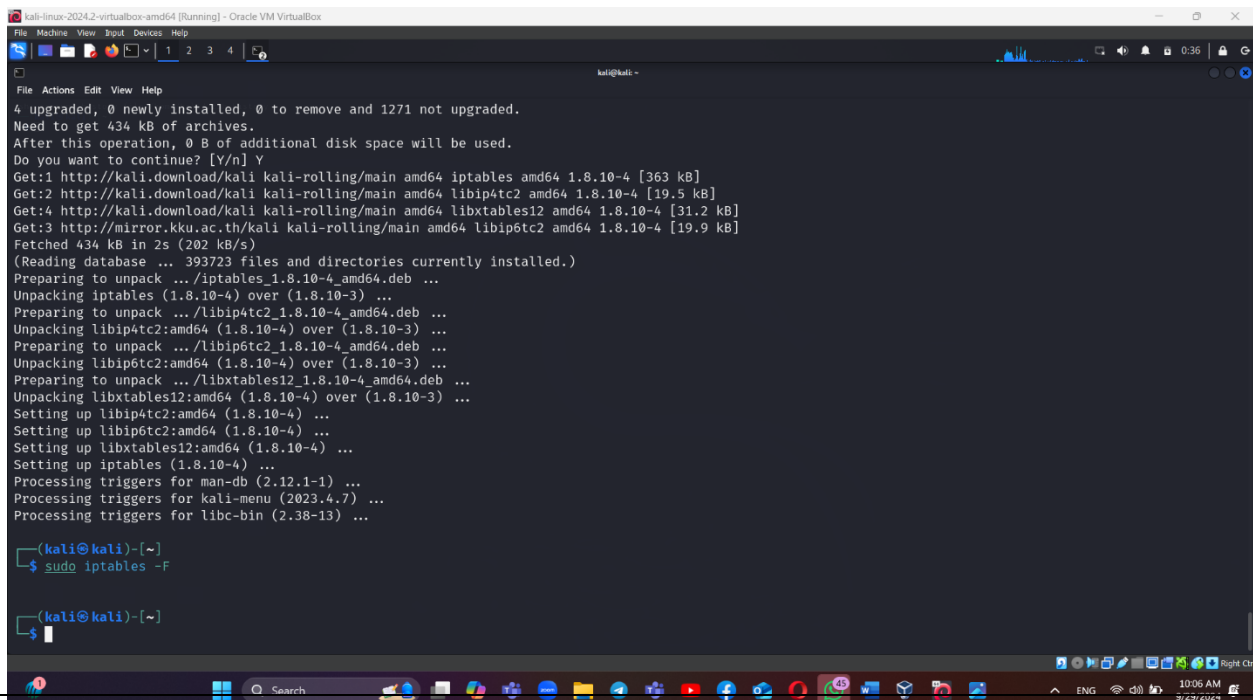
(kali@kali)~$ sudo apt-get install iptables
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

The following packages were automatically installed and are no longer required:
  ibverbs-providers libboost-iostreams1.83.0 libboost-thread1.83.0 libcephfs2 libgfrpc0 libgfrpc0 libglusterfs0 libibverbs1 libpython3.11-dev
  librados2 librdmacm164 python3-lib2to3 python3.11 python3.11-dev python3.11-minimal samba-ad-provision samba-dsdb-modules samba-vfs-modules
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libip4tc2 libip6tc2 libxtables12
The following packages will be upgraded:
  iptables libip4tc2 libip6tc2 libxtables12
4 upgraded, 0 newly installed, 0 to remove and 1271 not upgraded.
Need to get 434 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://kali.download/kali kali-rolling/main amd64 iptables amd64 1.8.10-4 [363 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libip4tc2 amd64 1.8.10-4 [19.5 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libxtables12 amd64 1.8.10-4 [31.2 kB]
Get:3 http://mirror.kku.ac.th/kali kali-rolling/main amd64 libip6tc2 amd64 1.8.10-4 [19.9 kB]
Fetched 434 kB in 2s (202 kB/s)
(Reading database ... 393723 files and directories currently installed.)
Preparing to unpack .../iptables_1.8.10-4_amd64.deb ...
Unpacking iptables (1.8.10-4) over (1.8.10-3) ...
Preparing to unpack .../libip4tc2_1.8.10-4_amd64.deb ...
Unpacking libip4tc2:amd64 (1.8.10-4) over (1.8.10-3) ...
Preparing to unpack .../libip6tc2_1.8.10-4_amd64.deb ...
Unpacking libip6tc2:amd64 (1.8.10-4) over (1.8.10-3) ...
Preparing to unpack .../libxtables12_1.8.10-4_amd64.deb ...
Unpacking libxtables12:amd64 (1.8.10-4) over (1.8.10-3) ...
Setting up libip4tc2:amd64 (1.8.10-4) ...
Setting up libip6tc2:amd64 (1.8.10-4) ...
Setting up libxtables12:amd64 (1.8.10-4) ...
Setting up iptables (1.8.10-4) ...
Processing triggers for man-db (2.12.1-1) ...
Processing triggers for kali-menu (2023.4.7) ...
Processing triggers for libc-bin (2.38-13) ...

(kali@kali)~$ sudo iptables -F
```

once I installed , before setting up new rules, its better to flush existing ones to avoid conflicts .

Using sudo iptables -F



```
kali@kali:~$ sudo iptables -F
4 upgraded, 0 newly installed, 0 to remove and 1271 not upgraded.
Need to get 434 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://kali.download/kali kali-rolling/main amd64 iptables amd64 1.8.10-4 [363 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libip4tc2 amd64 1.8.10-4 [19.5 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libxtables12 amd64 1.8.10-4 [31.2 kB]
Get:3 http://mirror.kku.ac.th/kali kali-rolling/main amd64 libip6tc2 amd64 1.8.10-4 [19.9 kB]
Fetched 434 kB in 2s (202 kB/s)
(Reading database ... 393723 files and directories currently installed.)
Preparing to unpack .../iptables_1.8.10-4_amd64.deb ...
Unpacking iptables (1.8.10-4) over (1.8.10-3) ...
Preparing to unpack .../libip4tc2_1.8.10-4_amd64.deb ...
Unpacking libip4tc2:amd64 (1.8.10-4) over (1.8.10-3) ...
Preparing to unpack .../libip6tc2_1.8.10-4_amd64.deb ...
Unpacking libip6tc2:amd64 (1.8.10-4) over (1.8.10-3) ...
Preparing to unpack .../libxtables12_1.8.10-4_amd64.deb ...
Unpacking libxtables12:amd64 (1.8.10-4) over (1.8.10-3) ...
Setting up libip4tc2:amd64 (1.8.10-4) ...
Setting up libip6tc2:amd64 (1.8.10-4) ...
Setting up libxtables12:amd64 (1.8.10-4) ...
Setting up iptables (1.8.10-4) ...
Processing triggers for man-db (2.12.1-1) ...
Processing triggers for kali-menu (2023.4.7) ...
Processing triggers for libc-bin (2.38-13) ...

(kali@kali)~$ sudo iptables -F
```

i. Web Server Security

Allow incoming traffic only on ports 80 (HTTP) and 443 (HTTPS) and block all other incoming traffic by default. I used the commands

Allow HTTP

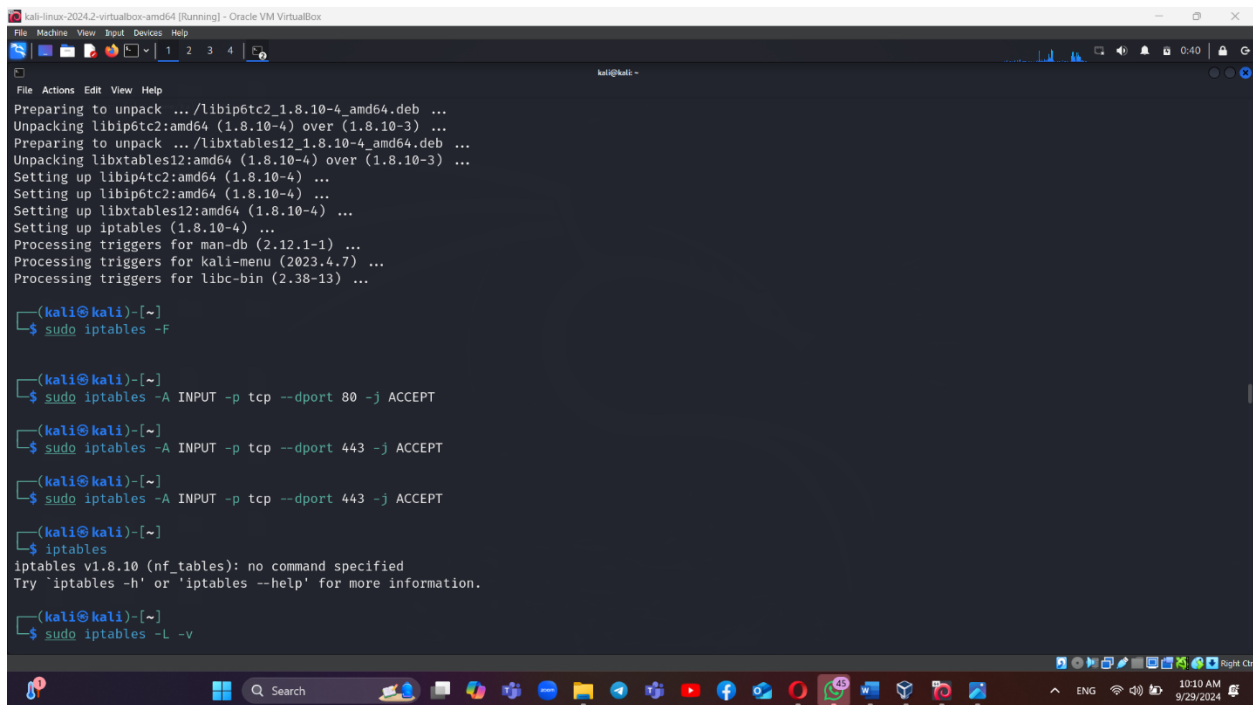
```
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

Allow HTTPS

```
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```

Block all other incoming traffic

```
sudo iptables -A INPUT -j DROP
```



```
kali@kali:~$ sudo apt-get install iptables
Preparing to unpack .../libip6tc2_1.8.10-4_amd64.deb ...
Unpacking libip6tc2:amd64 (1.8.10-4) over (1.8.10-3) ...
Preparing to unpack .../libxtables12_1.8.10-4_amd64.deb ...
Unpacking libxtables12:amd64 (1.8.10-4) over (1.8.10-3) ...
Setting up libip4tc2:amd64 (1.8.10-4) ...
Setting up libip6tc2:amd64 (1.8.10-4) ...
Setting up libxtables12:amd64 (1.8.10-4) ...
Setting up iptables (1.8.10-4) ...
Processing triggers for man-db (2.12.1-1) ...
Processing triggers for kali-menu (2023.4.7) ...
Processing triggers for libc-bin (2.38-13) ...

(kali@kali)-[~]
$ sudo iptables -F

(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

(kali@kali)-[~]
$ iptables
iptables v1.8.10 (nf_tables): no command specified
Try `iptables -h' or 'iptables --help' for more information.

(kali@kali)-[~]
$ sudo iptables -L -v
```

ii. Remote Administration Access

Allow SSH access (port 22) only from specific IP addresses of your trusted machines.

Allow SSH from a trusted IP address

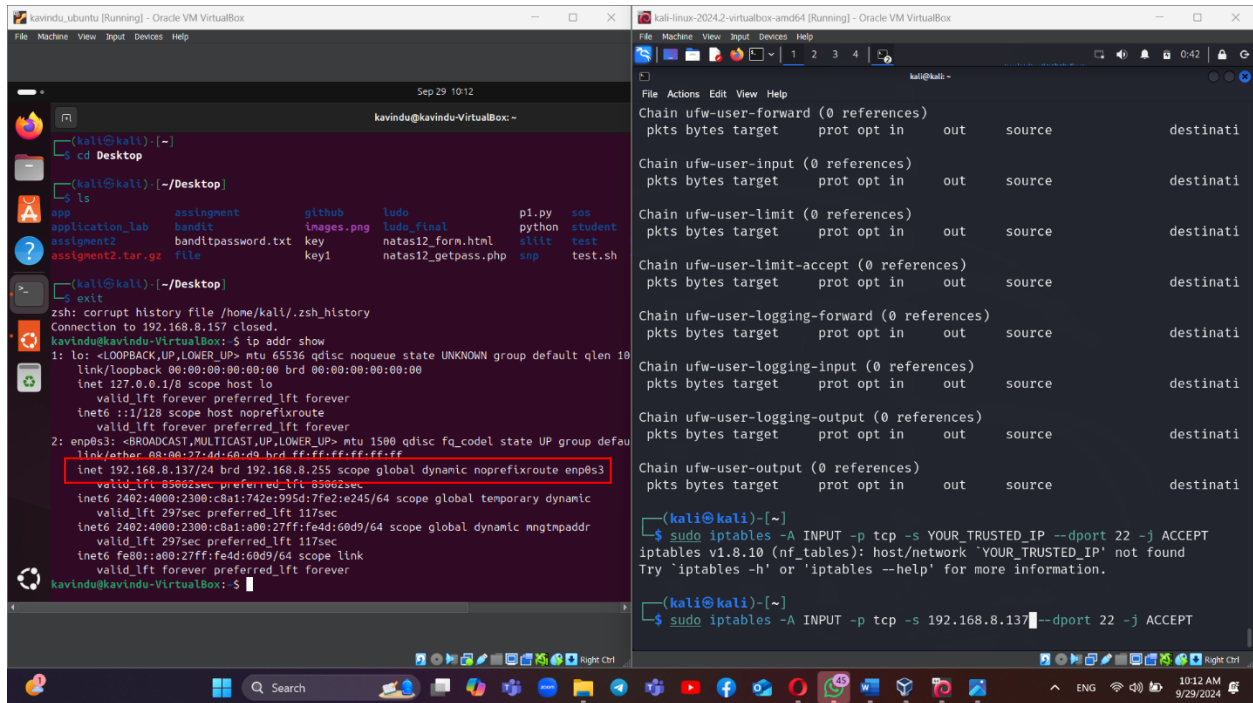
```
sudo iptables -A INPUT -p tcp -s 192.168.8.137 --dport 22 -j ACCEPT
```

Block all other SSH access

```
sudo iptables -A INPUT -p tcp --dport 22 -j DROP.
```

```
(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp -s 192.168.8.137 --dport 22 -j ACCEPT

(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 22 -j DROP
```



Ip address I chose is from other VM I was using.

iii. Allow Specific Applications

Allow traffic for specific applications using their port numbers (e.g., video conferencing app on port 443). For this I used,

Allow application traffic on port 443 (if needed, as it's already allowed for HTTPS)

sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

```
(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```

iv. Allow Pings (ICMP Echo Request)

Allow ping requests to your machine.

Allow incoming ICMP (ping)

sudo iptables -A INPUT -p icmp --icmp-type echo-request -j ACCEPT

```
(kavindu@kali)-[~]
$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j ACCEPT
```

```
(kavindu@kali)-[~]
$ iptables -L -v
iptables v1.8.10 (nf_tables): Could not fetch rule set generation id: Permission denied (you must be root)

(kavindu@kali)-[~]
$ sudo iptables -L -v

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination
 0      0 ACCEPT      icmp -- any    any    anywhere      anywhere      icmp
echo-request

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination

(kavindu@kali)-[~]
$
```

v. Printer Server Access

Allow printing traffic (port 9100) only from specific IP addresses within our local network.

As the machine I gave the ipaddress as 192.168.8.137

Allow printing traffic from a trusted IP address

sudo iptables -A INPUT -p tcp -s 192.168.8.137 --dport 9100 -j ACCEPT

Block all other access to the printer server

sudo iptables -A INPUT -p tcp --dport 9100 -j DROP

```
(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp -s 192.168.8.137 --dport 9100 -j ACCEPT

(kali@kali)-[~]
$ sudo iptables -A INPUT -p tcp --dport 9100 -j DROP
```

Now lets save these rules to ensure they persists after a reboot.

Save iptables rules

sudo iptables-save | sudo tee /etc/iptables/rules.v4

```
sudo iptables-save | sudo tee /etc/iptables/rules.v4
```

Now lets view these changes.

sudo iptables -L -v

```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
File Actions Edit View Help
-A INPUT -s 192.168.8.137/32 -p tcp -m tcp --dport 9100 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 9100 -j DROP
COMMIT
# Completed on Sun Sep 29 00:44:31 2024

(kali@kali)-[~]
$ sudo iptables -L -v -n

Chain INPUT (policy DROP 521 packets, 145K bytes)
pkts bytes target      prot opt in      out     source            destination
0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
    tcp dpt:80
0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
    tcp dpt:443
0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
    tcp dpt:443
0      0 ACCEPT     tcp  --  *      *       192.168.8.137     0.0.0.0/0
    tcp dpt:22
0      0 DROP       tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
    tcp dpt:22
0      0 ACCEPT     tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
    tcp dpt:443
0      0 ACCEPT     tcp  --  *      *       192.168.8.137     0.0.0.0/0
    tcp dpt:9100
0      0 DROP       tcp  --  *      *       0.0.0.0/0         0.0.0.0/0
    tcp dpt:9100

Chain FORWARD (policy DROP 0 packets, 0 bytes)
pkts bytes target      prot opt in      out     source            destination
```

10.Best Practices.

1. Change Default Network Settings

- Changing the host name : Default network settings may be predictable and can be exploited by attackers. Changing these settings can help improve security.

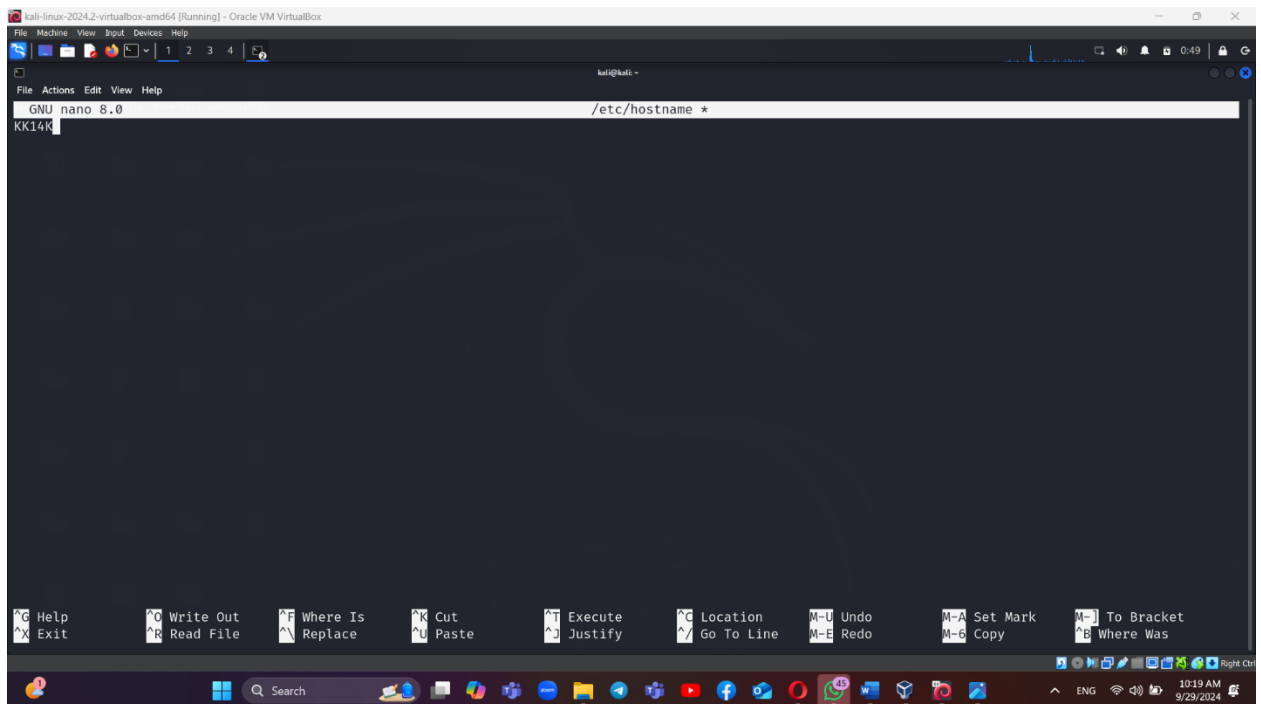
Implementation:

- Change the default hostname and ensure that it's not easily guessable.

To do that we have to edit **/etc/hostname** to set a new hostname.

So lets open the file using command **sudo vim /etc/hostname**

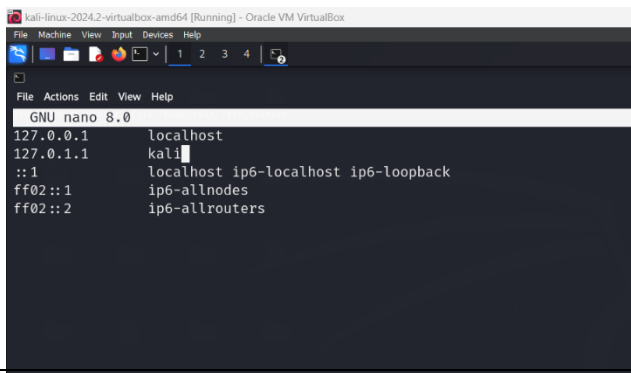
And changed it content to host name I want . I chose kk14k.



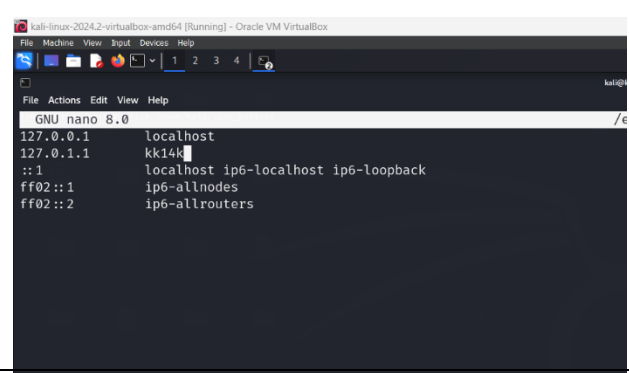
```
kali@kali:~$ sudo nano /etc/hostname
GNU nano 8.0 /etc/hostname *
KK14k
```

And now lets change the our **/etc/hosts** file to reflect the new hostname . I used the command **sudo nano /etc/hosts**.

I changed the line 127.0.1.1 kali to 127.0.1.1 kk14k



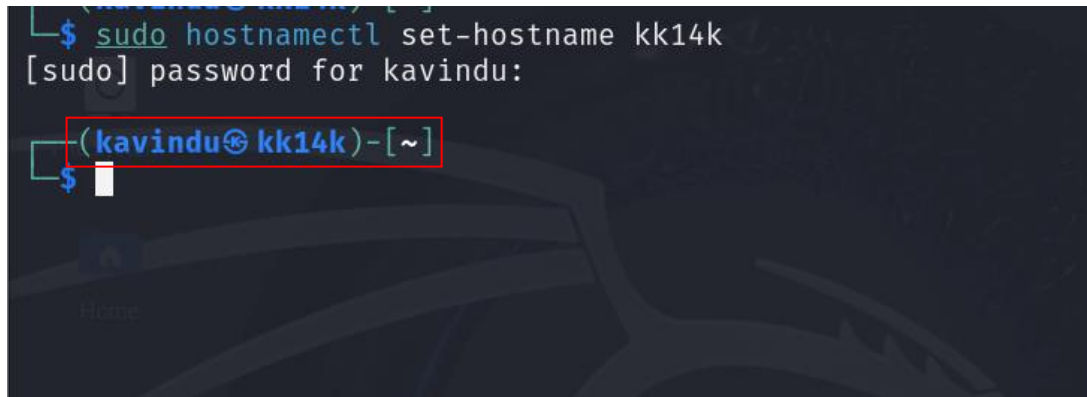
```
kali@kali:~$ sudo nano /etc/hosts
GNU nano 8.0 /etc/hosts
127.0.0.1    localhost
127.0.1.1    kali
::1         localhost ip6-localhost ip6-loopback
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters
```



```
kali@kali:~$ sudo nano /etc/hosts
GNU nano 8.0 /etc/hosts
127.0.0.1    localhost
127.0.1.1    kk14k
::1         localhost ip6-localhost ip6-loopback
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters
```

Now apply it , using the command without re booting ,

sudo hostnamectl set-hostname my-new-hostname

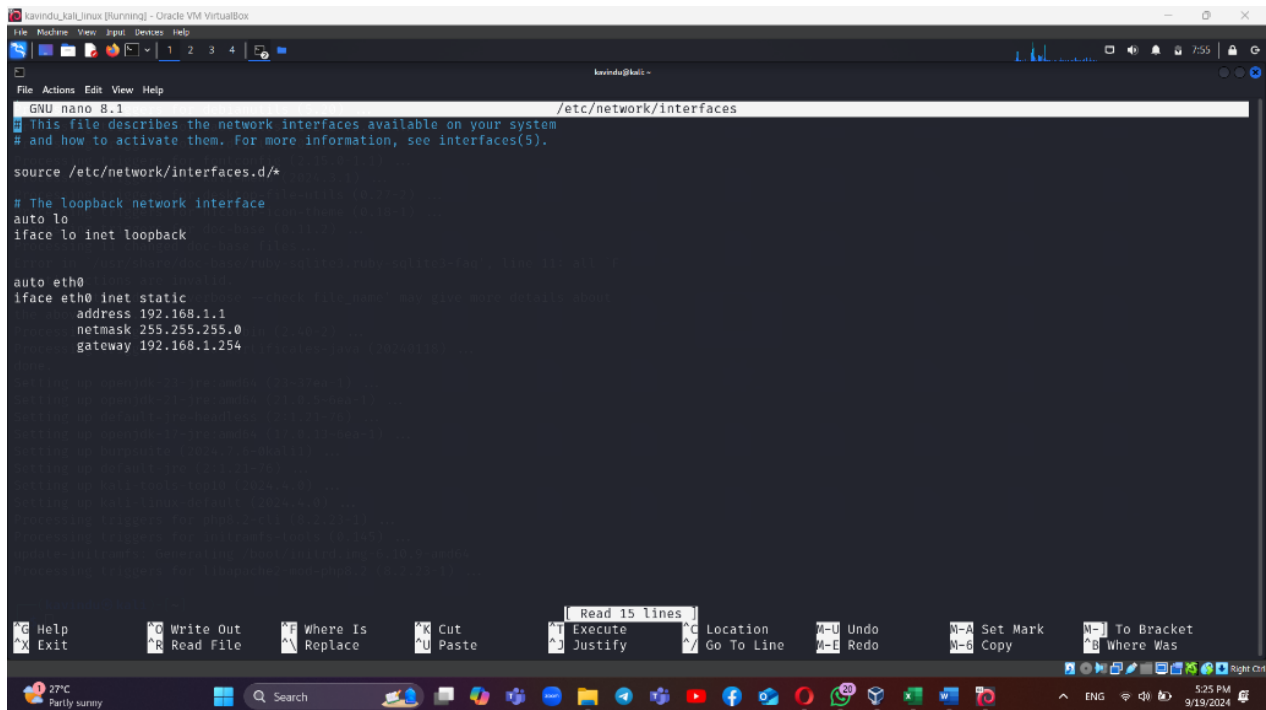


```
$ sudo hostnamectl set-hostname kk14k
[sudo] password for kavindu:
(kavindu@kk14k)-[~]
$
```

It has successfully changed.

- Update the network configuration files to assign a static IP if applicable, rather than using DHCP.

This has been implemented by me when configuring dhcp server earlier.



```
kavindu@kali: ~$ nano /etc/network/interfaces
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The ethernet interface
auto eth0
iface eth0 inet static
    address 192.168.1.1
    netmask 255.255.255.0
    gateway 192.168.1.254
```

2. Disable Unused Network Interfaces

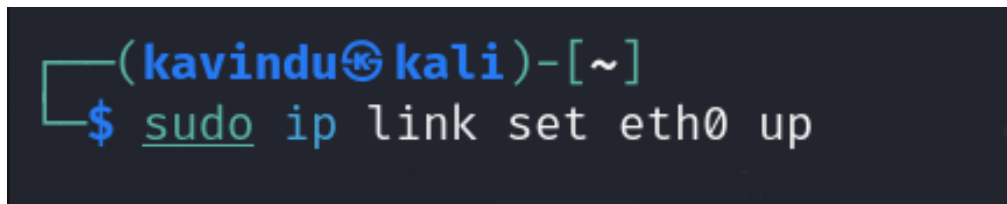
Disabling unused network interfaces minimizes the potential attack surface of your system.

Disable any interfaces that are not in use, such as Wi-Fi if we are connected via Ethernet. Use the command `ip link set <interface> down` to disable an interface (replace <interface> with the actual interface name, e.g., `eth0`).

Ensure these changes persist across reboots by modifying your network configuration files. I have used these commands previously, but in different manner to up my interface when using bridge adapter network.

The command I used was

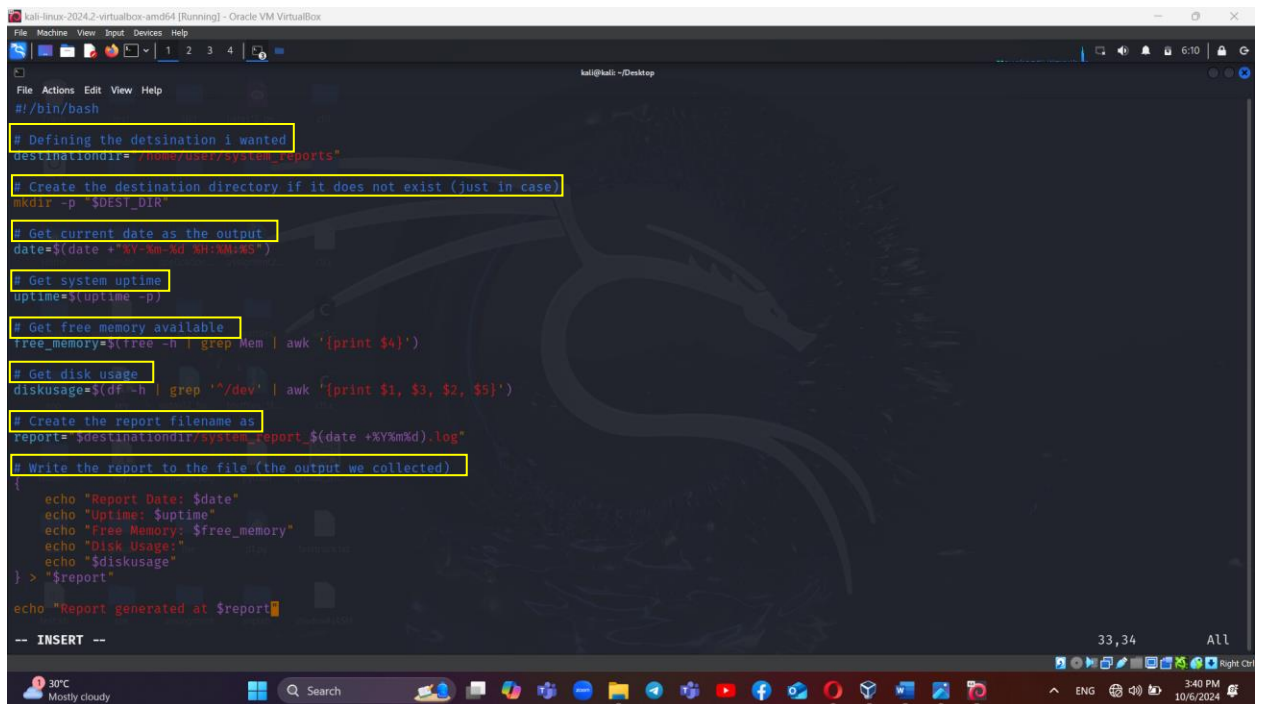
`sudo ip link set eth0 up` (not down) to make my interface up running again.



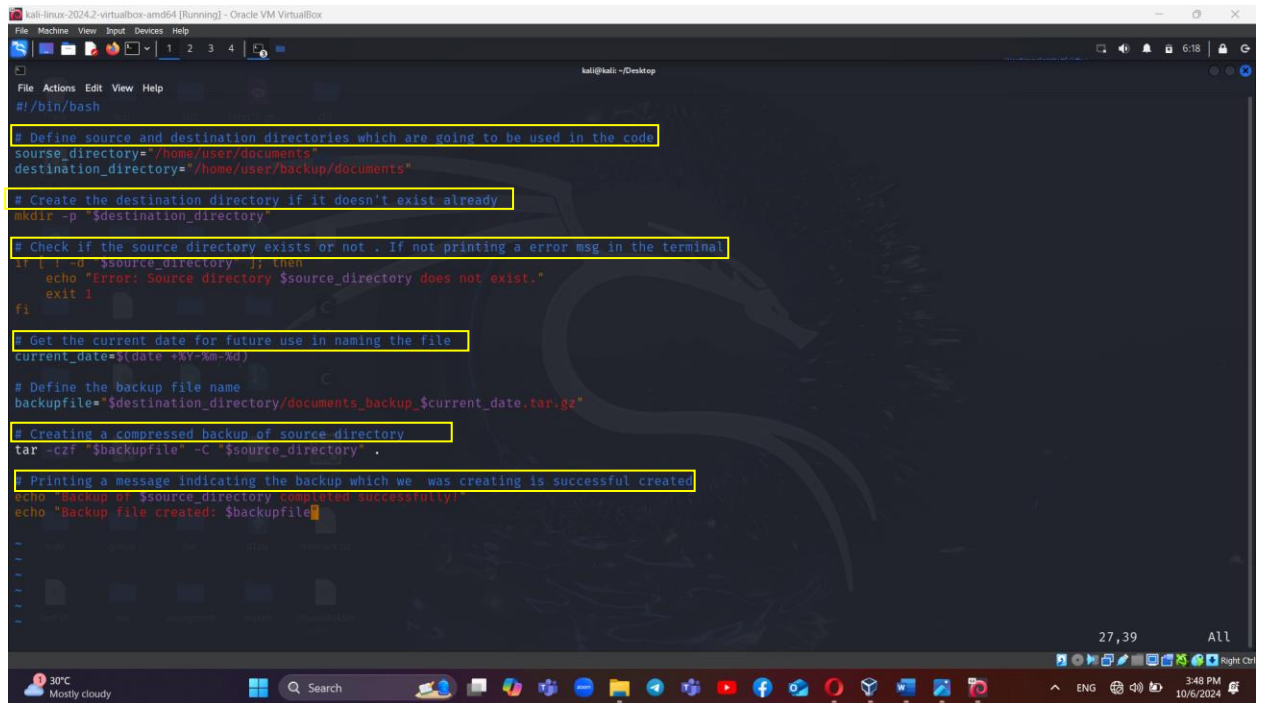
```
(kavindu@kali)-[~]  
$ sudo ip link set eth0 up
```

3. Adding comments and check errors of the scripts and codes and lines.

Adding comments will enhance the readability of the code. I have added comments scripts, files as #lines in above implementations.



```
#!/bin/bash  
  
# Defining the destination I wanted  
destinationdir="/home/user/system_reports"  
  
# Create the destination directory if it does not exist (just in case)  
mkdir -p $DEST_DIR  
  
# Get current date as the output  
date=$(date +"%Y-%m-%d %H:%M:%S")  
  
# Get system uptime  
uptime=$(uptime -p)  
  
# Get free memory available  
free_memory=$(free -h | grep Mem | awk '{print $4}')  
  
# Get disk usage  
diskusage=$(df -h | grep '/dev' | awk '{print $1, $3, $2, $5}')  
  
# Create the report filename as  
report="$destinationdir/system_report_$(date +"%Y%m%d").log"  
  
# Write the report to the file (the output we collected)  
{  
  echo "Report Date: $date"  
  echo "Uptime: $uptime"  
  echo "Free Memory: $free_memory"  
  echo "Disk Usage:"  
  echo "$diskusage"  
} > "$report"  
  
echo "Report generated at $report"  
  
-- INSERT --
```



```
kali@kali: ~/Desktop
# Define source and destination directories which are going to be used in the code
source_directory="/home/user/documents"
destination_directory="/home/user/backup/documents"

# Create the destination directory if it doesn't exist already
mkdir -p $destination_directory

# Check if the source directory exists or not . If not printing a error msg in the terminal
if [ ! -d $source_directory ]; then
    echo "Error: Source directory $source_directory does not exist."
    exit 1
fi

# Get the current date for future use in naming the file
current_date=$(date +%Y-%m-%d)

# Define the backup file name
backupfile="$destination_directory/documents_backup_$current_date.tar.gz"

# Creating a compressed backup of source directory
tar -czf $backupfile -C $source_directory .

# Printing a message indicating the backup which we was creating is successful created
echo "Backup of $source_directory completed successfully"
echo "Backup file created: $backupfile"
```

4.Configure a Firewall

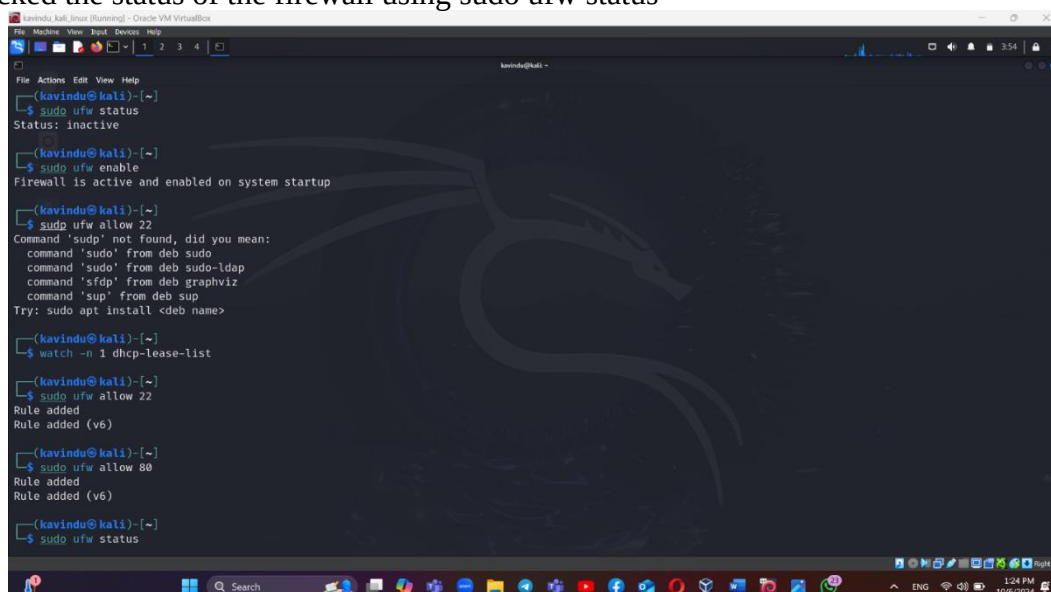
A firewall helps filter incoming and outgoing traffic based on predefined security rules, providing an additional layer of protection. Use iptables or ufw (Uncomplicated Firewall) to create rules that limit access to necessary ports only.

Installed ufw sudo using apt install ufw

Enable ufw and allow necessary ports (e.g., SSH on port 22, HTTP on port 80):

```
sudo ufw allow 22
sudo ufw allow 80
sudo ufw enable
```

Checked the status of the firewall using sudo ufw status



```
(kavindu@kali)-[~]
$ sudo ufw status
Status: inactive

(kavindu@kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

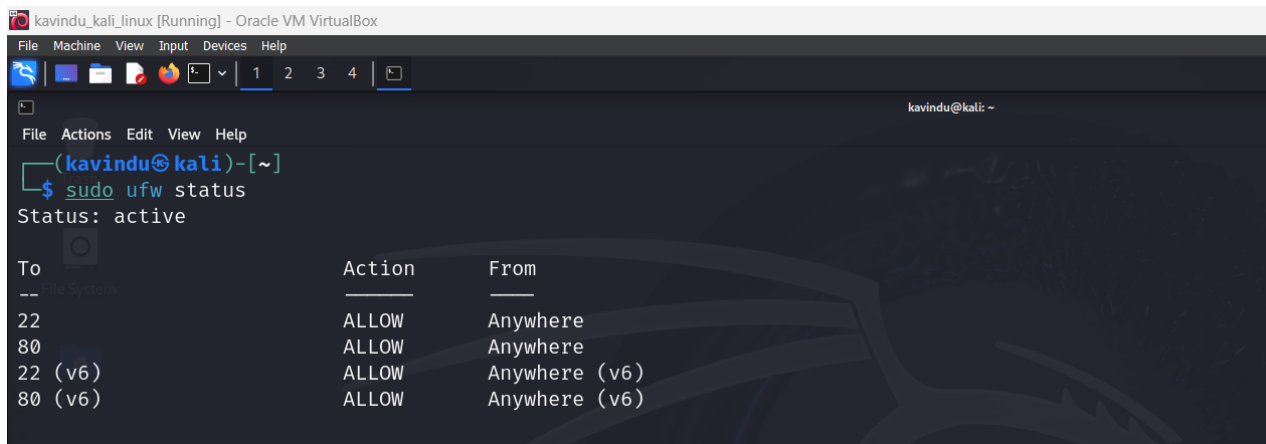
(kavindu@kali)-[~]
$ sudo ufw allow 22
Command 'sudp' not found, did you mean:
command 'sudo' from deb sudo
command 'sudo' from deb sudo-ldap
command 'sfop' from deb graphviz
command 'sup' from deb sup
Try: sudo apt install <deb name>

(kavindu@kali)-[~]
$ watch -n 1 dhcp-lease-list

(kavindu@kali)-[~]
$ sudo ufw allow 22
Rule added
Rule added (v6)

(kavindu@kali)-[~]
$ sudo ufw allow 80
Rule added
Rule added (v6)

(kavindu@kali)-[~]
$ sudo ufw status
```



```
kavindu_kali_linux [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kavindu@kali: ~
File Actions Edit View Help
(kavindu@kali)-[~]
$ sudo ufw status
Status: active

To Action From
--
22 ALLOW Anywhere
80 ALLOW Anywhere
22 (v6) ALLOW Anywhere (v6)
80 (v6) ALLOW Anywhere (v6)
```

5. Use Strong Passwords and Key-Based Authentication

Explanation: Using strong passwords and implementing key-based authentication for SSH access enhances security against brute-force attacks.

Therefore let's Disable password authentication for SSH and enforce key-based access.

Generate SSH keys using: `ssh-keygen`

Copy the public key to the server: `ssh-copy-id user@server_ip` (server ip we use)

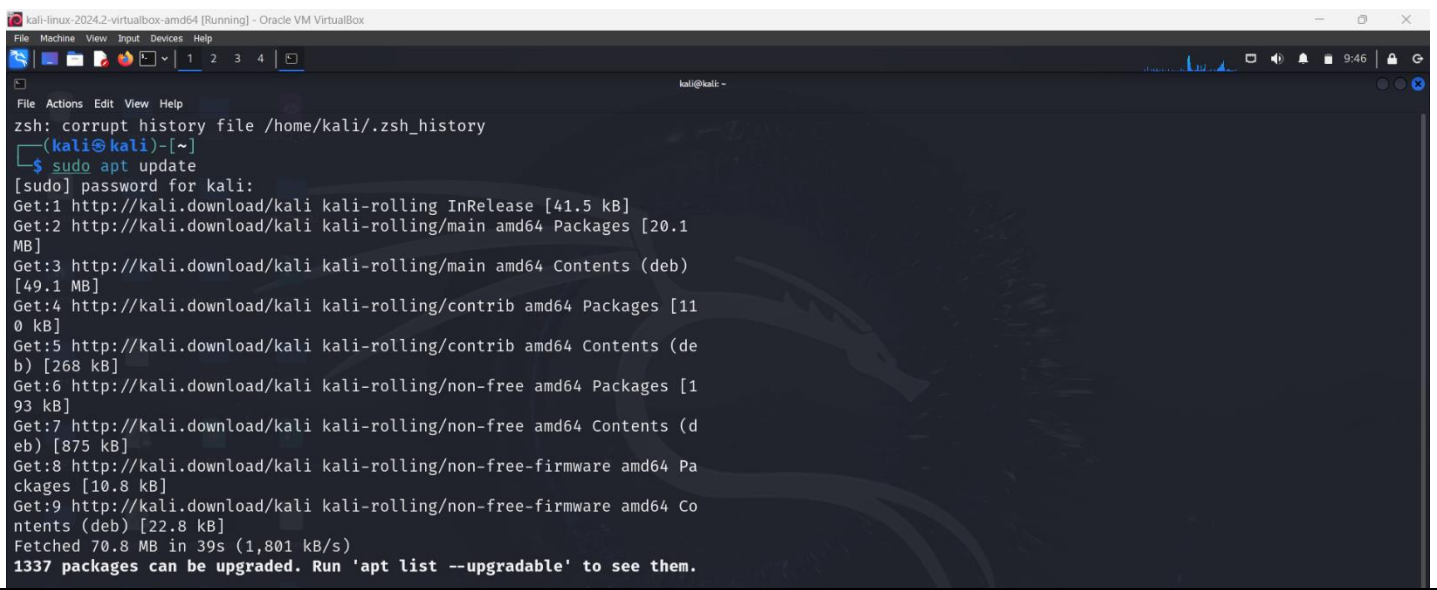
Edit `/etc/ssh/sshd_config` file to set `PasswordAuthentication no` and `PermitRootLogin no`.

Restart the SSH service `sudo systemctl restart sshd`

6. Regularly Update the System

Keeping the system and its packages up to date is crucial for patching known vulnerabilities. Set up automatic updates or regularly check for updates manually.

Install unattended-upgrades for automatic updates: `sudo apt update && sudo apt upgrade`



```
kali-linux-2024.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.1 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [49.1 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [110 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [268 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [193 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [875 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [10.8 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [22.8 kB]
Fetched 70.8 MB in 39s (1,801 kB/s)
1337 packages can be upgraded. Run 'apt list --upgradable' to see them.
```